



US012415991B2

(12) **United States Patent**
Novak et al.

(10) **Patent No.: US 12,415,991 B2**
(45) **Date of Patent: Sep. 16, 2025**

(54) **RECOMBINANT HEME THIOLATE
OXYGENASES**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 842 days.

(21) Appl. No.: **17/625,081**

(22) PCT Filed: **Jul. 6, 2020**

(86) PCT No.: **PCT/EP2020/069020**

§ 371 (c)(1),

(2) Date: **Jan. 5, 2022**

(87) PCT Pub. No.: **WO2021/005013**

PCT Pub. Date: **Jan. 14, 2021**

(65) **Prior Publication Data**

US 2022/0282227 A1 Sep. 8, 2022

(30) **Foreign Application Priority Data**

Jul. 5, 2019 (EP) 19184720

(51) **Int. Cl.**
C12N 9/08 (2006.01)

C12N 15/81 (2006.01)

(52) **U.S. Cl.**
CPC **C12N 9/0065** (2013.01); **C12N 15/815**
(2013.01); **C12N 2830/00** (2013.01); **C12Y**
111/02001 (2013.01)

(58) **Field of Classification Search**
CPC **C12N 9/0065**
See application file for complete search history.

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(57) **ABSTRACT**

The invention relates to polypeptides having peroxygenase activity and compositions comprising such polypeptides. The invention also relates to improved methods of producing such polypeptides in yeasts.

7 Claims, 24 Drawing Sheets

Specification includes a Sequence Listing.

(56)

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Fig. 1

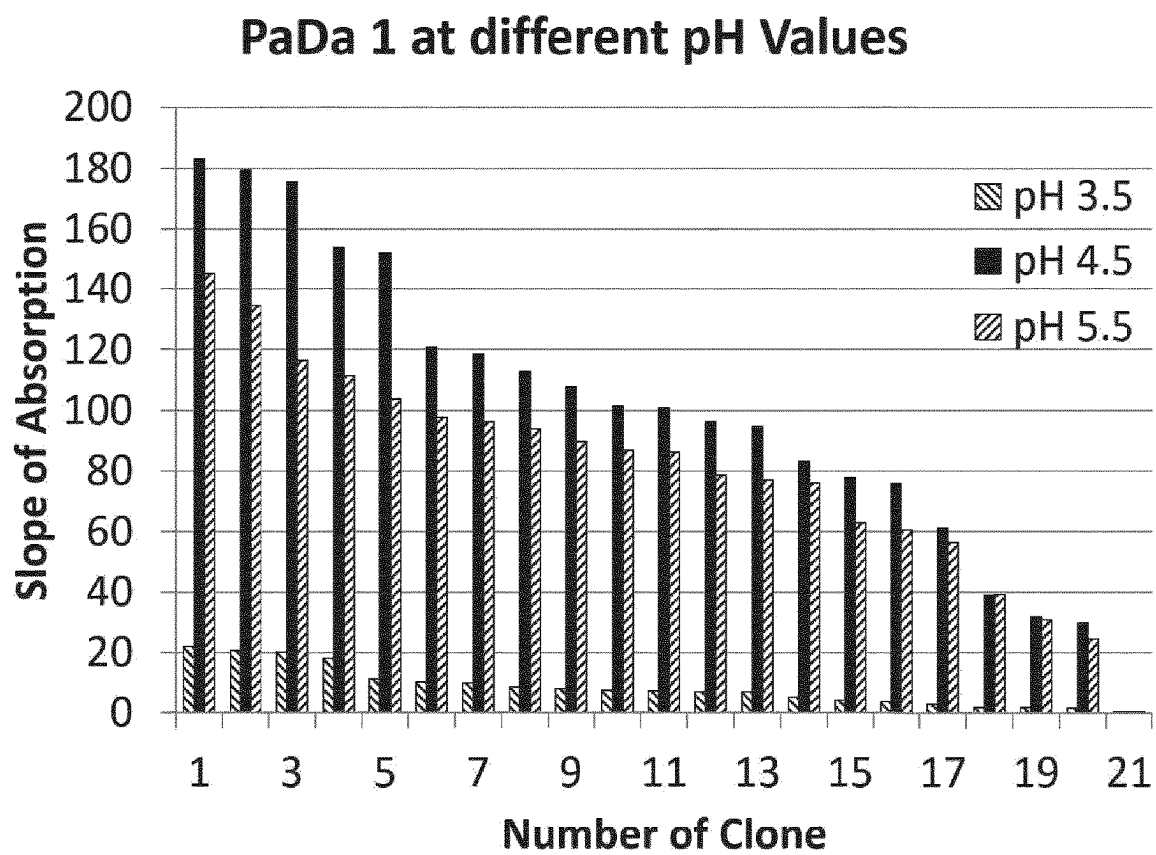


Fig. 2

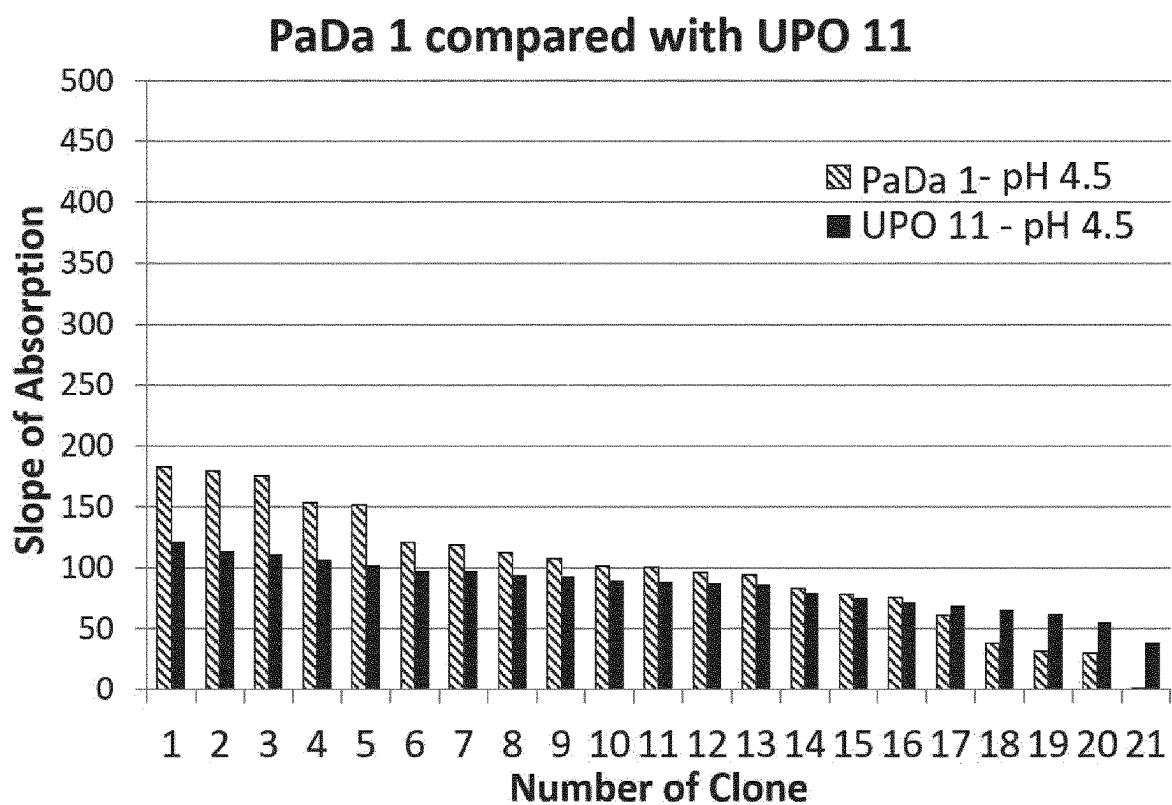


Fig. 3

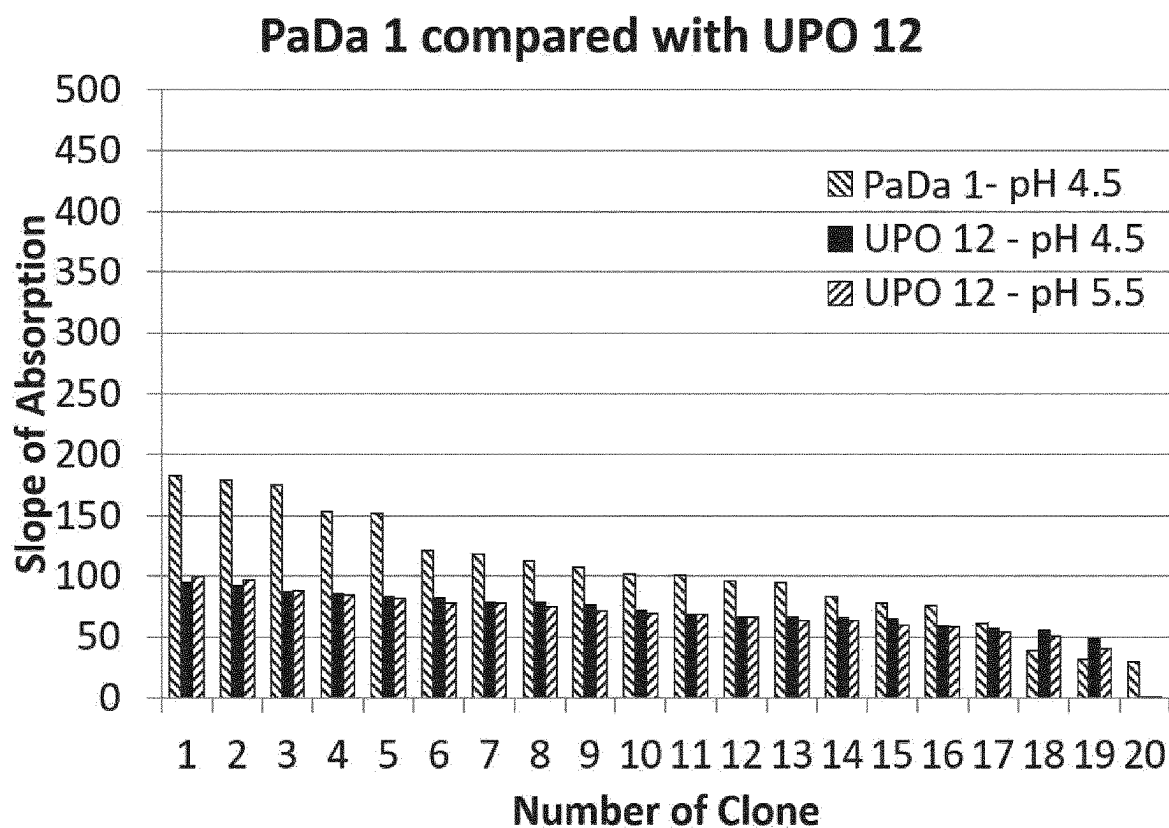


Fig. 4

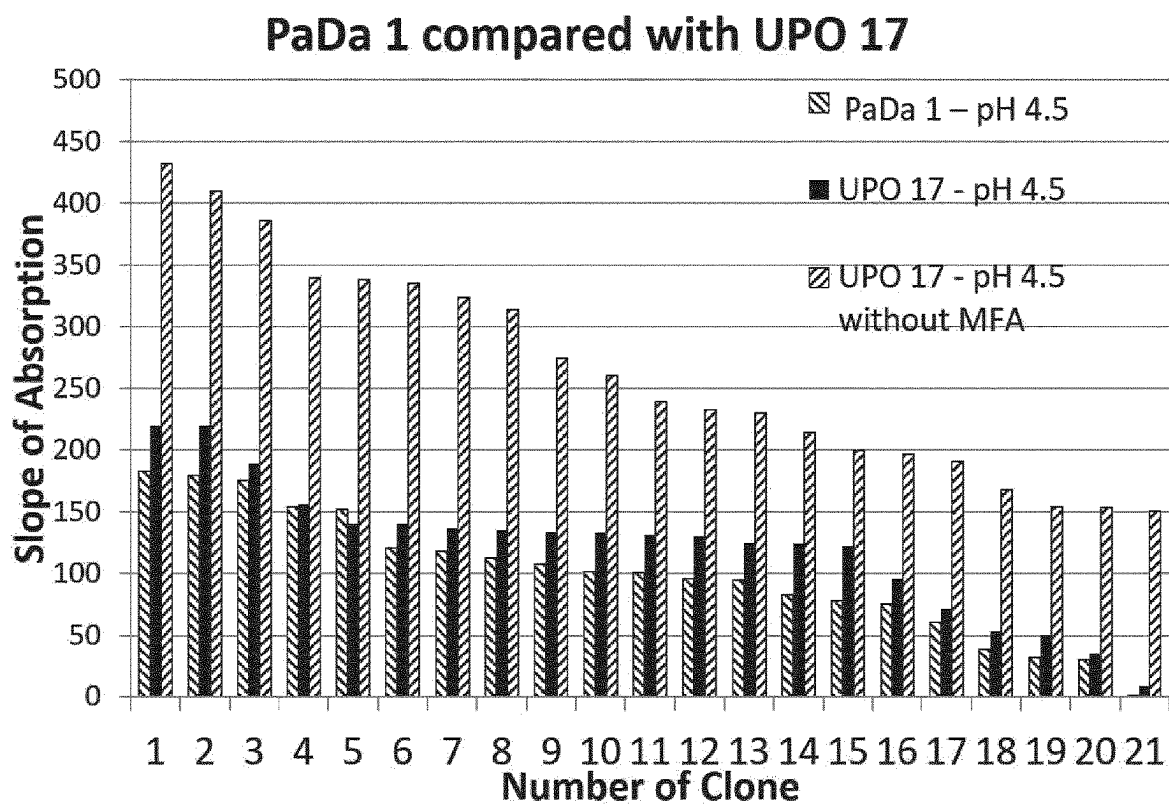


Fig. 5

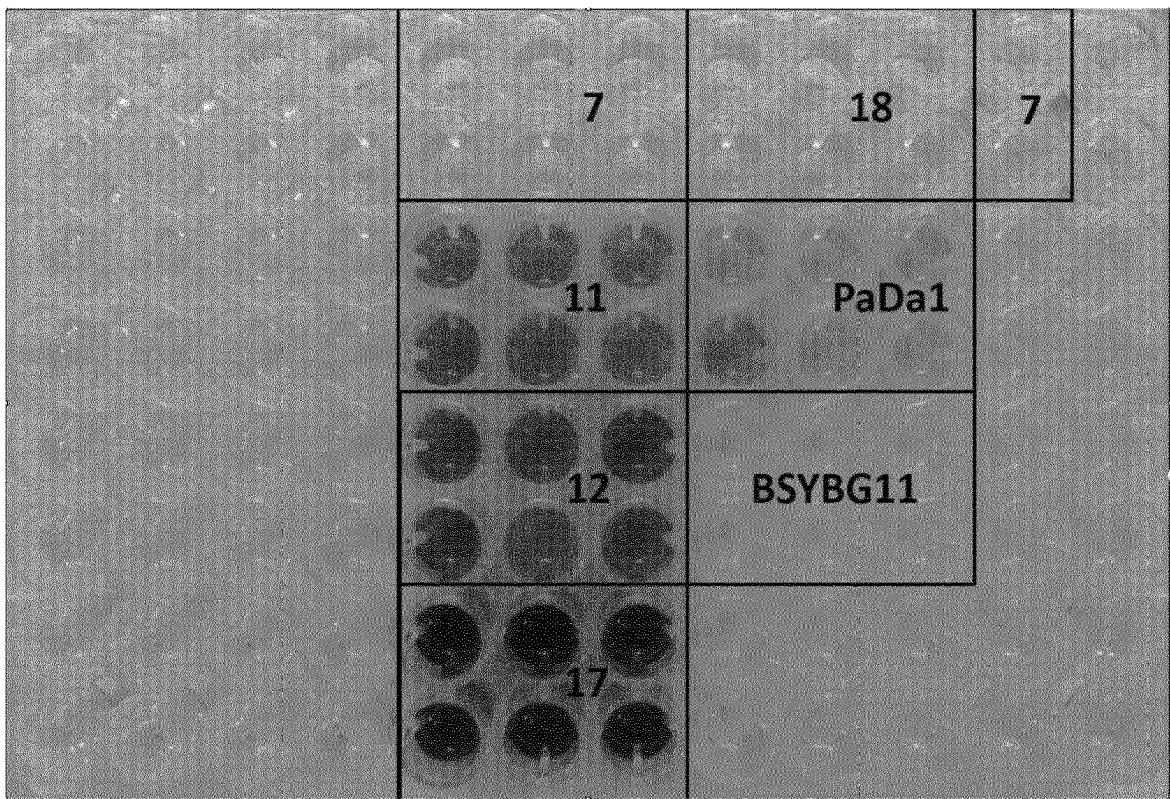


Fig. 6

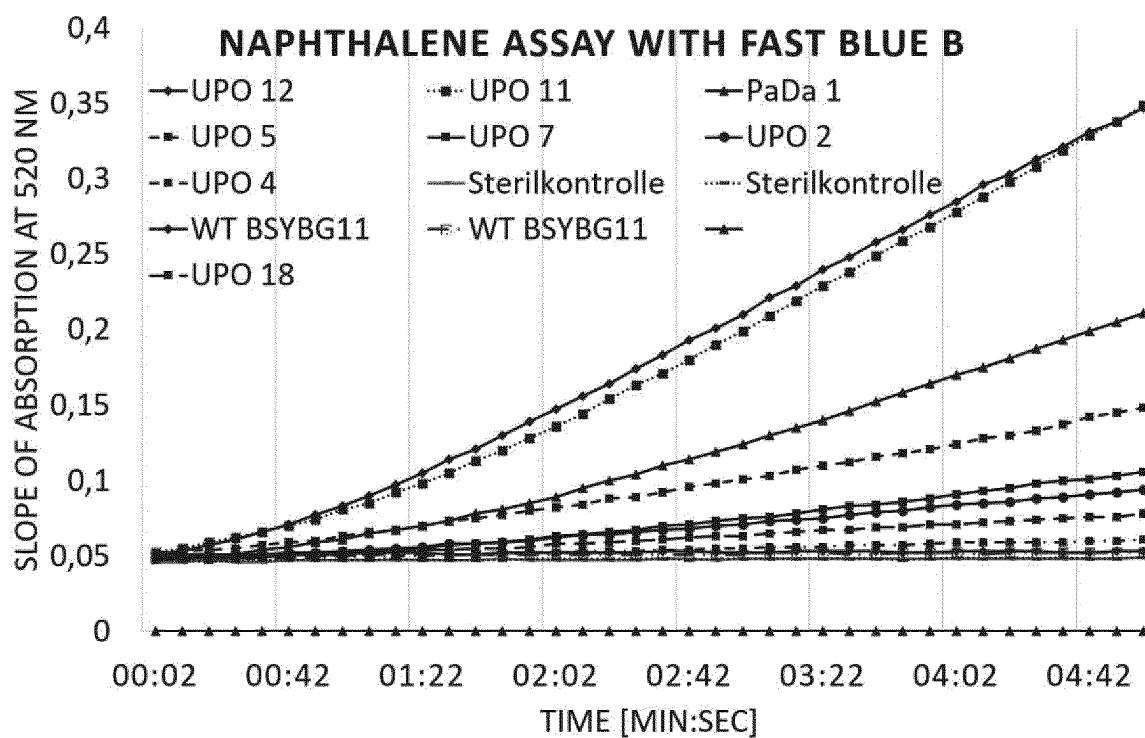


Fig. 7

UPO and CPO sequences

UPO1 mut, PaDa1 variant of *Aae*UPO 1 (SEQ ID NO:1)

MKYFPLFPTLVYAVGVVAFPDYASLAGLSQQELDAIPTLEAREPGLPPGPLENSSAKL
VNDEAHPWKPLRPGDIRGPCPGLNTLASHGYLPRNGVATPAQIINAVQEGFNFDNQAA
IFATYAAHLVDGNLITDLLSIGRKTRLTGPDPPPPASVGGLNEHGTFEGDASMTRGDFAF
FGNNHDFNETLFEQLVDYSNRFGGGKYNLTVAGELRFKRIQDSIATNP NFSFVDFRFF
TAYGETTFPANLFVDGRRDDGQLDMAARSFFQFSRMPDDFFRAPSPRSGTGVEVVV
QAHMPQGRNVGKINSYTVDP TSSDFSTPCLMYEKFVNITVKS LYPNPTVQLRKALNT
NLDFLFQGVAAAGCTQVFPYGRD*

UPO 2 (SEQ ID NO:2)

MARGAARFAVLIALFTHAAIAFPAYGSLAGLTREQLDEILPTLEIREPGKPPGPKDTSK
LVNDKAHPWKPVAPADIRGPCPGLNTLASHGWLPRNGIASPSEIITAVQEGFNMDNGL
AIFVTYAAHLVDGNILTDKLSIGGKTGLTGP NPPAPAI VGGLNTHAVFEGDTSMTRGDF
FFGNNHDFNETLFDEFVDFS NRFGAGKYNLTVAGEFRWQRIQDSIATNPEFSFVSPRF
FTAYAESTFPINFFIDGRQTDGQLDLTVARGFFQNSRMPDDFHRANGTRGTEGIDLVA
EAHPIEPGSNVGGVNNYVVDPTSADFSTFCLLYENFVNKTVKGLYPNPTGALRKALNT
NLGFFFSGISDSGCTQVFPY GK*

UPO 3 (SEQ ID NO:3)

MLKLFFVQTALLALSGTTFAYP SHMSLAGLTREQLDQIVPTLTFTPPPPPPAPLNDTSA
KLVNDPAHPWQPLRAGDIRGVCPGLNTLASHGYLPRNGIVTPNQIIEAAQDGFNMDNT
LARFLAYGTFLVDGNVVTNEMSIGSKSAATGPDPPAPAI VGGLDTHAVFEGDASMTRQ
DFFFGNNHDFNETLFDQFVEFSNRFGAGKYNLTVAGELRHQRIQQSIATNP NFTFVAP
RYFTAFAESAFAFPVDFFIDGRDSNGQLEMDVAR SFFQNSRFPDGGFRPNHSVTGEGSDV
VFAAHPIEPGRNVGGVNNYVLDPTSADFTTPCLLYTNFVNETIVGLYPSPTGDLRTALN
FYLNLFFEAFDNSEGSGCTQLFPY GQD*

UPO 4 (SEQ ID NO:4)

MFSLLNFVTLALACTWSALAFPS SYTSLGGLPREELDRILPSLQYRSPGAPPGPLKFNG
TKLVNDDQHPWKPLKHGDMRGPCPGLNTLASHGYLPRNGIATPVQIINAVQEGFNME
NSVARLVTYAAHLVDGNLVTDKLSIGGKSPLTGPSPPAPANAAGLNTHALFEGDVSM T

RADAFFGDNHSFNETLFDEFTAFSNQFGAGKYNLTVA AEYRFHRIQESIATNP NFSFVS
PRFFTAYAESVFPINFFIDGRQGDGQLDL DVARGFFQNM RMPDGFHRASIPTGLEGLA
EIASVHPISPGANVNGVNTYTFDPSSADFTTFCLLYVNFVNQTVRSLYPEPTGNLKKAL
KKNLEFLYGPFS DQCSQVFPY GKDN*

UPO 5 (SEQ ID NO:5)

MARLTFLVAVALALSSTTVAFPSYGLAGLSEAELDRIIPLLEARDACPPPGPLKNTSTK
LVNDKDHWPWKPLRDGDIRGPCPGLNTLASHGWLPRNGVATPAQIINAVQEGFNMGN
DLAVFVTYAAHLVDGNQVTDLLSIGGKTPQTGPDPQPAIVGGGLDTHAVFEGDASMTR
GDAFFGDNHSFNETQFDEFSAFSNKF GGGYYNLSVAAEFRWQRIQESIATNP NFSFISP
RYFTAYAESVFLVFFVDGRVSDGRLSLPNARGFFQNSQMPTDFFRPNQSIGLDVIGD
GISAIASAHPIAPGKNEG VGNYVLDPTSADFDHFCLLYINFVNQTVKSLYPNPTGVLRD
ALKRNLDNFYSPLNGSDCVQIFPYGK*

UPO 6 (SEQ ID NO:6)

MVQFTVILSLLLATGKALAFPQYGLAGLSERELEDILPRLHAVKPPPPPGPLNDTSTK
LVNNPAHPFLPQRNGDMRGPCPGLNTLASHGYLPRNGIATPAQIINAVQEGFNMGND
LAVFVTYAAFLVDGNQVTNLLSIGGKSSLTGPDPKPAIVGGGLDTHAVFEGDASMTRG
DAFFGDNHSFKENQFDEFIAFSNKF GGGKYNLTVASEFRWQRIQESTATNP NFSFISPR
YFTAYAESTFPITFFVDGRNEDGALS LDVARGFFQDSRMPNGFFRANQSIGLDIIGSLID
FIFEPHPIQPGGNQGRVNSYTVDPNSANFSQFCQLYQDFVNNTVKGLYPNPQGVLRD
NLNTNLGFFFSPLQGSGCPQVFPFGQ*

UPO 7 (SEQ ID NO:7)

MRFFSHLSIIPLLSLHGVLAFPSYGTLAGLSRSELEAILPTLKPGVPESPPGPLNDTSAK
LVNDKKHPWKPA GKKDIRGPCPGLNTLASHGWLPRDGVATPAQIVNAVQEGFNMGN
DLAVFVTYAAHLVDGNLITNLLSIGGKTDRTGPNPPPPAIVGGGLNTHAVFEGDASTTRA
DFFF GDNHSFNETLFDELTAFSNKF GGGFYNLSVAAEFRFQRIQDSIATNPQFDFISPR
YFTAYAESIFPLTFFIDGRDKSLHLD MNVARGFFQNSRFPDGGFFRSNTSITLDVIGGGID
YIFSKHPVPPGSNNGTVNSYTPNPNSADFTQFCKLYTDFVNITIRGLYPNAKGALLTAL
NKNLEYFYSPLVGSGCPQVPPFV*

UPO 8 (SEQ ID NO:8)

MARVFFAIAALLLAAKDVVSFPNYASLAGLSERELDEIIPQLTVRTLKPPGQMKNLT
KLVNDPAHPWIAPAPDDQRDPCPLNTLANHGYLPRDGIATPAQIVNAVQEGFNMAN
DIAVFVTYAAHLVDGNLLTDLLSIGGKSAKTGPNPPSPAIVGGLDTHAVFEGDASTTRG
DAFFGDNHSFNESLFDeltaFSNKFAGFYNLSVATEFRFQRIQDSIATNPQFSLISPR
YYTAYAESVFPVAFFVDGRETNGLNMTVARGFFQDGRMPNDFFRSNISWGLDLIGE
GIGFIFTPHPIEPTNNGTLNSYTLDPNSADFSDFCKLYTDFVNVTVRGLYPNATGPLL
NALNQNLDDFFGPLGDQGCTQVPAFV*

UPO 9 (SEQ ID NO:9)

MKLNIFSTTLAIGLVSAGAHYHQQDVVANGTEGVWIAPTD TDYRGPCPMMNTLANHG
FLPRDGKNLTEYNVVKGLNDGLNFNKSLATIMFQQAI PASPAYPNATFFTLNDLNRHN
VLEHDGSISRSDAYYGNNYIFNQ TIFDTTKAYWPSETLTAQHLIDGKMFRQIVSRSTNP
NYTFSATTQQFSLGEMAAPIVAFGD KYVVTANRTL VESWIENERLPTELGW RKPVEEIL
LSDITYVTEVLGNLTSLYSTVIITPNPDSLAKRQMGHWGQSI*

UPO 10 (SEQ ID NO:10)

MKTTTTLLCLAAALTQTYAFPQQGAPHPLWSPPGPNDVRAPCPMLNTLANHGYLPH
NGKDITERHTINALYNALGIEEELAIYLHQEAVTTNPAPNATTFSLNDLSRHDILEHDAS
LSRQDAYFGDNHDFNQ TIFDETRSYWTSPIIDVKQAAVSRQARVNTSMATNPNYTMS
ELGDSFSYGETAAYIIVLGDKEKGLVNR SRVEYLFENERLPDLGWSRAKENITFDDLS
TMLQRIINATGGEMDFRATIALPRLVYIYYEEA*

UPO 11 (SEQ ID NO:11)

MKTTTTLLFLVGALTQTHAFPQQGVPHPLWSPPGPNDVRAPCPMLNTLANHGFLPH
NGKNITQQHTINALYNALGIDAELATYLHQEAVTTNPVPNATTFSLNDLSRHDILEHDA
SLSRQDAFFGDNHDFNQ TIFNQTRSYWTSPIIDVKQAALARQARVNTSMATNPNYTM
SELGDAFSYGETAAYIIVLGDKEAGLVNR SRVEYLFENERLPVELGWSRARENITFDDLS
TMLNKIINATGGESEFERELAKRGGVHVGVWR*

UPO 12 (SEQ ID NO:12)

MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDVRAPCPMLNTLANHGFLPHDGK
GITVNTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLSRHNILEHDASLSR
QDSYFGPADVFNEAVFNQTKSFWTGD IIDVQMAANARIVRLT SNLTNPEYSLSDLGS
AFSIGESAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLST
QIINAQHFPQSPGKVEKRGDTRCPYGYH*

UPO 13 (SEQ ID NO:13)

MKTLFLLTLAAFTPVFAGFDTWAPPGPYDVRAPCPMLNTLANHGFLPHDGHEITREQ
TENALFDALHIDKMLGSSLFDFAMTTNPVANSTTFSLNDLGHNHNVLEHDASLSRSDA
YFGNTLTFNQTVFDETKSYWTDETVTIEMASNARLARIKTSNATNPTYSMSELGNGFT
KGESAAYVVIFGDKISGTVPRAWVEWLFEIALKTQPSTPSIKPTQTPSSPTRLLLLKRLGR
QLMLIVPRPIRLRVLRNTPPLRLITKNKPREMAPNLLILAVHKRATSMQKR*

UPO 14 (SEQ ID NO:14)

MRTSLLPALAAVSPVLAGFDTWAPPGPYDVRGPCPMLNTLTNHGFFPHDGQDIDRET
TENALFDALHVNKT LASFLRADAYHG SVLAFNHTIFEETKSYWTDETVTLKMAADARY
YRIKSSQATNPTYQMSELGDAFTYGESAAYVVLFGDKESQTVPRSWVEWLFEKEQLP
QHLGWKRPATSFELNDLDKFMALIQNYTQEIEEPSCESRKQRRKPRGPSHFGE*

UPO 15 (SEQ ID NO:15)

MAKFSTLFAFSALAIQAIALPQYRSLAGLSERELEGILPRLNVVTPPPSPPGPPNDTSVK
LVNDAAHPFMPLQDGDIRGPCPGLNTLASHGYLPRNGIATPAQIINAVQDGFSDNGL
ATLLAYATMLVDGNPLTNLMSIGGKSPLTGMDPPQPAIVGGLDTHAVFEGDASMTRA
DFFFGDNHSFNQTLFNQFANFSNQFGDGNYNLT TAEYRFFRIQQSIAENPQFSFISPR
FFTAYFESAFPLVFFVDGRQADGQLSVENATSFFRDMQFPDDFHRADGSQTADLVNN
AATAIFSAHPMQPGGNNGTVNSYTFDPNSANFTEGCKLYTDFVNNVVVPLYPTPQGA
LKVNLNANLGFLFSTFSNCTQVFPYQQ*

UPO 16 (SEQ ID NO:16)

MAKFSTLLALSVLAIQAVAFPQHQLAGLTERELEDLLPRFKPVVPPPPPGPPKDTSVK
LVNDKDHPLYEPLRKGDIRGPCGLNTLASHGYLPRNGVVTPAQIINAVQDGFMDNEL
AILLAYSTMLTDGNVVNTLMSIGQKTPLTGPDPAPPAIVGGLNTHGTFEGDAGLTRAD
FFFGDNHHSFNQTLFNEFVEFSNKFGGGVYNQTVAAEYRFFRIQQSTAENPTFTFVTPR
FVTAYRESVFPFIFFDGRKADGQLSMKDAFGFFNESRMPDGFHRADGSKTADLVGN
ASDAIFAAHPVQPGANAGKVNTYTPDPNSPTDDCGLYETFVNLMVKQYPNPQGVLRT
NLNLNLGFFFQGFPGCTQLFPFGQ*

UPO 17 (SEQ ID NO:17)

MLGIRLVSLAFTGSALAELDFSKWKTRQPGELRAPCPAMNSLANHGFQIRDGKNITV
EGLTPVLKEVFHLSHELAFTVSQGLGLFTALDPSKGVFTLQDLTDRHNVFEHDASLSRE
DAKFGGDQSVLHKGQFQKFMDHFKGEKYISFEAAAKARYAMVQDSRKRNPFTYDV
THRITSYGETIKYLRTIVEPSTGKCPVDWIKILFEQERLPYNEGWRPPTNELSGFSLASE
VLELALITPEKLPVDECLGKGKGKGNCKRRRSYLG I*

UPO 18 (SEQ ID NO:18)

MNPFLLKLAVLALVTAPLAGAFPSHRSLGGLSSEQLDRIPTLKVAPPEGPPPPQDDTS
TRLVDDADHPFMPAGPNDMRGPCGLNTLASHGYLPRNGIATPAQVINATMQGFNM
EFLAKFVTYAAFLVDGNPITNLMSIGGKSDLTGEDPPDPATVGGLNTHAVFEGDASM
TRADAFFGDHHSFNQTLWDGFVDFSNRFGAGKYNLTVATELRIQRIQDSIATNPQFSF
VSPRIFITAYAESTFPINFFIDGRQQDGQLDLDAISFFRDMRYPSGFFRAPKPMGVEGIE
TIIAAHPIPAGANNGAVNTYTPDPHSGDFNSFCTVYTNFVNETIRGLYPSPTGILKDSL N
RNLDLFLHDFVSGCPQIFPWGR*

CPO 19 (SEQ ID NO:19)

MFSKVLPFVGAVAALPHSVRQEPGSGIGYPYDNNTLPYVAPGPTDSRAPCPALNALAN
HGYIPHDGRAISRETLQNAFLNHMGIANSVIELALTNAFVVCEYVTGSDCGDSLNLTL
LAEPHAFEHDHSFSRKDYKQGVANSNDFIDNRNFD AETFQTSLDVVAGKTHFDYADM
NEIRLQRESLSNELDFPGWFTESKPIQNVESGFIFALVSDFNLPDNDENPLVRIDWWKY
WFTNESFPYHLGWHPSPAREIEFVTSASSAVLAASVTSTPSSLPSGAIGPGA EAVPLS
FASTMTPFLLATNAPYYAQDPTLRPQRQA*

CPO 20 (SEQ ID NO:20)

MFSKILPLVGVAALPHWLQLRQEPNSGIGYPYDNHTKPYVHPGPHDSRAPCPALNA
LANHGYIPHNGRAITKENLQNAFLEHMGIGNSVIALALTNAFVVCEYVTGQDCGDTLVN
LTLLSEPHAFEHDHSFSRKDYKQGVSNFNEIVDNRNFDLSTFETSLDVVAGQTHFGYA
EMNQIRLQRESLSNEADFPGWFTESKPIQEVEAGFIFALVSDFNLPDNDENPLVRVDW
WKYWFINESFPYHLGWHPPTPAREIEFVTSASSAILAAVTSTPSSLPSGAIGPGA EAVP
LSFASTMTFP LLATDIPYFAHPTLGPNDKREAAPAPAATTSTATFKNPYLEPIGTQDIK
NQAYVSSKAAAMSSAMAVNKARSL*

UPO 21 (SEQ ID NO:21)

MKYFPLFPTLVFAARVVAFPAYASLAGLSQQELDAIIPTLEAREPGLPPGPLENSSAKLV
NDEAHPWKPLRPGDIRGPCPGLNTLASHGYLPRNGVATPVQIINAVQEGLNFDNQAAV
FATYAAHLVDGNLITDLLSIGRKT RLTPDP PPPASVGGLNEHGT FEGDASMT RGD AF
FGNNHDFNETLFEQLVDYSNRFGGGKYNLTVAGELRFKRIQDSIATNP NFSFVD FRFF
TAYGETTFPANLFVDGRRDDGQLDMAARSFFQFSRMPDDFFRAPSPRSGTGVEVVI
QAHPMQPGRNVGKINSYTV DPTSSDFSTPCLMYEKFVNITVKS LYPNPTVHVRKALNT
NL DFFFQGVAAAGCTQVFPYGRD

UPO22 (SEQ ID NO:22)

MKL VYLSSAVAFGSAIADTAPWEGPGPN DVRGPCPMLNTLANHGFLPHDGKNIHVNK
TVDALSSALNIDPELGSFLHSFAVTANPQPNATWWNLDHLSRHNILEHDASLSRQDAY
FGAPDVFNEAVFNQTKSYWTGDVITLQMAANARLARLMTSNLTNPEYSMSDLGSSFS
IGESVAYVAILGSKETRTPKAYVEYLF EKERLPYELGFKKAETPMTETDLGNLMDELIS
LQHFPQSPGKIAKR SERPSEKRAEKRCPFH

UPO23 (SEQ ID NO:23)

MKTATLLFLAAGLTQTQAFPSQGAAPHLPWSPPGPNDVRAPCPMLNTLANHGYLP
HNGKNITEQHTINALYNALGIDAELSAFLHQEAVTTNPTPNATTFSLNDLSRHDILEHD
ASLSRQDAYFGDNHDFNQTFDETRSYWTSPIIDVKQAALSRQARVNTSMATNP NYT
MSELGASFSYGETAAYIIVLGDKENGLVNR SRVEYLFENERLPLDLGWTRAKENITFDD
LRTMLNRIVNATGGESEFDRELAKRGGVHVGRWRGY

UPO24 (SEQ ID NO:24)

MKTTTLLCLAAALTQTYAFPQQGAPHPLPWSPPGPNDVRAPCPMLNTLANHGYPH
NGKDITERHTINALYNALGIEEELAIYLHQEAVTTNPAPNATTFSLNDLSRHDILEHDAS
LSRQDAYFGDNHDFNQTFDETRSYWTSPIIDVKQAAVSRQARVNTSMATNPNYTMS
ELGDSFSYGETAAYIIVLGDKEKGLVNRSRVEYLFENERLPLDLGWSRAKENITFDDLS
TMLQRIINATGGESEFDRELA KRGGVHVGSWRG

UPO25 (SEQ ID NO:25)

MKTTPLLFFAAGLAQTHAFPSQGGAPHPLPWSPPGPNDVRAPCPMLNTLANHGYPH
HNGKDITEQHTINALYNALGIDAEATY LHQEAVTTNPAPNATTFSLNDLSRHDILEHD
ASLSRQDAFFGDNHDFNQTFDETRSYWTSPIIDVMQAALSRQARVDTSMATNPNYT
MSELGASFSYGETAAYIIVLGDKENGLVNRSRVEYLFENERLPLDLGWTRAKENITFDD
LSTMLNRIVNATGGESEFDRELA KRGGVHVGVKWRGY

UPO12_23E12 (SEQ ID NO:30)

MKSLSFSLALGFGSTLVYSAPSPFSGWQAPGPNDVRAPCPMLNTLANHGFLPHDGK
GITVNKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLSRHNILEHDASLSR
QDSYFGPADVFNEAVFNQTKSFWTGDII DVQMAANARIVRLLTSNLTNPEYSLSDLGS
AFSIGESAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLST
QIINAQHFPQSPGKVEKRGDTRCPYGYH

UPO12_11G3 (SEQ ID NO:31)

MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDVRAPCPMLNTLANHGFLPHDGK
GITVNKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLSRHNILEHDASLSR
QDSYFGPADVFNEAVFNQTKSFWTGDIIYVQMAANARIVRLLTSNLTNPEYSLSDLGS
AFSIGESAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLST
QIINAQHFPQSPGKVEKRGDTRCPYGYH

UPO12_8G3 (SEQ ID NO:32)

MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDVRAPCPMLNTLANHGFLPHDGK
GITVNKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLSRHNILEHDASLSR
QDSYFGPADVFNEAVFNQTKSFWTGD IIDVQMAANARIVRLT SNLTNPEYSLSDLGS
AFSIGESAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGD LST
QIINAQHFPQSPGKVEKRGDTRSPYGYH

UPO12_11H12 (SEQ ID NO:33)

MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDVRAPCPMLNTLANHGFLPHDGK
GITVNKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLSRHNILEHDASLSR
QDSYFGPADVFNEAVFNQTKSFWTGD IIDVQMAANARIVRLT SNLTNPEYSLSDLGS
AFSIGESAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGD LST
QIINAQHFPQSPGKVEKRGDTR

UPO12_13A2 (SEQ ID NO:34)

MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDVRAPCPMLNTLANHGFLPHDGK
GITVNKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLSRHNILEHDASLSR
QDSYFGPADVFNEAVFNQTKSFWTGD IIDVQMAANARIVRLT SNLTNPEYSLSDLGS
AFSIGESAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGD LST
QIINAQHFPQSPGKVEKRGNTRCPYGYH

UPO12_18G3 (SEQ ID NO:35)

MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDVRAPCPMLNTLANHGFLPHDGK
GITVNKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLSRHNILEHDASLSR
QDSYFGPADVFNEAVFNQTKSFWTGD IIDVQMAANARIVRLT SNLTNPEYSLSDLGS
AFSIGESAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGD LST
QIINAQHFPQSPGKVEKRGITRCPYGYH

UPO12_20H11 (SEQ ID NO:36)

MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDVRAPCPMLNTLANHGFPHDGGK
GITVNKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLSRHNILEHDASLSR
QDSYFGPADVFNEAVFNQTKSFWTGDIIQVQMAANARIVRLTSLTNPEYSLSDLGS
AFSIGESAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLDST
QIINAQHFPQSPGKV

POX27_OTB02684.1 [*Hypoxylon* sp. CI-4A] (SEQ ID NO:37)

MKSVQLSALIAFGAKAVYSFPSANAPWSGPGTDDVRGPCPMLNTLANHGFPHSGKG
ITVNKTIDALNAGLNMEADLAALLFDFAVTTNPTPNASYFDLDHLTRHNILEHDASISR
QDSYFGRADILNEAVFNQTKSYWTGELVDIQMAANARVARLMTSLTNPEYSLSDTG
SVFSIGESAAYVGILGDKVSGTVPKTWLIYLFQERLPYELGFKRPVDPFTEDDLFNMS
EAIRDAQHFPQDIGKVTKRGNKARCPHGYCIEVL

POX30_GAP92448.1 [*Rosellinia necatrix*] (SEQ ID NO:38)

MKLTTLLFPAVVLGAACPYGTFKPEEPTDTRGVCPMLNALANHGFLLPRDGRNINENQ
TVTALNNALNLTPDFGRFLFTAGRLSNPKPNSTTFDLNHLDRHNLFEHDGSLSRQDA
HFGQWSRFNQTVWNWTMQYWTGDILDVQMVANGRAQRHTRSNLTNPDYALSVVGY
DFSVAENAALLSIIGDKVTQTCPKKFVDYLFVNEELPYSVGWKKSELPIALEDLIRTFRDI
ELATAFPAPPPPDNSGEIFA

POX32_OTB09996.1 [*Daldinia* sp. EC12] (SEQ ID NO:39)

MKLTFMSSSVTLGSAVAAYPTSWEAPGPNDVRGPCPMLNTLANHGFPHDGNINIV
NNTAEALSKGLNLAWELGVDLHDFAVMANPQPNATTFDLHLSRHNILEHDGSLSR
QDAHFGPPDVFNEAVFNQTVSYWTGDVVTMQMAANARLARLMTSLTNPEYSLSDL
GSGFSIGESVVYLLVLGNKDTAEAPKNYLEYWFRNERLPYELGWERPENVIMTGDDL
NAMDKLVTLQHFPQSPGKITSDEKASAKLAGKRHLFH

POX34_RYP66388.1 [*Monosporascus* sp. CRB-8-3] (SEQ ID NO:40)

MKFELAATILAAGTASAFRLKARDTYDWHPPAYGDVRGPCPMLNTLANHGYLPRNGK
DITENRTIEALGTALSIDSELAQLLFEQAITTNPAPNATTFSLNDLVRHNILEHDASLSRV
DFYFGNPQPFNQTVFAQTRSYWTTPIIDVQQAANARWARVETSNATNPNTLSTLGER
FSYGESAAYIVILGNKITGTVPRDWVEYLFENERLPLEIGWTRRTGSITRNDLEDVMQQI
YAATPNNNATTNSWRGNPRALHMAVRASA

POX39 XP_020060613.1 [Aspergillus aculeatus ATCC 16872] (SEQ ID NO:41)

MRYFVLACAPLLYAVTLAFPRADYVSEGKLPAGHPPLDWKPAGLGDARAPCPMLNTL
ANHGYLPHDGGKDITKAHTIAALHSALNIDRELAQYLFQEALTTNPAANATTFSLNDLSR
HNILEHDASLSRLDYYFGDNHDFNQAIQFDQTRQHWPDPIITVQAAANAREARVRTSNA
TNPTFTLSELGTAFGYGETAAYIIILGNKTSGLVDRSWVEYLFENERLPVELGWTRHEE
AVSMDDLEGMMQEVINATGHAEVVKRELVRRGDLHVGRRA

Fig. 7 continued

Fig. 8

		Peroxygenase	Peroxidase	Peroxygenase : Peroxidase
		Naphthalene -Fast Blue Assay	ABTS Assay	
Construct	Clone	$\Delta\text{ABS}(520\text{nm})/\text{min}/\mu\text{l}$	$\Delta\text{ABS}(405\text{nm})/\text{min}/\mu\text{l}$	Ratio between Activities
UPO 12	12-C0	2,07	4,43	0,47
UPO 12	12-A0	2,05	4,59	0,45
UPO 11	11-A0	1,98	4,35	0,45
UPO 11	11-C0	1,97	4,14	0,48
UPO 11	11-B0	1,85	4,27	0,43
PaDa1	1-B0	1,05	3,86	0,27
PaDa1	1-C0	0,94	3,84	0,24
PaDa1	1-A0	0,81	4,45	0,18
UPO 5	5-A0	0,63	0,04	15,83
UPO 2	2-A0	0,3	0,05	6,36
UPO 2	2-B0	0,28	0,03	10,63
UPO 4	4-B0	0,09	0,01	7
UPO 4	4-A0	0,08	0,03	3,13
UPO 16	16-A0	0,03	0,01	2,25
UPO 15	15-A0	0,02	0,05	0,43
UPO 17	17-A0	0,01	23,29	0
UPO 17	17-B0	0,01	19,45	0
UPO 17	17-C0	0,01	19,01	0

Fig. 9

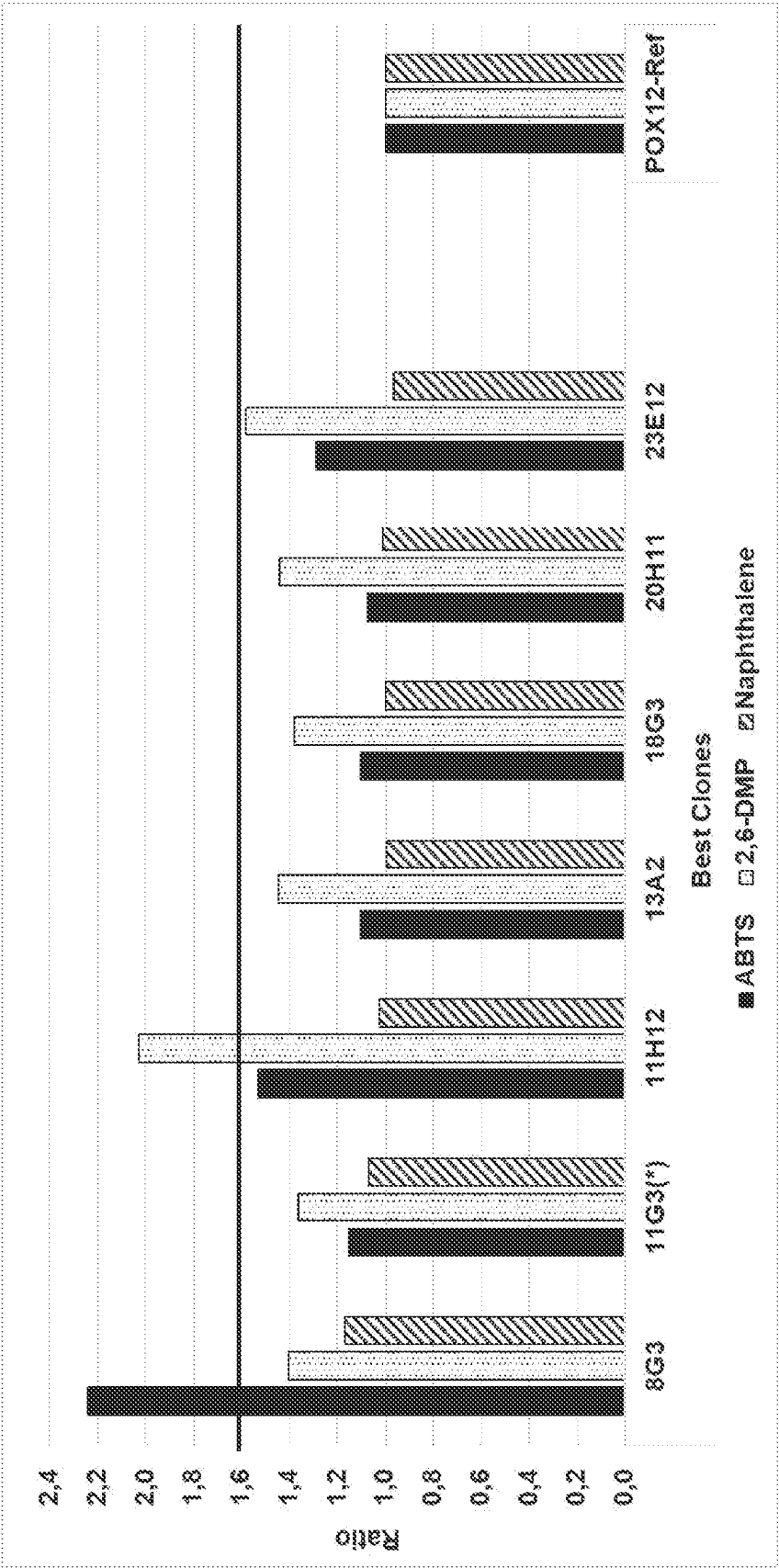


Fig. 10

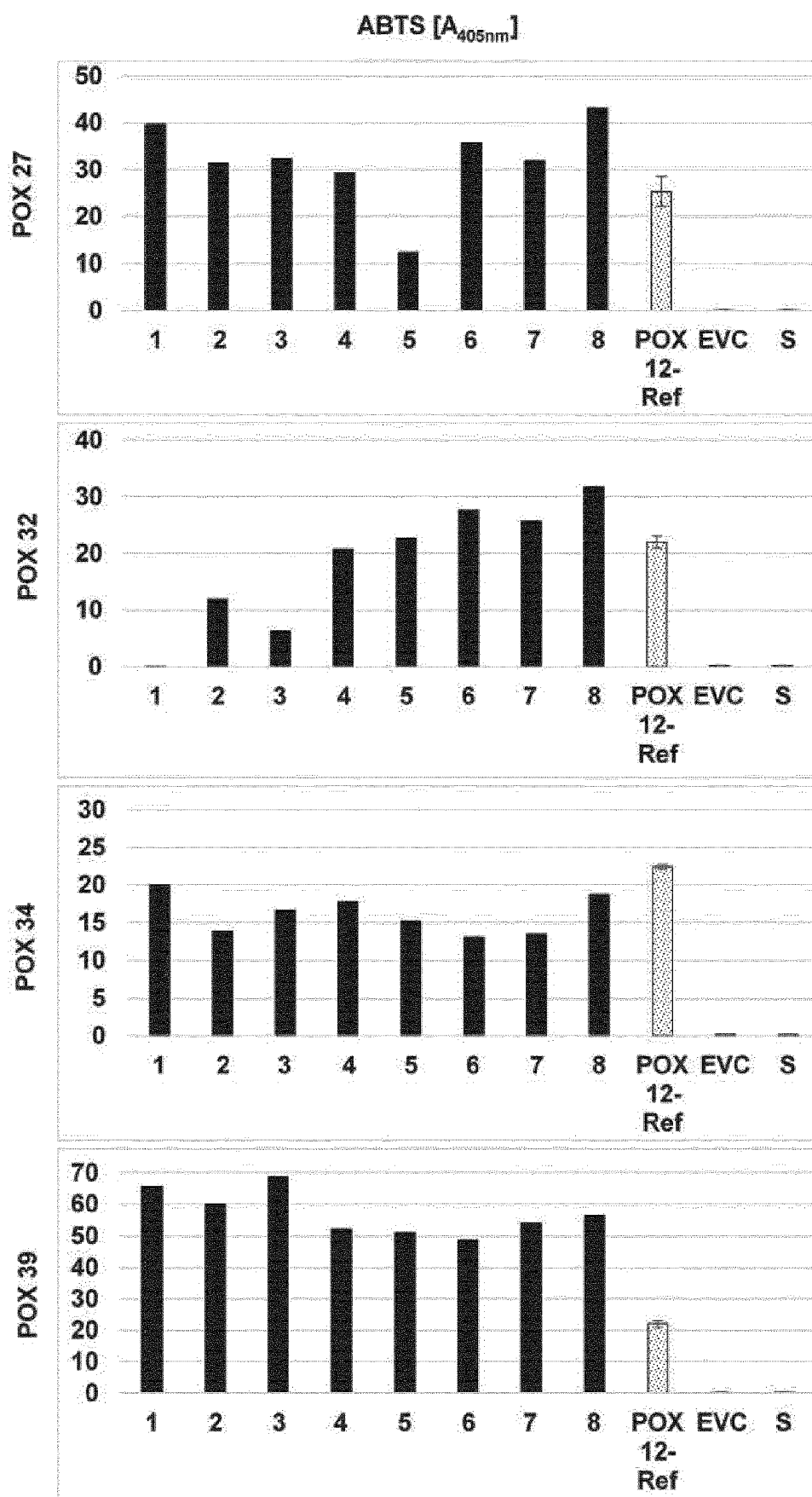


Fig. 10 continued

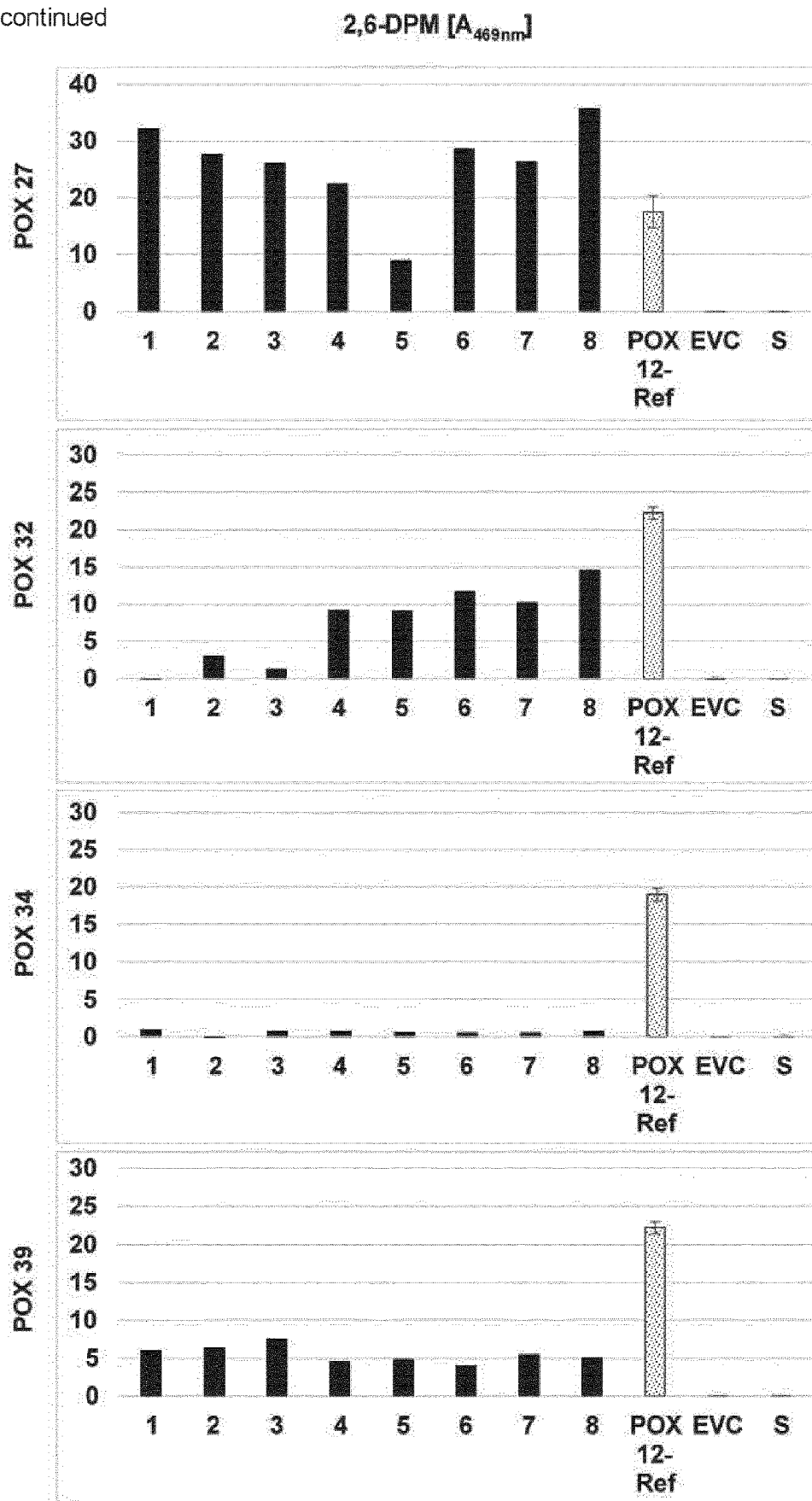


Fig. 10 continued

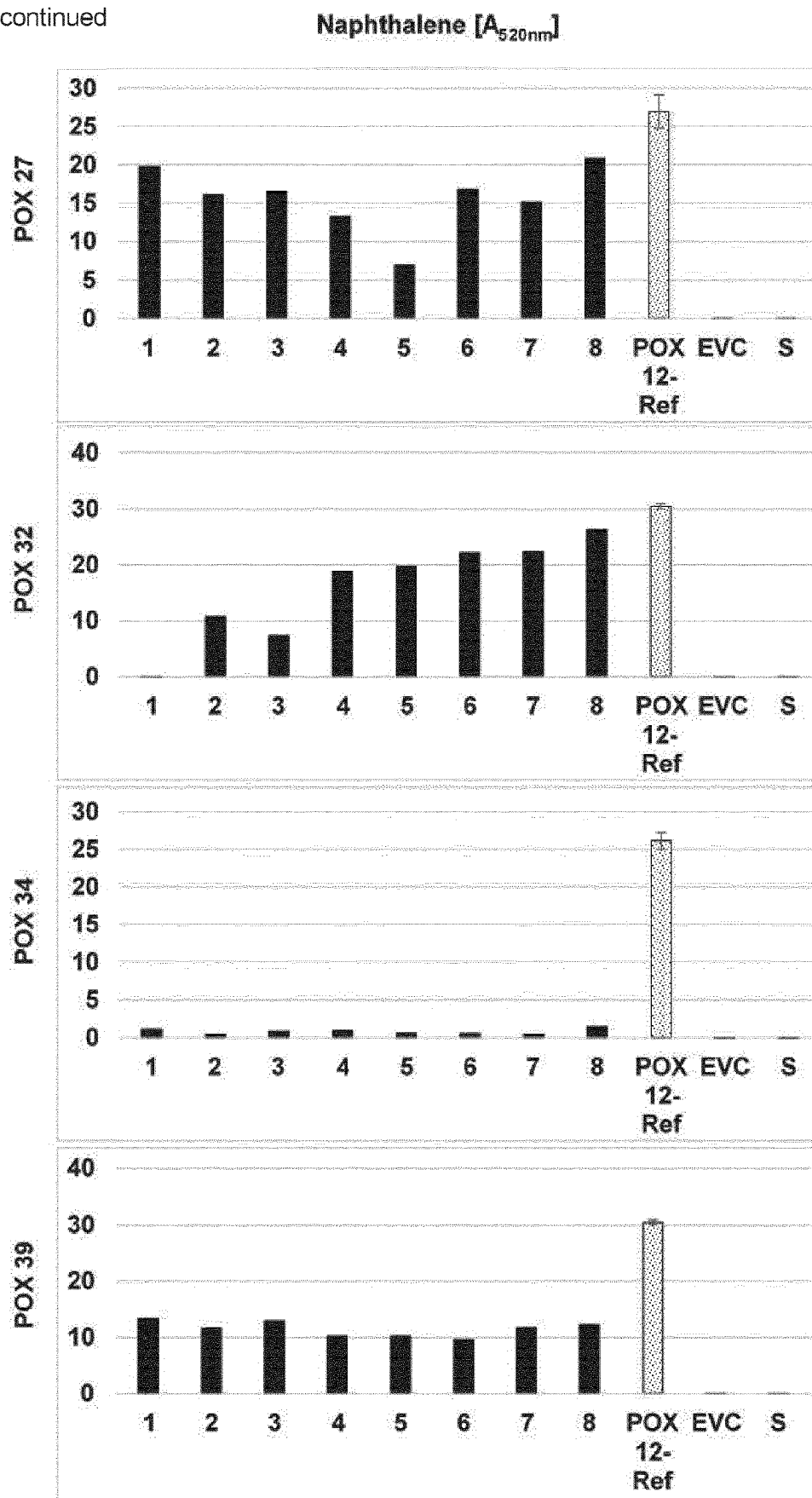


Fig. 11

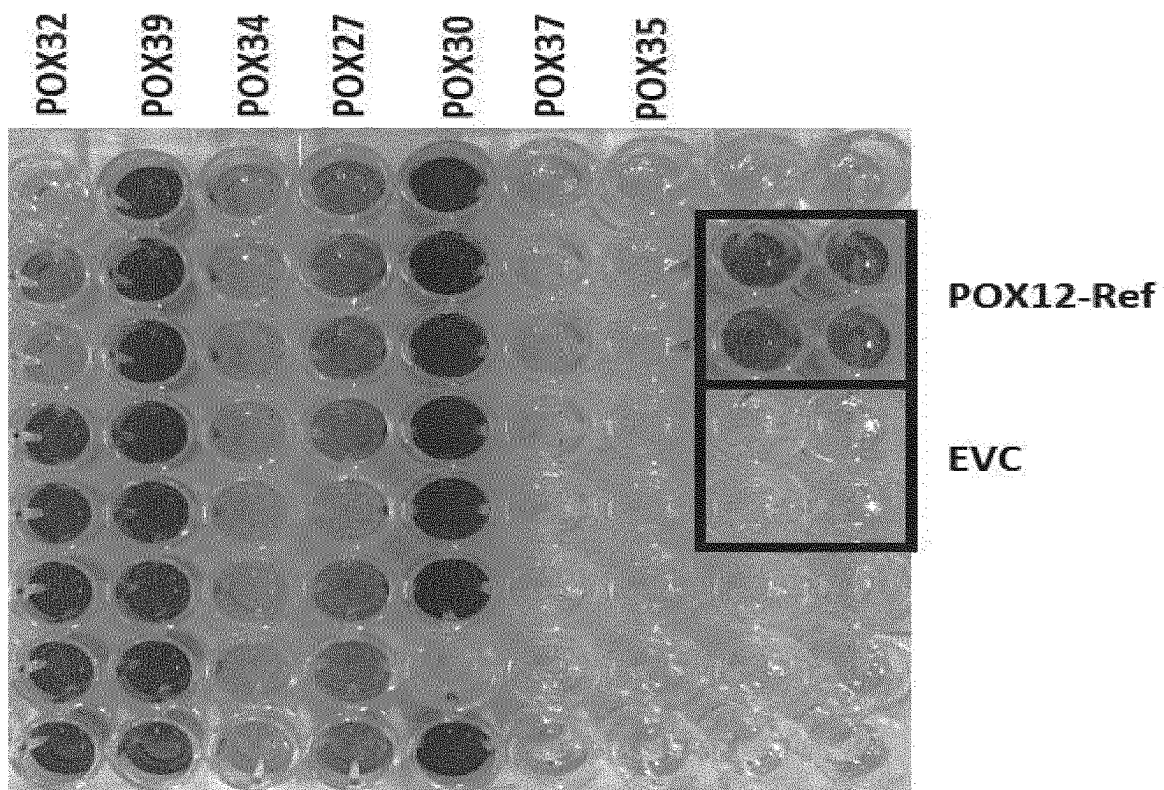


Fig. 12

Alignment UPO12 variants to UPO12 wild type ("POX12")

```
POX12_23E12      MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDRAPCPMLNTLANHGFLPHDGKGITV 60
POX12_11G3       MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDRAPCPMLNTLANHGFLPHDGKGITV 60
POX12_20H11      MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDRAPCPMLNTLANHGFLPHDGKGITV 60
POX12_18G3       MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDRAPCPMLNTLANHGFLPHDGKGITV 60
POX12_13A2       MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDRAPCPMLNTLANHGFLPHDGKGITV 60
POX12            MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDRAPCPMLNTLANHGFLPHDGKGITV 60
POX12_8G3        MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDRAPCPMLNTLANHGFLPHDGKGITV 60
POX12_11H12      MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDRAPCPMLNTLANHGFLPHDGKGITV 60
***** * *****

POX12_23E12      NKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLNRHNILEHDASLSRQDSY 120
POX12_11G3       NKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLNRHNILEHDASLSRQDSY 120
POX12_20H11      NKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLNRHNILEHDASLSRQDSY 120
POX12_18G3       NKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLNRHNILEHDASLSRQDSY 120
POX12_13A2       NKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLNRHNILEHDASLSRQDSY 120
POX12            NKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLNRHNILEHDASLSRQDSY 120
POX12_8G3        NKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLNRHNILEHDASLSRQDSY 120
POX12_11H12      NKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLNRHNILEHDASLSRQDSY 120
***** *****

POX12_23E12      FGPADVFNEAVFNQTKSFWTGDII DVQMAANARIVRLLT SNLTNPEYSLSDLGSAFSIGE 180
POX12_11G3       FGPADVFNEAVFNQTKSFWTGDII VVQMAANARIVRLLT SNLTNPEYSLSDLGSAFSIGE 180
POX12_20H11      FGPADVFNEAVFNQTKSFWTGDII DVQMAANARIVRLLT SNLTNPEYSLSDLGSAFSIGE 180
POX12_18G3       FGPADVFNEAVFNQTKSFWTGDII DVQMAANARIVRLLT SNLTNPEYSLSDLGSAFSIGE 180
POX12_13A2       FGPADVFNEAVFNQTKSFWTGDII DVQMAANARIVRLLT SNLTNPEYSLSDLGSAFSIGE 180
POX12            FGPADVFNEAVFNQTKSFWTGDII DVQMAANARIVRLLT SNLTNPEYSLSDLGSAFSIGE 180
POX12_8G3        FGPADVFNEAVFNQTKSFWTGDII DVQMAANARIVRLLT SNLTNPEYSLSDLGSAFSIGE 180
POX12_11H12      FGPADVFNEAVFNQTKSFWTGDII DVQMAANARIVRLLT SNLTNPEYSLSDLGSAFSIGE 180
***** *****

POX12_23E12      SAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLDLSTQI INAQH 240
POX12_11G3       SAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLDLSTQI INAQH 240
POX12_20H11      SAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLDLSTQI INAQH 240
POX12_18G3       SAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLDLSTQI INAQH 240
POX12_13A2       SAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLDLSTQI INAQH 240
POX12            SAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLDLSTQI INAQH 240
POX12_8G3        SAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLDLSTQI INAQH 240
POX12_11H12      SAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLDLSTQI INAQH 240
***** *****

POX12_23E12      FPQSPGKVEKRGDTRCPYGYH 261
POX12_11G3       FPQSPGKVEKRGDTRCPYGYH 261
POX12_20H11      FPQSPGKVXKRGDTRCPYGYH 261
POX12_18G3       FPQSPGKVEKRGITRCPYGYH 261
POX12_13A2       FPQSPGKVEKRGNTRCPYGYH 261
POX12            FPQSPGKVEKRGDTRCPYGYH 261
POX12_8G3        FPQSPGKVEKRGDTRSPYGYH 261
POX12_11H12      FPQSPGKVEKRGDTRXPYGYH 261
***** * * * *
```

POX 23E12 = SEQ ID NO:30
POX12 11G3 = SEQ ID NO:31
POX12 20H11 = SEQ ID NO:36
POX12 18G3 = SEQ ID NO:35
POX12 13A2 = SEQ ID NO:34
POX12 = SEQ ID NO:12
POX12 8G3 = SEQ ID NO:32
POX12 11H12 = SEQ ID NO:33

Fig. 13

Clustalw Alignment (UPO12 (SEQ ID NO:12) and UPO27 (SEQ ID NO:37), full length protein sequence)

```
UPO12_(SEQ      MKSLSFSLALGFGSTLVYSAPSPSSGWQAPGPNDRVAPCPMLNTLANHGFLPHDGKGITV      60
UPO27_(SEQ      MKSVQLSALIAFGAKAVYSFPSANAPWSGPGTDDVRGPCPMLNTLANHGFLPHSGKGITV      60
                ***:.:*  :.***.  *** **  .:  *.**  :***.*****.*****.*****

UPO12_(SEQ      NKTIDALGSALNIDANLSTLLFGFAATTNPQPNATFFDLHLRHNILEHDASLSRQDSY      120
UPO27_(SEQ      NKTIDALNAGLNMEADLAALLFDFAVTTNPTPNASYFDLHLTRHNILEHDASISRQDSY      120
                *****.:.**.:*:*:*:*:*.*.***  ***.:*****.*****.*****.*****

UPO12_(SEQ      FGPADVFN EAVFNQTKSFWTGDII DVQMAANARIVRLTSLTNPEYSLSDLGSAFSIGE      180
UPO27_(SEQ      FGRADILNEAVFNQTKSYWTGELVDIQMAANARVARLMTSLTNPEYSLSDTGSVFSIGE      180
                **  *:.*****.***:.*:*:*:*:*:*.*.*****.***.*****

UPO12_(SEQ      SAAYIGILGDKKSATVPKSWVEYLFENERLPYELGFKRPNDPFTTDDLGLSTQIINAQH      240
UPO27_(SEQ      SAAYVGILGDKVSGTVPKTLIYLFEQERLPYELGFKRPVDPFTEDDLFNMSEAIRDAQH      240
                ****:*****  *.***:.*:  ***:*****.*****  ***  ***  :.*  *  :***

UPO12_(SEQ      FPQSPGKVEKRGD-TRCPYGYH*---      261
UPO27_(SEQ      FPQDIGKVTGRGNKARCPHGYCIEVL      266
                ***.  ***  ***:  :***:**
```

Percent Identity Matrix - created by Clustal2.1

#

```
1: UPO12__SEQ  100.00   72.41
2: UPO27__SEQ   72.41  100.00
```

Clustalw Alignment (UPO12 (SEQ ID NO:12) and UPO27 (SEQ ID NO:37), without predicted signal sequence)

```
UPO12      APSPSSGWQAPGPNDRVAPCPMLNTLANHGFLPHDGKGITVNKTIDALGSALNIDANLST  60
UPO27      FPSANAPWSGPGTDDVRGPCPMLNTLANHGFLPHSGKGITVNKTIDALNAGLNMEADLAA  60
                **  .:  *.**  :***.*****.*****.*****.*****.*****.*****.*****

UPO12      LLFGFAATTNPQPNATFFDLHLRHNILEHDASLSRQDSYFGPADVFNEAVFNQTKSEW  120
UPO27      LLFDFAVTTNPTPNASYFDLHLTRHNILEHDASISRQDSYFGRADILNEAVFNQTKSYW  120
                ***.***.***  ***.:*****.*****.*****.*****  ***.:*****.***

UPO12      TGDII DVQMAANARIVRLTSLTNPEYSLSDLGSAFSIGESAAYIGILGDKKSATVPKS  180
UPO27      TGELVDIQMAANARVARLMTSLTNPEYSLSDTGSVFSIGESAAYVGILGDKVSGTVPKT  180
                ***:.*:*:*:*:*:*.*.*****.*****  ***.*****.*****  *.***:

UPO12      WVEYLFENERLPYELGFKRPNDPFTTDDLGLSTQIINAQHFPQSPGKVEKRGD-TRCPY  239
UPO27      WLIYLFEQERLPYELGFKRPVDPFTEDDLFNMSEAIRDAQHFPQDIGKVTGRGNKARCPH  240
                *:  ***:*****.*****  ***  ***  :.*  *  :*****.  ***  ***:  :***:

UPO12      GYH*---  242
UPO27      GYCIEVL  247
                **
```

Percent Identity Matrix - created by Clustal2.1

#

```
1: UPO12      100.00   74.38
2: UPO27      74.38  100.00
```

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RECOMBINANT HEME THIOLATE OXYGENASES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is the U.S. national stage of International Patent Application No. PCT/EP2020/069020, filed on Jul. 6, 2020 and entitled RECOMBINANT HEME THIOLATE OXYGENASES, which claims the benefit of priority under 35 U.S.C. § 119 from European Patent Application No. 19184720.1, filed Jul. 5, 2019. The disclosures of the foregoing applications are incorporated herein by reference in their entirety.

SEQUENCE LISTING

The entire content of a Sequence Listing titled "Sequence_Listing.txt," created on Jan. 3, 2022 and having a size of 97 kilobytes, which has been submitted in electronic form in connection with the present application, is hereby incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

The present invention relates to recombinant polypeptides having peroxxygenase activity, their encoding polynucleotides, expression vectors and recombinant host cells comprising such polynucleotides or vectors. The present invention also relates to the use of the recombinant polypeptides as catalysts.

BACKGROUND ART

In the field of synthetic chemistry is oxygenation of organic molecules one of the major tasks. Oxygen-transfer-ring enzymes can be used to solve this task through biocatalysis. In addition to cytochrome P450 enzymes, flavin dependent monooxygenases or di-iron dioxygenases, unspecific peroxxygenases, called UPOs, or also POX due to their peroxidase activity, have the ability to transfer oxygen selectively to a wide range of substrates, such as polycyclic aromatic hydrocarbons, heterocycles, benzene derivatives, alkenes as well as linear and cyclic alkanes. Other reactions catalyzed by UPOs include double bond epoxidations, dealkylations, oxidation of inorganic halides as well as organic hetero atoms and also typical peroxidase reactions including radical based polymerization. Further they can even use pesticides or complex drug molecules as substrate. As such, applications of UPOs can be numerous; they reach from pharmaceutical production to environmental applications, including environmental problems caused by industry. For example, transformation of pollutants through peroxidases can result in reduction of toxicity or bioavailability. Also, removal of pollutants from water can be achieved.

UPOs belong to the peroxidase-peroxxygenase group with haloperoxidases such as the *Caldariomyces fumago* chloroperoxidase (CfuCPO) as first and long known representative. Recombinant production of CfuCPO is also possible in suitable hosts such as *Aspergillus* spp., but no successful recombinant CfuCPO expression in yeast was published so far. CfuCPO has been the only heme-thiolate peroxidase characterized on the protein level for almost 50 years, before more versatile heme thiolate peroxidases were discovered. In the past decade a new subgroup of enzymes accepting especially aromatic substrates was described. One typical example for aromatic substrate conversion is the formation

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of 1- and 2-naphthol with naphthalene as substrate. 1-naphthol plays an important role in the production of pharmaceuticals, herbicides, and others. The first aromatic peroxxygenase (AaeUPO) described was derived from the mushroom *Agrocybe aegerita* oxidizing similar substrates as CfuCPO, typical peroxidase substrates and aromatic alcohols and aldehydes. AaeUPO has the unique ability to epoxidize and hydroxylate aromatic rings efficiently by using hydrogen peroxide as oxygen donor.

In spite of the high technological potential and interest in these new secreted heme thiolate enzymes, their recombinant expression showed to be challenging and largely unsolved. Bormann et al. (2015) reported that attempts to express CfuCPO in *Escherichia coli*, *S. cerevisiae*, or *Pichia pastoris* (*Komagataella phaffi*) did not yield active enzyme. Although recombinant expression was successful in *Aspergillus niger*, the enzyme levels of few mg/L were significantly lower than with the native host. Molina-Espeja and M. Alcalde (2014) for the first time reported recombinant expression and engineering of AaeUPO in *S. cerevisiae*, still with very low enzyme yields of less than 0.01 mg/L but the first successful overexpression of a fungal heme thiolate peroxxygenase by a yeast species. Employing the native signal sequence of AaeUPO1 resulted in 2-times higher secretion efficiency than the signal sequence of the *S. cerevisiae* mating factor alpha. No other UPOs could be functionally expressed as secreted enzymes by any yeast species and in general very few recombinant UPOs (expressed by filamentous fungi) were known so far. Directed evolution of AaeUPO1 (also named UPO1) in *S. cerevisiae* resulted in a mutant enzyme with increased activity and higher recombinant production yields (up to 217 mg/L in *P. pastoris*). This mutated sequence was also the first UPO which was successfully expressed by *P. pastoris* (Molina et al (2015)) using a methanol inducible AOX1 promoter and secreted to the culture supernatant. The AOX1 promoter is not a derepressed promoter and relies on methanol to obtain significant expression levels. Without the addition of methanol, the AOX1 promoter shows far less than 1% activity. No other UPO had been overexpressed by *P. pastoris* before and no natural heme thiolate peroxxygenase with UPO or CPO activity had been successfully expressed by *P. pastoris*.

WO2008/119780 discloses polypeptides having peroxxygenase activity. The polypeptides may be produced recombinantly in *Aspergillus oryzae*.

Thus, there is still the need for an effective expression system for producing novel unspecific peroxxygenase enzymes (UPOs) in high yields and high enzyme activity.

SUMMARY OF INVENTION

It is the objective of the present invention, to provide novel recombinant polypeptides with peroxxygenase activity, showing at least complementary activities and properties to known native UPOs and the UPO1 variants developed by Molina et al (2015).

It is further a specific objective of the present invention to provide polypeptides and polypeptide preparations having increased peroxxygenase activity compared to the respective native UPOs, and to provide means and methods of their production in yeast cells.

The problem is solved by the present invention.

According to the invention, there is provided a method for producing a polypeptide having peroxxygenase activity, comprising:

a. cultivating a yeast cell in a medium conducive for the production of said polypeptide, wherein the yeast cell

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comprises a polynucleotide comprising a nucleic acid sequence encoding said polypeptide operably linked to a derepressed promoter sequence which is functional in methylotrophic yeasts, and

b. isolating said polypeptide from the cultivation medium. Specifically, the derepressed promoter sequence is a methanol-independent promoter.

A further embodiment relates to the method as described herein, wherein said promoter is an engineered or synthetic promoter variant.

A further embodiment relates to the method as described herein, wherein the promoter is a CTA1 (PDC) or FMD promoter.

A further embodiment relates to the method as described herein, wherein the expression and/or secretion is increased by co-expression of helper proteins.

A further embodiment relates to the method as described herein, wherein the helper protein is PDI.

A further embodiment relates to the method as described herein, wherein said yeast cell is a *Pichia pastoris* (*Komagaella phaffii*) cell.

A further embodiment relates to the method as described herein, wherein said polypeptide is obtained in a yield of about 1 mg/L, 10 mg/L, 50 mg/L, or of about 100 mg/L, or of about 250 mg/L.

Specifically, employing the method described herein, said polypeptide having peroxygenase activity, specifically a heme thiolate peroxygenase such as any of the unspecific peroxygenases (UPOs) described herein, is expressed at a yield of at least 250 mg/L.

A further embodiment relates to the method as described herein, wherein said polypeptide is obtained in the culture supernatant in a titer of about 300 mg/L, or of about 0.5 g/L, or of about 1 g/L.

A further embodiment relates to the method as described herein, wherein the polypeptide having peroxygenase activity comprises an MF-alpha signal sequence ("mating factor alpha" signal sequence).

Further provided herein is a method of producing a polypeptide having peroxygenase activity, comprising:

a. cultivating a methylotrophic yeast cell, preferably *Pichia pastoris*, in a medium conducive for the production of said polypeptide, wherein the yeast cell comprises a polynucleotide comprising a nucleic acid sequence encoding said polypeptide operably linked to a promoter sequence which is functional in methylotrophic yeasts, and

b. isolating said polypeptide from the cultivation medium.

Specifically, said promoter is an engineered or synthetic promoter variant. Specifically, said promoter is a CTA1 (PDC), FMD or AOX1 promoter.

Specifically, expression and/or secretion of the polypeptide having peroxygenase activity is increased by co-expression of helper proteins, preferably PDI.

Specifically, the polypeptide having peroxygenase activity comprises a MF-alpha signal sequence ("mating factor alpha" signal sequence).

Specifically, the yeast cell is a *Pichia pastoris* (*Komagaella phaffii*) cell.

One embodiment of the invention relates to a polypeptide which has a peroxygenase activity which is obtained by a method as described herein.

One embodiment of the invention relates to a polypeptide having peroxygenase activity selected from the group consisting of a polypeptide comprising an amino acid sequence having at least 70% sequence identity to the polypeptide of SEQ ID NO:1 (UPO1 mut), SEQ ID NO:2 (UPO2), SEQ ID

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NO:4 (UPO4), SEQ ID NO:5 (UPO5), SEQ ID NO:7 (UPO7), SEQ ID NO:11 (UPO11), SEQ ID NO:12 (UPO12), SEQ ID NO:17 (UPO17), SEQ ID NO:18 (UPO18), SEQ ID NO:19 (UPO19), SEQ ID NO:22 (UPO22), SEQ ID NO:23 (UPO23), SEQ ID NO:24 (UPO24), or SEQ ID NO:25 (UPO25).

One embodiment of the invention relates to a polypeptide having peroxygenase activity comprising an amino acid sequence having at least 75%, 80%, 85%, 90%, 95%, 97%, 98%, or 99% identity to the polypeptide of SEQ ID NO:1 (UPO1 mut), SEQ ID NO:2 (UPO2), SEQ ID NO:4 (UPO4), SEQ ID NO:5 (UPO5), SEQ ID NO:7 (UPO7), SEQ ID NO:11 (UPO11), SEQ ID NO:12 (UPO12), SEQ ID NO:17 (UPO17), SEQ ID NO:18 (UPO18), SEQ ID NO:19 (UPO19), SEQ ID NO:22 (UPO22), SEQ ID NO:23 (UPO23), SEQ ID NO:24 (UPO24), or SEQ ID NO:25 (UPO25).

One embodiment of the invention relates to a polypeptide having peroxygenase activity comprising or consisting of the amino acid sequence of SEQ ID NO:1 (UPO1 mut), SEQ ID NO:2 (UPO2), SEQ ID NO:4 (UPO4), SEQ ID NO:5 (UPO5), SEQ ID NO:7 (UPO7), SEQ ID NO:11 (UPO11), SEQ ID NO:12 (UPO12), SEQ ID NO:17 (UPO17), SEQ ID NO:18 (UPO18), SEQ ID NO:19 (UPO19), SEQ ID NO:22 (UPO22), SEQ ID NO:23 (UPO23), SEQ ID NO:24 (UPO24), or SEQ ID NO:25 (UPO25).

One embodiment of the invention relates to a polypeptide comprising an amino acid sequence having at least 70% sequence identity to the polypeptide of SEQ ID NO:12 (UPO12).

One embodiment of the invention relates to a polypeptide as described herein having increased peroxygenase activity when compared to a control peroxygenase (SEQ ID NO:1 (UPO1)), wherein the activity is about 10-fold, 20-fold, or 50-fold when measured in an ABTS assay.

One embodiment of the invention relates to the use of a polypeptide having peroxygenase activity as defined herein as peroxygenase, specifically as a catalyst in organic synthesis processes, polymerization processes, drug metabolite production, environmental application, application in consumer products,

One embodiment of the invention relates to a recombinant polypeptide having peroxygenase activity and peroxidase activity, wherein the ratio between peroxidase activity and peroxygenase activity is about 1:1, 1:2, 1:3, 1:4, or 1:5 when peroxidase activity is expressed as ABTS units and peroxygenase activity is expressed as naphthalene units.

One embodiment of the invention relates to a recombinant polypeptide having peroxidase activity, with said peroxidase is active in a broad range of pH activity as determined by an ABTS assay.

One embodiment of the invention relates to a recombinant polypeptide having peroxygenase activity and peroxidase activity, wherein the KM value for hydrogen peroxide is about 1 mM or lower.

One embodiment of the invention relates to a polypeptide having peroxygenase activity and comprising an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, or 99% sequence identity to the polypeptide of SEQ ID NO:12 and comprising at least one amino acid modification, and wherein the polypeptide has increased peroxygenase activity when compared to UPO12 (SEQ ID NO:12). Preferably, said modification is at least one amino acid substitution in the sequence of SEQ ID NO:12. Specifically, the peroxygenase activity is about 1.0-fold, 1.1-fold, 1.2-fold, 1.3-fold, 1.4-fold, 1.5-fold, 1.6-fold, 1.7-fold,

1.8-fold, 1.9-fold, or 2.0-fold or more increased when measured in an ABTS assay and/or 2,6-DMP assay.

A specific embodiment of the invention relates to a polypeptide having peroxxygenase activity and comprising an amino acid sequence having at least 70% sequence identity to the polypeptide of SEQ ID NO:12 and comprising one or more amino acid substitutions in the C-terminal region of SEQ ID NO:12 ranging from positions 130 to 261, preferably positions 145 to 261, of SEQ ID NO:12, wherein the polypeptide has increased peroxxygenase activity when compared to UPO12 (SEQ ID NO:12).

Specifically, provided herein is a modified unspecific peroxxygenase (UPO) comprising an amino acid sequence having at least 70% sequence identity to the polypeptide of SEQ ID NO:12 and having increased peroxxygenase activity as compared to the unmodified wild-type UPO12, wherein the modification is a modification of at least one amino acid corresponding to any one of amino acids 145-261 of the unspecific peroxxygenase of SEQ ID NO:12.

Specifically, the modified unspecific peroxxygenase comprises a sequence having at least about 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, or 99% identity to SEQ ID NO:12.

According to a specific embodiment, the modification is a modification of at least one amino acid corresponding to any one or more of amino acids C256, D253, E249, and/or D145 of the unspecific peroxxygenase of SEQ ID NO:12. Specifically, the modified UPO comprises at least a mutation corresponding to C256S, D253N, D253I, and/or D145Y.

According to a further specific embodiment, the modification comprises introduction of a stop codon, preferably by an amino acid substitution, and/or comprises deletion of one or more amino acids, preferably at the C-terminus. Specifically, introduction of a stop codon is at a position corresponding to C256 or E249 of SEQ ID NO:12, in other words a modification corresponding to C256X or E249X, see for example SEQ ID NO:33 and SEQ ID NO:36.

According to a further specific embodiment, the modification comprises fusion to one or more N-terminal and/or C-terminal tags. Specific examples of such tags include but are not limited to fluorescent tags, such as a GFP tag or m-Cherry tag, and/or His-tags.

Specifically, the modified UPO comprises SEQ ID NO:31, SEQ ID NO:32, SEQ ID NO:33, SEQ ID NO:34, SEQ ID NO:35, or SEQ ID NO:36, or an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, or 99% identity to SEQ ID NO:31, SEQ ID NO:32, SEQ ID NO:33, SEQ ID NO:34, SEQ ID NO:35, or SEQ ID NO:36.

A further specific embodiment of the invention relates to a modified unspecific peroxxygenase (UPO) having increased peroxxygenase activity as compared to the unmodified wild-type UPO12, wherein the modified UPO comprises SEQ ID NO:30, or an amino acid sequence having at least 70%, 75%, 80%, 85%, 90% or 95% sequence identity to SEQ ID NO:30. Specifically, said UPO comprises a modification of at least one amino acid corresponding to the amino acid at position S24 of the UPO of SEQ ID NO:12. Specifically, said modification is an amino acid substitution corresponding to S24F.

According to a specific embodiment, the modified UPO comprising SEQ ID NO:30, or an amino acid sequence having at least 70% sequence identity to SEQ ID NO:30, comprises an additional modification of at least one amino acid corresponding to any one or more of amino acids at positions C256, D253, E249, and/or D145 of the UPO of SEQ ID NO:12. More specifically, said UPO further com-

prises one or more mutations corresponding to C256S, C256X, E249X, D253N, D253I, and/or D145Y.

Specifically, the peroxxygenase activity of the modified UPO described herein is about 1.0-fold, 1.1-fold, 1.2-fold, 1.3-fold, 1.4-fold, 1.5-fold, 1.6-fold, 1.7-fold, 1.8-fold, 1.9-fold, or 2.0-fold or more increased when measured in an ABTS assay and/or in a DMP assay as described herein.

One embodiment of the invention relates to an isolated polypeptide having peroxxygenase activity, wherein the polypeptide comprises SEQ ID NO:37 (POX27 or UPO27), SEQ ID NO:38 (POX30 or UPO30), SEQ ID NO:39 (POX32 or UPO32), SEQ ID NO:40 (POX34 or UPO34) or SEQ ID NO:41 (POX39 or UPO39), or an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, or 99% identity to comprises SEQ ID NO:37 (POX27), SEQ ID NO:38 (POX30), SEQ ID NO:39 (POX32), SEQ ID NO:40 (POX34) or SEQ ID NO:41 (POX39).

Further provided herein is the use of the polypeptides having peroxidase activity described herein as peroxxygenase, specifically they are used in a method employing a biocatalyst having peroxxygenase activity, which is the polypeptide having peroxidase activity as described herein.

Specifically, the isolated polypeptide having peroxxygenase activity, wherein the polypeptide comprises SEQ ID NO:37 (POX27 or UPO27), SEQ ID NO:38 (POX30 or UPO30), SEQ ID NO:39 (POX32 or UPO32), SEQ ID NO:40 (POX34 or UPO34) or SEQ ID NO:41 (POX39 or UPO39), or an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, or 99% identity to comprises SEQ ID NO:37 (POX27), SEQ ID NO:38 (POX30), SEQ ID NO:39 (POX32), SEQ ID NO:40 (POX34) or SEQ ID NO:41 (POX39) is used as peroxxygenase.

Specifically, the polypeptides described herein comprising or consisting of the amino acid sequence of SEQ ID NO:1 (UPO1 mut), SEQ ID NO:2 (UPO2), SEQ ID NO:4 (UPO4), SEQ ID NO:5 (UPO5), SEQ ID NO:7 (UPO7), SEQ ID NO:11 (UPO11), SEQ ID NO:12 (UPO12), SEQ ID NO:17 (UPO17), SEQ ID NO:18 (UPO18), SEQ ID NO:19 (UPO19), SEQ ID NO:22 (UPO22), SEQ ID NO:23 (UPO23), SEQ ID NO:24 (UPO24), or SEQ ID NO:25 (UPO25) or the polypeptides described herein comprising an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, or 99% identity to the polypeptide of SEQ ID NO:1 (UPO1 mut), SEQ ID NO:2 (UPO2), SEQ ID NO:4 (UPO4), SEQ ID NO:5 (UPO5), SEQ ID NO:7 (UPO7), SEQ ID NO:11 (UPO11), SEQ ID NO:12 (UPO12), SEQ ID NO:17 (UPO17), SEQ ID NO:18 (UPO18), SEQ ID NO:19 (UPO19), SEQ ID NO:22 (UPO22), SEQ ID NO:23 (UPO23), SEQ ID NO:24 (UPO24), or SEQ ID NO:25 (UPO25) are used as peroxxygenase.

Specifically, the newly identified peroxxygenase UPO27 (SEQ ID NO:37) has about 72% sequence identity to the peroxxygenase UPO12 (SEQ ID NO:12). Without the signal sequence, UPO27 has about 74% sequence identity to UPO12.

According to a specific embodiment of the invention, the modified unspecific peroxxygenases described herein, and the isolated polypeptides having peroxxygenase activity described herein are produced according to the method described herein.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1: Activity landscapes of 21 different PaDa1 mutant transformants at different pH values. While the mutant

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samples converted ABTS only really slow at pH 3.5, the conversion was more than 10 times faster at pH 4.5 for several clones.

FIG. 2: Comparison of the slope of absorption of the PaDa1 mutant of AaeUPO1 variant PaDa1 (UPO1) and UPO 11 both at pH 4.5.

FIG. 3: Comparison of the slope of absorption of the PaDa1 mutant of AaeUPO1 variant PaDa1 (UPO1) and UPO 12, both at pH 4.5. Also, UPO 12 at pH 5.5 is compared and shown.

FIG. 4: Comparison of the slope of absorption of the PaDa1 mutant, UPO 17 and UPO 17 without mating factor alpha but with the native signal. All constructs were measured at pH 4.5.

FIG. 5: ABTS peroxidase assay to compare PaDa1 mutant of AaeUPO1 containing the evolved signal peptide for secretion with the PaDa1 mutant of AaeUPO1 containing a native signal as well as with the UPOs 7, 8, 11, 12 (linked to the short mating factor alpha signal) and UPO17 with its native signal peptide. As a control BSYBG11 was applied on the same microtiter plate. A dark color can be observed for UPO 17, a little less dark for UPO 12 and 11, indicating either high specific peroxidase activity and/or high expression especially for UPO17, but also for other UPOs. The PaDa1 mutant of AaeUPO1 with evolved signal showed low intensity coloring (indicating low expression in this specific experiment), the PaDa1 mutant of AaeUPO1 with the native signal sequence showed no coloring that could be observed with the eye. UPOs 7 and 18 as well as empty control strain BSYBG11 also showed no coloring in the peroxidase assay. The assay solution was performed in 200 mM citrate buffer at pH 4.5.

FIG. 6: Diagram of different UPOs converting naphthalene by oxygenation, followed by hydroxy naphthol detection with fast blue, measured photometrically by absorption at 520 nm over 5 minutes.

FIG. 7: Sequence of selected constructs

FIG. 8: Comparison of peroxygenase and peroxidase activities of selected constructs in the Naphthalene—Fast Blue assay and ABTS assay.

FIG. 9: Activity of selected UPO12 variants in relation to wild type UPO12 (clone 1G). Substrates: ABTS, 2,6-DMP, naphthalene; Cultivation: 96 hours in shake flask (48 hours growth/derepression, 48 hours MeOH induction).

FIG. 10: Novel POXs (POX27, POX32, POX34, POX39). Screening results of 8 clones per enzyme. Substrates: ABTS, 2,6-DMP, naphthalene; Cultivation: 96 hours DWP cultivation (48 hours growth/de-repression, 48 hours MeOH induction).

FIG. 11: ABTS-Assay (2.0 mM H₂O₂) results novel POXs (POX27, POX32, POX34, POX39). Eight clones of each variant studied using an 8-fold H₂O₂ access.

FIG. 12: ClustalW Alignment of wildtype UPO12 and UPO12 variants.

FIG. 13: ClustalW Alignment of newly identified peroxygenase UPO27(POX27) and wildtype UPO12.

DESCRIPTION OF EMBODIMENTS

Unless indicated or defined otherwise, all terms used herein have their usual meaning in the art, which will be clear to the skilled person.

The terms “comprise”, “contain”, “have” and “include” as used herein can be used synonymously and shall be understood as an open definition, allowing further members or parts or elements. “Consisting” is considered as a closest

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definition without further elements of the consisting definition feature. Thus “comprising” is broader and contains the “consisting” definition.

The term “about” as used herein refers to the same value or a value differing by +/-5% of the given value.

As used herein and in the claims, the singular form, for example “a”, “an” and “the” includes the plural, unless the context clearly dictates otherwise.

Peroxidases are divided in four superfamilies, the peroxidase catalase superfamily, the peroxidase-cyclooxygenase superfamily, the peroxidase-chlorite dismutase superfamily and the peroxidase-peroxygenase superfamily (Zámocký et al. 2015).

Peroxidases carry iron (III) protoporphyrin IX as prosthetic group and in general catalyze the oxidation of various organic and inorganic compounds and the reduction of peroxides as H₂O₂.

According to Zámocký et al. (2015) peroxidases catalyze four different reactions:



In Reaction 1 electron donors (AH₂) are oxidized to radicals (AH) while H₂O₂ is reduced to water. Reaction 2 shows halides as two-electron donors (X⁻). These are oxidized to hypohalous acids (HOX). The third reaction shows the release of oxygen when a second hydrogen peroxide is used as electron donor. The fourth reaction shows the introduction of oxygen functionalities into organic molecules. Reactions 1 and 2 are common peroxidation reactions, Reaction 3 can be observed only in few heme peroxidases and Reaction 4 is a peroxygenation like reaction, additionally to their peroxidative activity, found in UPOs. The peroxygenase activity reaction mechanism of UPOs is similar to the peroxide shunt pathway of cytochrome P450 enzymes (Zámocký et al. 2015) and bacterial intracellular P450 peroxygenases such as OleT.

According to phylogenetic analyses UPO sequences consist of the motifs (PCP---EGD---R---E) required for the enzyme activity. Both, UPO and CPO have the PCP motif which is required for catalytic activity. The distal cavity of both enzymes consists of a negatively charged glutamic acid residue, which is stabilized by histidine in case of CPO and arginine in case of AaeUPO. In LfuCPO (=CfuCPO) this H105 is involved in the mechanism of its peroxidase function, participating in the cleavage of hydrogen peroxide. The third required motif for catalytic activity in AaeUPO is EGD, which is EHD in CPO. The extended conserved motif for AaeUPO is -PCP-EGD-R--E, and for MroUPO and CPO is -PGP-EHD-E. According to Faiza et al. 2019, most of the putative fungal UPOs reside in Basidiomycota phylum of fungal kingdom. Interestingly MroUPO was placed along with the LfuCPO and some other CPO sequences in the phylogenetic tree. Two new motifs were identified namely, the S [IL] G motif located between the PCP and the EGD motifs and SXXRXD motif present after the EGD motif, except in MroUPO. According to their analysis a II UPOs consist Ile in S [IL] G motif except three species: *Jaapia argillacea* mucI33604, *Mixia osmundae* iam14324, and *Sphaerulina musiva* so2202, which contain Leu in place of Ile. This motif was predicted to be relevant for specific substrate selectivity. Thr55 in AaeUPO was predicted to be

a critical amino acid residue possibly responsible for driving the functional divergence of UPOs from the CPOs.

Only few wild-type UPOs, including isolated enzymes from *Coprinellus radians*, *Marasmius rotula* and *A. aegerita* have been characterized biochemically. Although more UPOs have been identified based on sequence similarities, these proteins were not isolated and biochemically characterized in detail yet.

So far UPOs were excluded from different possible industrial applications due to missing suitable heterologous expression system. Attempts to functionally express native UPOs in *P. pastoris* failed or showed nearly undetectable levels of expression (Molina-Espeja et al., 2015) and isolation of such recombinant enzyme from the culture supernatant was not feasible (Molina-Espeja et al. 2015). A wild-type peroxygenase of *C. cinerea* was expressed heterologously in *A. oryzae* (Babot et al., 2013). In one case expression of stable, soluble AaeUPOs in *S. cerevisiae* and *P. pastoris* was brought to an acceptable level through directed evolution over several generations. The activity was measured mainly through ABTS assays with 0.3 mM ABTS and 2 mM hydrogen peroxide (Molina-Espeja et al., 2015).

Further studies showed that there is a similarity of around 30% identity from the sequences of the unspecific peroxygenases AaP and CrP to the sequence of the chloroperoxidase of *C. fumago* (CfuCPO or LfuCPO). This similarity is located at the N-terminus and comprises the proximal heme-binding region, while the C-terminus is differing completely (Pecyna et al., 2009).

A Blast search of selected sequences of possible unspecific peroxygenases against the sequence of this chloroperoxidase showed similar results with a maximum identity of 25%, but all sequences contained the conserved cysteine residue of the PCP motif that is found in the peroxygenases AaP and CrP as well as in the chloroperoxidase where it serves as fifth heme ligand and has the position Cys29 (Pecyna et al. 2009).

The following alignment, created with Clustal Omega, shows the conserved sequence motifs described above:

	(SEQ ID NO: 26)	
31	YDNNTLPYVAPGPTDSRAPCPALNALANHGYPHDG	90
	(SEQ ID NO: 27)	
	VNDKDHWPWKLPKPCDTRGFCPLNLTLASHGYLPRNG	118
	proximal heme binding	
	(SEQ ID NO: 28)	
91	LALTNAFVVC-EY-----VTGSDCGDSLVLNLTLLAEPAFPHDHSFSEKDYKQGI	138
	(SEQ ID NO: 29)	
119	FATYEAHLMVGNLLTDLISIGRKTPLTGPDLP-PPANIGGLSEHGLFEGDASMTREGDAFFG	177
	heme propionates environment	

Alignment of the AaP and the CrP peroxygenase with the chloroperoxidase of *C. fumago* (CfuCPO) demonstrated that the substrate binding is different. Although some epoxidation activity was described for LfuCPO in comparison to UPOs, CPOs are usually not able to epoxidize aromatic rings or to hydroxylate alkanes with the same efficiency.

Thus, it was an object of the invention to evaluate the *Pichia* system for achieving high yields and titers of new UPOs. The present invention therefore relates to reproducible expression of novel UPOs by the robust and efficient expression system *P. pastoris* as folded and functional enzymes. The recombinant UPOs of the present invention

also showed improved technical properties compared to previously described recombinant UPOs and they can be expressed by secretion by yeast.

The present invention also relates to nucleic acid constructs comprising an isolated polynucleotide of the present invention operably linked to one or more (several) control sequences that direct the expression of the coding sequence in a suitable host cell under conditions compatible with the control sequences. An isolated polynucleotide encoding a polypeptide of the present invention may be manipulated in a variety of ways to provide for expression of the polypeptide. Manipulation of the polynucleotide's sequence prior to its insertion into a vector may be desirable or necessary depending on the expression vector. The techniques for modifying polynucleotide sequences utilizing recombinant DNA methods are well known in the art. The control sequence may be an appropriate promoter sequence, a nucleotide sequence that is recognized by a host cell for expression of a polynucleotide encoding a polypeptide of the present invention. The promoter sequence contains transcriptional control sequences that mediate the expression of the polypeptide. The promoter may be any nucleotide sequence that shows transcriptional activity in the host cell of choice including mutant, truncated, and hybrid promoters, and may be obtained from genes encoding extracellular or intracellular polypeptides either homologous or heterologous to the host cell.

There are several methods to determine the activity of peroxygenases based on different hydrogen donors, such as guaiacol, pyrogallol, ABTS (2,2'-Azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) diammonium salt), 4-methoxy- α -naphthol and phenol plus amino-antipyrine, or 2,6-DMP (Yuan & Jiang, 2002). Among them, ABTS is a widely used substrate in the spectrophotometric determination of peroxidase and peroxygenase activity because the method is sensitive and the chromogenic products are stable (Pütter & Becker, 1983; Yuan & Jiang, 2002).

The peroxygenase activity of the polypeptides having peroxygenase activity described herein, specifically the

UPOs described herein, is preferably determined using an ABTS assay or a 2,6-DMP assay.

The ABTS Assay (2,2'-Azino-bis(3-Ethylbenzothiazoline-6-Sulfonic Acid) (ABTS) Enzymatic Assay) is a colorimetric assay based on the ABTS cation radical formation and is well-known in the art, described for example in Pütter & Becker, 1983. The radical formation is catalyzed by the reduction of HRP in the presence of hydrogen peroxide.

According to a specific example, the ABTS assay is carried out analogously as described by Morawski et al. (2000) for horse radish peroxidase (HRP). The ABTS assay may be performed with variable parameters, including vary-

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ing concentration of the buffer at different pH values. As ABTS assay solution 440 mg 2,2'-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) in NaOAc may be mixed with buffer and 30% H₂O₂. The cell culture supernatant is mixed with the assay solution and the increase in absorption at 405 nm is measured to determine the peroxidase and/or peroxygenase activity.

The 2,6-DMP assay, or DMP assay in short, is another preferred activity assay that is used to detect and measure peroxygenase activity of the polypeptides described herein. In this method, 2,6-dimethoxyphenol and hydrogen peroxide are used as co-substrates in a nonspecific peroxygenase-catalyzed reaction leading to the formation of a colored product.

To determine an increased activity, a benchmark is also measured in the activity assay. The benchmark may for example be the wild-type polypeptide, not comprising any of the modifications described herein, or the PaDa1 mutant (of AaeUPO1). The benchmark is measured under the same conditions as the polypeptide of interest for which an increased activity shall be determined.

Surprisingly, the modified unspecific peroxygenases described herein comprise an increased peroxygenase activity of about at least 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5 or 10-fold increased activity as determined by ABTS assay and/or by DMP assay. Surprisingly, the largest group of the UPO12 variants described herein comprising improved peroxygenase activity were found to have a mutation at the C-terminus of the POX12 (UPO12) protein sequence.

The term "C-terminus" (also known as the carboxyl-terminus, carboxy-terminus, C-terminal tail, C-terminal end, or COOH-terminus) as used herein refers to the end of an amino acid chain (protein or polypeptide) comprising a free carboxyl group (—COOH). The C-terminus may comprise any of 5, 10, 15, 20, 25, 50, 100, 150, or 200 amino acids, or any number in between.

Specifically, the term C-terminus as used herein with reference to the modified UPOs described herein, refers to a sequence of amino acids corresponding to amino acids 145-261 of UPO12, preferably amino acids 230 to 261, or even more preferably to the amino acids from position 240 or 250 to 261, of the UPO of SEQ ID NO:12. Specifically, the sequence corresponding to the C-terminus of SEQ ID NO:12 is not necessarily identical to the C-terminus of SEQ ID NO:12 but shares at least about 70, 75, 80, 85, 90, or 95% sequence identity.

Specifically, the modified UPOs described herein comprise one or more amino acid modifications at positions corresponding to S24, C256, D253, E249, and/or D145 of SEQ ID NO:12. The position of the amino acid modification may not be identical to positions S24, C256, D253, E249, and/or D145 of SEQ ID NO:12, but it is functionally equivalent to said positions. Identification of functionally equivalent positions is readily available to a person skilled in the art, for example by employing structural alignments.

The polypeptides having peroxygenase activity described herein, specifically the UPOs described herein, including the modified UPOs described herein, can be used in various applications. Specifically, the polypeptides described herein are employed in oxyfunctionalization reactions, oxidative defunctionalization reactions and/or oxidative polymerization reactions. Industrial applications of the UPOs described herein and isolated polypeptides comprising peroxygenase activity are numerous; they reach from pharmaceutical production to environmental applications, including environ-

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mental problems caused by industry. For example, transformation of pollutants using the UPOs described herein can result in reduction of toxicity or bioavailability. Also, removal of pollutants from water can be achieved.

The polypeptides having peroxygenase activity described herein, specifically the UPOs described herein, including the modified UPOs described herein, may be further modified, such modifications including for example insertion or deletion of post-translational modification sites, insertion or deletion of targeting signals (e.g.: leader peptides), fusion to tags, linker peptides, proteins or protein fragments facilitating their processing such as purification or detection or enhancing their stability.

The term "nucleic acid construct" as used herein refers to a nucleic acid molecule, either single- or double-stranded, which is isolated from a naturally occurring gene or which is modified to contain segments of nucleic acids in a manner that would not otherwise exist in nature or which is synthetic. The term nucleic acid construct is synonymous with the term "expression cassette" when the nucleic acid construct contains the control sequences required for expression of a coding sequence of the present invention.

The term "control sequences" is defined herein to include all components necessary for the expression of a polynucleotide encoding a polypeptide of the present invention. Each control sequence may be native or foreign to the nucleotide sequence encoding the polypeptide or native or foreign to each other. Such control sequences include, but are not limited to, a leader, linker peptides causing ribosomal skipping, polyadenylation sequence, pro-peptide sequence, promoter, signal peptide sequence, and transcription terminator. At a minimum, the control sequences include a promoter, and transcriptional and translational stop signals. The control sequences may be provided with linkers for the purpose of introducing specific restriction sites facilitating ligation of the control sequences with the coding region of the nucleotide sequence encoding a polypeptide.

The term "operably linked" denotes herein a configuration in which a control sequence is placed at an appropriate position relative to the coding sequence of the polynucleotide sequence such that the control sequence directs the expression of the coding sequence of a polypeptide.

The term "expression" includes any step involved in the production of the polypeptide including, but not limited to, transcription, post-transcriptional modification, translation, post-translational modification, and secretion.

The term "expression vector" is defined herein as a linear or circular DNA molecule that comprises a polynucleotide encoding a polypeptide of the present invention and is operably linked to additional nucleotides that provide for its expression.

The term "functional variant" or "functionally active variant" also includes naturally occurring allelic variants, as well as mutants or any other non-naturally occurring variants of the UPOs described herein. As is known in the art, an allelic variant is an alternate form of a nucleic acid or peptide that is characterized as having a substitution, deletion, or addition of one or nucleotides or more amino acids that does essentially not alter the biological function of the nucleic acid or polypeptide.

Functional variants may be obtained by sequence alterations in the polypeptide or the nucleotide sequence, e.g. by one or more point mutations, wherein the sequence alterations retains or improves a function of the unaltered polypeptide or the nucleotide sequence, when used in combination of the invention. Such sequence alterations can include,

but are not limited to, (conservative) substitutions, additions, deletions, mutations and insertions.

A point mutation is particularly understood as the engineering of a polynucleotide that results in the expression of an amino acid sequence that differs from the non-engineered amino acid sequence in the substitution or exchange, deletion or 5 insertion of one or more single (non-consecutive) or doublets of amino acids for different amino acids.

The term “heterologous” as used herein with respect to a nucleotide or amino acid sequence or protein, specifically the UPOs and promoters described herein, refers to a compound which is foreign, i.e. “exogenous”, such as not found in nature, to a given host cell. The heterologous nucleotide sequence may also be expressed in an unnatural, e.g., greater than expected or greater than naturally found, amount in the cell. Specifically, heterologous nucleotide sequences are those not found in the same relationship to a host cell in nature (i.e., “not natively associated”). Any recombinant or artificial nucleotide sequence is understood to be heterologous. An example of a heterologous polynucleotide or nucleic acid molecule comprises a nucleotide sequence not natively associated with a promoter, e.g., to obtain a hybrid promoter, or operably linked to a coding sequence, as described herein. As a result, a hybrid or chimeric polynucleotide may be obtained. A further example of a heterologous compound is a UPO-encoding polynucleotide or gene operably linked to a transcriptional control element, e.g., a promoter, to which an endogenous, naturally-occurring POI coding sequence is not normally operably linked.

“Sequence identity” as described herein is defined as the percentage of nucleotides or amino acid residues in a candidate sequence that are identical with the nucleotides or amino acid residues in the specific nucleotide or polypeptide sequence to be compared (the “parent sequence”), after aligning the sequence and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared.

The term “operably linked” as used herein refers to the association of nucleotide sequences on a single nucleic acid molecule, e.g. the vector, plasmid or chromosome, in a way such that the function of one or more nucleotide sequences is affected by at least one other nucleotide sequence present on said nucleic acid molecule. For example, a promoter is operably linked with a coding sequence encoding a UPO described herein, when it is capable of effecting the expression of that coding sequence. Specifically, such nucleic acids operably linked to each other may be immediately linked, i.e. without further elements or nucleic acid sequences in between or may be indirectly linked with spacer sequences or other sequences in between.

The term “host cell”, as used herein, includes any cell type that is susceptible to transformation, transfection, transduction, and the like with a nucleic acid construct or expression vector comprising a polynucleotide of the present invention.

Specifically, the host yeast cells are maintained under conditions allowing expression and/or secretion of the peroxigenases described herein.

In one aspect the host cell is a yeast cell. “Yeast” as used herein includes *ascosporogenous* yeast (*Endomycetales*), *basidiosporogenous* yeast, and yeast belonging to the Fungi *Imperfecti* (*Blastomycetes*). In one aspect, the yeast host cell is a *Candida*, *Hansenula*, *Kluyveromyces*, *Pichia*, *Saccha-*

romyces, *chizosaccharomyces*, or *Yarrowia* cell. In a further aspect, the yeast host cell is a *Pichia pastoris* cell.

Specifically, the methylotrophic yeasts *Komagataella* (*Pichia*) *pastoris*, *Komagataella* (*Pichia*) *phaffii* (Pp), *Komagataella kurtzmanii*, *Ogataea* (*Hansenula*) *polymorpha* (Hp), *Candida boidinii* (Cb) and *Ogataea* (*Pichia*) *methanolica* (Pm) have been established as efficient alternative production strains. These strains make it possible to achieve high expression rates for heterologous proteins with a high cell density. Of the aforementioned four yeast species, *P. pastoris* (*Komagataella phaffii*) has in the meantime been used most commonly for heterologous protein production.

The term “methylotrophic yeast cells”, as used herein, includes yeast cells capable of growing on culture media containing as carbon source substances with only one carbon atom, for example methanol.

The term “promoter” as used herein refers to an expression control element that permits binding of RNA or DNA polymerase and the initiation of transcription.

“Derepressing conditions”, as used in culturing the yeast cells according to one aspect, means that the yeast cells are first cultured in the presence of a repressing carbon source (e.g. glucose) until this carbon source has been mostly or entirely consumed. After reducing the concentration of the repressing carbon source (e.g. glucose), the cells are in derepressing conditions with respect to the repressing carbon source and glucose, respectively. The strength of the repression effects may depend on the type of carbon source and on specific growth rates.

Derepressed promoter sequences are activated by derepression upon carbon source limitation and depletion and not upon induction by methanol.

The derepressed and methanol-independent promoters used according to the present invention display at least 10% activity in a suitable environment that does not comprise methanol. Preferably, such promoters comprise at least 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, or 70% under derepressing conditions and without the addition of methanol.

In contrast, methanol-dependent promoter sequences, such as the AOX1 promoter, display less than 1% activity, typically less than 0.1% or even less, without the addition of methanol to the cell culture.

In a yeast host, useful promoters are for example, AOX1, PDC, and PDF, FMD and FDH or FLD promoters and peroxisomal catalase gene promoters of different methylotrophic yeast as well as for example promoters of genes, coding for peroxisomal proteins. According to a preferred embodiment, the PDC or FMD promoter is used in the method described herein.

The term “signal peptide”, as used herein, refers to a peptide linked to the C-terminus or N-terminus of the polypeptide, which controls the secretion of the polypeptide. The signal sequence used may be a polynucleotide which codes for an amino acid sequence which initiates the transport of a protein through the membrane of the endoplasmic reticulum (ER). The nucleic acid sequence of these signal sequences may correspond to the natural sequence of the original host cell or may be codon-optimized. The non-limited examples of the signal sequence include native fungal plant or animal protein signal sequences, MF-alpha (“mating factor alpha” signal sequence), the OST1 signal peptide, the signal sequence of the CBH2 protein from *Trichoderma reesei*, the signal sequence of the xylanase A from *Thermomyces lanuginosus*, Kl killer toxin signal, the signal peptide for invertase secretion, the signal sequence of the killer toxin from *Kluyveromyces lactis*, the signal

sequence of the killer toxin from *Pichia acaciae*, the signal sequence of the killer toxin from *Hanseniaspora uvarum* and from *Pichia (Hansenula) anomala* or variants thereof and signal sequences of proteins exposed at the surface of *P. pastoris*. In one aspect, the preferred signal sequence is MF-alpha ("mating factor alpha" signal sequence). According to a further preferred aspect, the signal sequence is a signal sequence from *Podospira anserine*.

A suitable expression system is for example disclosed in WO2017/109082.

On aspects related to the selection and codon optimization of sequences, expression system and confirmation of the activity of the enzymes. Different new enzyme sequences showing a clear difference to so far known enzymes were identified and provided.

The term "cell culture" or "cultivation" ("culturing" is herein synonymously used), also termed "fermentation", with respect to a host cell line is meant to be the maintenance of yeast cells in an artificial, e.g., an in vitro environment, under conditions favoring growth, differentiation or continued viability, in an active or quiescent state, of the cells, specifically in a controlled bioreactor according to methods known in the industry. When cultivating, a cell culture is brought into contact with the cell culture media in a culture vessel or with substrate under conditions suitable to support cultivation of the cell culture and expression and/or secretion of the peroxygenases described herein. Specifically, a culture medium is used to culture cells according to standard cell culture techniques that are well-known in the art for cultivating or growing yeast cells.

Cell culture may be a batch process or a fed-batch process. A batch process is a cultivation mode in which all the nutrients necessary for cultivation of the cells, and optionally including the substrates necessary for production of the carbonyl compounds described herein, are contained in the initial culture medium, without additional supply of further nutrients during fermentation. In a fed-batch process, a feeding phase takes place after the batch phase. In the feeding phase one or more nutrients, such as the substrate described herein, are supplied to the culture by feeding. In certain embodiments, the method described herein is a fed-batch process. Specifically, a host cell transformed with a nucleic acid construct encoding the polypeptides described herein, specifically the UPOs as described herein, is cultured in a growth phase medium and transitioned to an induction phase medium in order to produce the polypeptides described herein.

In another embodiment, host cells described herein are cultivated in continuous mode, e.g. a chemostat. A continuous fermentation process is characterized by a defined, constant and continuous rate of feeding of fresh culture medium into the bioreactor, whereby culture broth is at the same time removed from the bioreactor at the same defined, constant and continuous removal rate. By keeping culture medium, feeding rate and removal rate at the same constant level, the cultivation parameters and conditions in the bioreactor remain constant.

Suitable cultivation techniques may encompass cultivation in a bioreactor starting with a batch phase, followed by a short exponential fed batch phase at high specific growth rate, further followed by a fed batch phase at a low specific growth rate. Another suitable cultivation technique may encompass a batch phase followed by a continuous cultivation phase at a low dilution rate.

It is preferred to cultivate the host cell line as described herein in a bioreactor under growth conditions to obtain a cell density of at least about 1 g/L, 5 g/L or 10 g/L cell dry

weight, more preferably at least 20 g/L cell dry weight, preferably at least 50 g/L cell dry weight. It is advantageous to provide for such yields of biomass production on a pilot or industrial scale.

The term "mutation" as used herein has its ordinary meaning in the art. A mutation may comprise a point mutation, or refer to areas of sequences, in particular changing contiguous or non-contiguous amino acid sequences. Specifically, a mutation is a point mutation, which is herein understood as a mutation to alter one or more (but only a few) contiguous amino acids, e.g. 1, or 2, or 3 amino acids, which are substituted, inserted or deleted at one position in an amino acid sequence. Amino acid substitutions may be conservative amino acid substitutions or non-conservative amino acid substitutions. Conservative substitutions are those that take place within a family of amino acids that are related in their side chains and chemical properties. Examples of such families are amino acids with basic side chains, with acidic side chains, with non-polar aliphatic side chains, with non-polar aromatic side chains, with uncharged polar side chains, with small side chains, with large side chains etc.

A point mutation is particularly understood as the engineering of a polynucleotide that results in the expression of an amino acid sequence that differs from the non-engineered amino acid sequence in the substitution or exchange, deletion or insertion of one or more single (non-consecutive) or doublets of amino acids for different amino acids.

The term "functional variant" or "functionally active variant" also includes naturally occurring allelic variants, as well as mutants or any other non-naturally occurring variants. As is known in the art, an allelic variant is an alternate form of a nucleic acid or peptide that is characterized as having a substitution, deletion, or addition of one or nucleotides or more amino acids that does essentially not alter the biological function of the nucleic acid or polypeptide. Functional variants may be obtained by sequence alterations in the polypeptide or the nucleotide sequence, e.g. by one or more point mutations, wherein the sequence alterations retain or improve a function of the unaltered polypeptide or the nucleotide sequence, when used in combination of the invention. Such sequence alterations can include, but are not limited to, (conservative) substitutions, additions, deletions, mutations and insertions.

In one aspect as described herein, several UPOs from basidiomycetes and ascomycetes were identified and studied. In Table 1 the constructs that have been tested are listed with their associated accession numbers.

TABLE 1

Tested UPO and CPO candidates		
Candidate	Accession number	Notes
UPO 1mut	B9W4V6	PaDa I mutant
UPO 2	KDR72024.1	Unspecific peroxygenase
UPO 3	KJA13294.1	Unspecific peroxygenase
UPO 4	XP_006458802	Unspecific peroxygenase
UPO 5	KIK06072.1	Unspecific peroxygenase
UPO 6	KIJ31387.1	Unspecific peroxygenase
UPO 7	KIM43689.1	Unspecific peroxygenase
UPO 8	KJA24696.1	Unspecific peroxygenase
UPO 9	ESZ93716.1	Unspecific peroxygenase
UPO 10	CAK39169.1	Unspecific peroxygenase
UPO 11	OJJ73116.1	Unspecific peroxygenase
UPO 12	OTA57433.1	Unspecific peroxygenase
UPO 13	XP_001225194.1	Unspecific peroxygenase
UPO 14	XP_001219540.1	Unspecific peroxygenase
UPO 15	KIJ30163.1	Unspecific peroxygenase

TABLE 1-continued

Tested UPO and CPO candidates		
Candidate	Accession number	Notes
UPO 16	KIJ46203.1	Unspecific peroxygenase
UPO 17	XP_001911526.1	Unspecific peroxygenase
UPO 18	XP_006459044.1	Unspecific peroxygenase
CPO 19	CAA28172	Chloroperoxidase
CPO 20	AJA36817	Chloroperoxidase
UPO21*	CAV28569.1	Unspecific peroxygenase
UPO22	OTB17553.1	Unspecific peroxygenase
UPO23	GAQ45152.1	Unspecific peroxygenase
UPO24	XP_001390900.2	Unspecific peroxygenase
UPO25	GAA88053.1	Unspecific peroxygenase

In one aspect, the polypeptide comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, or 99% identity to the polypeptide of SEQ ID NO:11 (UPO11).

In one aspect, the polypeptide comprises an amino acid sequence having at least 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, or 99% identity to the polypeptide of SEQ ID NO:12 (UPO12).

In one aspect, the polypeptide comprises an amino acid sequence having at least 80%, 85%, 90%, 95%, 97%, 98% or 99% identity to the polypeptide of SEQ ID NO:17 (UPO17).

In one aspect, the polypeptide comprises an amino acid sequence having at least 95% identity to the polypeptide of and SEQ ID NO:23 (UPO23).

In one embodiment as described herein, using *Pichia pastoris* as expression system with a methanol-independent PDC promoter and the engineered gene/protein sequence as described herein more than 200 mg/L secreted enzyme were obtained. One aspect provides for yields of 0.5 g/L or even 1 g/L of the desired enzyme. This yield came close to secreted UPO concentrations observed in native hosts.

EXAMPLES

The Examples which follow are set forth to aid in the understanding of the invention but are not intended to, and should not be construed to limit the scope of the invention in any way. The Examples do not include detailed descriptions of conventional methods, e.g., cloning, transfection, and basic aspects of methods for expressing proteins in microbial host cells. Such methods are well known to those of ordinary skill in the art.

Materials and Methods

Sequence Selection Procedure

Sequences described in databases were analyzed or potential peroxygenase activity using various free available sequence databases, e.g. genbank at NCBI with the data sets nonredundant or patdb Google Patent search, Canadian Patents Database, Patentscope, Espacenet, and DPMA.

Searches were done based by blast searches using previously published sequences with known or claimed activities as input.

Signal BLAST and SignalP were used for analyzing all those sequences individually in order to find out if a hypothetical protein is potentially secreted and to identify predictable signal sequence cleavage sites, enabling the replacement of native signal peptides by others such as the signal sequence of the *S. cerevisiae* mating factor alpha.

A Multiple Sequence Alignment as well as a phylogenetic tree were obtained by Clustal Omega analysis, which uses the neighbor-joining method for the phylogenetic tree.

Out of this huge amount of data a matrix for choosing sequences was generated. Two big groups were identified depending on the similarity, with each of them having several hotspots containing notably high similarities as it showed up, in the heatmap, which supported the final decision process.

Final sequence selections were made based on the sequence comparisons in order to stay in distance from previously known and/or characterized UPOs, for reflecting a broad coverage of sequence diversity in the phylogenetic tree and to cover a broad sequence diversity which also might reflect functional diversity. However also known heme thiolate peroxygenases such as the AaeUPO1 variant PaDa1 and the CfUPO were included in the performed expression studies. The evolved AaeUPO1 variant served as a positive control for expression and activity tests, while the CPO was used as one of the negative controls, since functional secretory expression by *Pichia pastoris* was reported to have failed in other labs before.

Sequencing Genes of Interest

Single colonies that (due to the colony PCR) were likely to contain the cloned peroxygenase reading frames, cloned into the expression vector, inter alia, were streaked out on LB Zeocin plates and incubated at 37° C. overnight to amplify the recombinant plasmid.

A miniprep of plasmid DNA was done with Promega's "Wizard® Plus SV Minipreps DNA Purification System" as described herein.

For sequence verification and analysis of potential errors produced by DNA synthesis or PCR amplification/cloning, isolated plasmid DNA was sent for Sanger sequencing of the DNA. Therefore at least 1,200 ng of DNA plus 3 µL of 10 µM forward or reverse primer, respectively, were brought to a total volume of 15 µL with dH₂O.

P. pastoris Transformation

For the transformation of electrocompetent cells and for genomic integration of the expression cassettes the vectors were linearized by a single cut using SmaI. Deviating from standard protocols only 0.5 µL of enzyme were used and the incubation time was increased to three hours after checking in the enzyme manufacturers description that the used restriction enzyme has no star activity.

The linearized expression cassettes were desalted by dialysis using filter discs floating on water, before the DNA was used for *Pichia* transformation.

For one transformation 40 µL of ready to use electrocompetent *Pichia pastoris* BSYBG11 cells (Table 6, Bisy GmbH, Austria) and around 1 µg of linearized plasmid DNA were used.

First of all the competent cells were defrosted on ice and the cuvettes were cooled. Then the competent cells and the plasmid DNA were pooled in the cooled cuvettes and kept on ice for at least 10 minutes. Afterwards electroporation was carried out with a voltage of 1.5 kV, followed by the addition of regeneration medium (YPD/1 M Sorbitol, 1:1 (v/v)).

The mixture containing transformants and regeneration medium was transferred to Eppendorf tubes (0) and regenerated for 2 hours at 30° C. and 700 rpm, followed by a centrifugation step (1 min, full speed). The supernatant was reduced to 100 µL, the cell pellet was resuspended therein and plated on LB-Zeocin plates. The plates were incubated at 30° C. for two days.

Cultivation

Cultivation was carried out in deep well plates either as one or as two-day(s) induction.

Two-Day Induction:

Single colonies of the transformed *Pichia pastoris* cells (from 0) were picked with sterile toothpicks. Then they were transferred to the wells of the deep-well-plates, containing 300 μ L BMD1 per well, and incubated at 28° C. with 320 rpm for 36-60 hours.

After this incubation they were induced by methanol adding 250 μ L of BMM2 per well and incubated again. 12, 24 and 36 hours later 50 μ L of BMM10 were added per well.

12 hours after the last addition of BMM10 the deep-well plates were centrifuged for 10 minutes at high speed. The supernatant containing the secreted enzyme was used for the assays described herein.

One Day Induction

The one-day induction protocol was following the same procedure as the two-day induction protocol but the cells were only induced for the first two times, followed by the harvest already on the next day.

Flask Cultivation

The Flask cultivation was carried out as follows:

450 mL of BMD1% were inoculated with the transformed *Pichia pastoris* as described above in a 2.5 L Ultra Yield Flask (UYF).

The flasks were incubated for 3 days at 28° C. and 100 rpm.

After incubation the induction was started with 50 mL of BMM1). Every 12 hours 5 mL of 100% methanol were added for three times.

The day after the last induction the culture was harvested by centrifugation in 500 mL tubes for 15 minutes at 8,000 rpm. The enzyme was in the supernatant. Cells were removed by centrifugation. The supernatants were filtered through a membrane with a pore size of 0.45 μ m and stored at 4° C.

The concentration of the enzyme in the supernatant was evaluated by centrifugation with Vivaspin columns with a 10 kD cutoff.

Bioreactor Cultivation

To scale up enzyme production Sartorius 5 L bioreactors were used for cultivations.

The bioreactor cultivations were based on Invitrogen's™ "*Pichia* Fermentation Process Guidelines". In detail the cultivation was done as follows:

The pre-culture I, consisting of 50 mL BMGY in 250 mL baffled flask with some cell material of transformants grown on an agar plate, was incubated at 110 rpm, 28° C. and about 50% humidity for about 60 hours.

After the incubation an aliquot of preculture I was used to inoculate the pre-culture II (200 mL BMGY in 1 L baffled flask) to an OD600 of 3.0. After about four hours the 3.5 L BSM medium in the 5 L bioreactor was inoculated to an OD600 of about 1.0 (as measured with the same photometer). The glycerol batch phase lasted for 22 hours until the entire carbon source was consumed.

The standard conditions in a non-optimized bioreactor cultivation was: 28° C., pH 6.0, min. stirring at 500 rpm, min. dO₂ of 30% (cascade setting) and 4 L/min airflow.

During the glycerol fed-batch phase the culture was fed constantly with 26 mL/h/L (L . . . liter of start volume; 3.5 L:91 mL/h) 50% glycerol with PTM1 and biotin (both 12 mL/L fed-batch medium) for 6 hours. During night the culture was fed with 2.6 mL/h/L fed-batch medium.

On the next morning the glycerol-feed was turned off and after 30 minutes 100% methanol was added to the bioreactor

culture to a final methanol concentration of 1%. After consumption a constant methanol feed was set to 3 mL/h/L (L . . . liter of start volume; 3.5 L:10.5 mL/h) pure methanol (without PTM1 or biotin). This flow rate was kept for 30 hours.

Finally, the culture was harvested in 1 L centrifuge tubes at 8,000 rpm and the supernatant was transferred into clean bottles and stored at 4° C. until future use.

Activity Assays

Well described standard assays for measuring the activity of unspecific peroxygenases were applied for proving the peroxidase and/or peroxygenase activities of the supernatants of the performed cultivations.

ABTS Assay

The ABTS assay was carried out analogously as described by Morawski et al. (2000) for horse radish peroxidase (HRP). The ABTS assay was performed with variable parameters, including varying concentration of the buffer at different pH values.

TABLE 2

Assay Solution	
Variable parameters	Range used
supernatant	2-15 μ L
buffer	NaOAc or citrate-phosphate-buffer
pH value of buffer	2.5-7.0
molarity of buffer	100-200 mM

For one 96-well plate 20 mL assay solution were prepared. Therefore 1 mL 20 $\times$ ABTS stock solution (440 mg 2,2'-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) in 50 mL 50 mM NaOAc) was mixed with 19 mL buffer and 1.75 μ L 30% H₂O₂. The assay solution was kept on ice.

15 μ L of the supernatant was mixed with 140 μ L of the assay solution and the increase in absorption at 405 nm was measured with the plate reader.

The first screening was carried out with a buffer concentration of 50 mM and pH values of 3.5, 4.5 and 5.5 respectively. As benchmark the PaDa1 mutant (of Aae-UPO1) was measured under the same conditions as the new constructs. The new UPO constructs were measured after secretion employing their natural signal peptide as well as the alternative mating factor alpha signal peptide. The measurement was carried out for 15 minutes.

2,6-DMP Assay

The 2,6-DMP (2,6-dimethoxyphenol) assay was done similar to the assay described by E. Breslmayr et al. (2018) for lytic polysaccharide monooxygenases and by P. Molina-Espeja et al. (2016). The 2,6-DMP assay was performed with potassium phosphate (KPi) buffer pH 7.0.

For one 96-well plate 20 mL assay solution were prepared. Therefore 2 mL 2,6-DMP stock solution (100 mM, 154 mg in 10 mL ddH₂O, heated to 60° C. for better solubility) was mixed with 2 mL KPi buffer (1.0 M, pH 7.0), 16 mL ddH₂O and 0.5 μ L 33% H₂O₂. The assay solution was kept on ice.

15 μ L of the supernatant was mixed with 185 μ L of the assay solution and the increase in absorption at 469 nm was measured with the plate reader. The measurement was carried out for 9 minutes.

Naphthalene Assay

This assay for aromatic peroxygenases was done similar to the Naphthalene-Fast Blue-Assay described by Gröbe et al. (2011). For one 96-well plate 20 mL of assay solution were prepared.

TABLE 3

Assay Solution	
Variable parameters	Range used
supernatant	2-15 μ L
buffer	NaOAc or citrate-phosphate-buffer
pH value of buffer	4.5-5.5
molarity of buffer	100-200 mM
Naphthalene stock solution	1-2 mL
Fast Blue stock solution	1-2 mL

4 mM naphthalene stock solution: 5 mg naphthalene, in 10 mL acetone

2 mM Fast Blue stock solution: 9.5 mg Fast Blue B salt/10 mL dH₂O

For one 96-well plate 20 mL assay solution were prepared, containing 10 mL citrate-phosphate-buffer, 1-2 mL 4 mM naphthalene stock solution and the same amount of 2 mM Fast Blue stock, 2 μ L 30% H₂O₂. Then dH₂O was added to obtain a final volume of 20 mL.

30 μ L of the supernatant was mixed with 150 μ L of assay solution and the increase in absorption at 520 nm was measured with plate reader.

Filter Assay

The filter assay was done similar to the ABTS assay described above. For proof of concept, HRP secreting *Pichia pastoris* BSYBG11 strains were used as positive control and wild type BSYBG11 strains as negative control.

The positive and negative controls were streaked out on agar plates containing Zeocin to get single colonies. The plates also contained methanol for induction. A filter paper was laid on the plate, so that the colonies stick to it. This filter was transferred to an empty petri dish with the colonies looking down. This should help to keep the colonies where they are, not washing them off. Then 100 μ L of assay solution, as described above, were carefully pipetted onto the filter. The assay was incubated at room temperature and controlled every five minutes.

After the concept was proven by seeing green color development due to ABTS oxidation on agar plates by HRP producing cells, the same assay was conducted with UPO 1, 12 and 17 secreting *P. pastoris* BSYBG11 as positive control and wild type *P. pastoris* BSYBG11 as negative control.

Plate Assay

It was assumed that a colony secreting an active UPO would be surrounded by a greenish halo, similar to the filter paper assay. To have a proof of concept the plate assay was carried out with an HRP (horse radish peroxidase) secreting *P. pastoris* BSYBG11 transformant as positive control and a *P. pastoris* BSYBG11 wild type as negative control.

The plates were made with buffered minimal medium containing 1% of methanol, sorbitol or glucose respectively. H₂O₂, 30% (Table 5) was added to final concentrations of 43.3 μ L*L⁻¹, 87.5 μ L*L⁻¹ and 175 μ L*L⁻¹, respectively.

The positive and negative controls were streaked out and the plates were incubated at 28° C. for two days. After that time single colonies should be formed. The plates were evaluated by visual inspection.

Volumetric Peroxidase Activity

Peroxidase activity measurements were performed in plate readers, normalizing by respective assay volumes in the plates. For the calculation of units, the layer thickness was calculated according to Formula 1. In Formula 1 the "h" value corresponds to the layer thickness "d".

Formula 1: The layer thickness was calculated depending on the total volume per well.

$$\pi * [(h^3 * \tan^2 \varphi) / 3 * h^2 * \pi * \tan \varphi * r^2] - V = 0$$

After determining the layer thickness, the units were calculated with Formula 2.

Formula 2: Calculation of volumetric peroxidase activity.

$$U = (\Delta A \Delta t^{-1} * V_{tot} * D) / (V_{sample} * \epsilon_{405} * d)$$

U Units per mL [μ mol*ml⁻¹*min⁻¹]

V_{tot} total assay volume [mL]

$\Delta A \Delta t^{-1}$ change in absorption per time [$\Delta A(405) * \text{min}^{-1}$]

D dilution factor of the sample

d layer thickness [cm]

v_{sample} sample volume [mL]

ϵ_{405} extinction coefficient at 405 nm [36,000 mL * μ mol⁻¹*cm⁻¹]

Bioconversions

As there are many known substrates that are converted by UPOs just a few exemplifying substrates were tested to proof that the new UPOs are active and able to convert those model substrates. To verify possible bio-conversions HPLC measurements were carried out.

To be able to carry out HPLC measurement including control samples also supernatants from *P. pastoris* BSYBG11 cultures, grown as negative control and even pure substrate in assay buffer (without enzymes) were applied on the 96-well plate. Furthermore, transformed strains expressing two other intracellular enzymes, human Cytochrome P450C9 and 3A4, were used as benchmark and control.

The following substrates of interest have been tested: Chlorzoxazone, testosterone, clopidogrel, diclofenac, dextromethorphan, estriol, ethionamide, ibuprofen, lidocaine, and moclobemide.

Bioconversions were carried out in 96-well deep well plates. The assay buffer consisted of 20 mL of 200 mM citrate-phosphate buffer at pH 4.7 containing 2 μ L of 30% (w/w) H₂O₂.

The refreshing buffer consisted of 20 mL of 200 mM citrate-phosphate buffer at pH 4.7 containing 200 μ L of 30% (w/w) H₂O₂.

Each well contained 100 μ L of supernatant, 100 μ L of assay buffer and 4 μ L of stock substrate solution (100 mM). The deep well plates were incubated at 28° C. and 320 rpm for 15 hours. 0.5 μ L of refreshing buffer were added per well. The deep well plates were re-incubated for another 6 hours. To stop the conversion 150 μ L of an acetonitrile/methanol (1:1) mixture was added.

For sample preparation, the polypropylene microtiter plates were centrifuged for 20 minutes at 4000 rpm and 4° C. 100 μ L of reaction supernatant was transferred into a fresh polypropylene microtiter plate. The new plate was used for measurements by HPLC.

The applied HPLC parameters are listed in Table 4.

The analyses were done on an Agilent 1200 series HPLC system (Agilent technologies, Santa Clara, California, USA) coupled with a mass spectrometer detector (MSD) containing an electron spray ionization unit.

TABLE 4

HPLC-MS parameters		
Parameters	Column	
	(Kinetex ® 2.6 μ m C18 100 Å, LC Column 50 × 4.6 mm)	
LC	Injection volume (μ L)	10
	Flow rate (ml min ⁻¹)	1
	Column temp. (° C.)	25
	Solvent A1	H ₂ O-0.1% acetic acid
API-ES	Solvent B2	ACN
	Gas temp. (° C.)	350
	Gas flow (L min ⁻¹)	12
	Nebulizer (psi)	35
	Quad temp. (° C.)	350
	Polarity	Positive

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Materials
Chemicals

TABLE 5

List of used chemicals.	
Name	Provider
LB-Medium	Carl Roth, GmbH, Karlsruhe, Germany
10x FastDigest Buffer	Thermo Scientific Inc., Massachusetts, USA
5x Q5 [®] Reaction Buffer	New England Biolabs, Ipswich, MA, United States
ABTS	Sigma-Aldrich Chemie GmbH, Missouri, USA
Agar-Agar	Carl Roth, GmbH, Karlsruhe, Germany
BD Bacto [™] yeast extract	Becton, Dickinson and Company, Sparks, MD, USA
Biotin	Sigma-Aldrich Chemie GmbH, Missouri, USA
Biozym LE agarose	Biozym Scientific, Hessisch Oldendorf, Germany
dATP, dGTP (10 mM)	Thermo Scientific Inc., Massachusetts, USA
dCTP, dTTP (20 mM)	Thermo Scientific Inc., Massachusetts, USA
D-Glucose-monohydrate	Carl Roth, GmbH, Karlsruhe, Germany
dNTPs	Thermo Scientific Inc., Massachusetts, USA
D-Sorbit	Carl Roth, GmbH, Karlsruhe, Germany
FastBlue B	Sigma-Aldrich Chemie GmbH, Missouri, USA
FastDigest Green Buffer	Thermo Scientific Inc., Massachusetts, USA
Hydrogen peroxide	Carl Roth, GmbH, Karlsruhe, Germany
K ₂ HPO ₄	Carl Roth, GmbH, Karlsruhe, Germany
KH ₂ PO ₄	Carl Roth, GmbH, Karlsruhe, Germany
Methanol	Sigma-Aldrich Chemie GmbH, Missouri, USA
MgCl ₂ (25 mM)	Thermo Scientific Inc., Massachusetts, USA
Naphthalene	Sigma-Aldrich Chemie GmbH, Missouri, USA
Trichloroacetic acid (TCA)	Sigma-Aldrich Chemie GmbH, Missouri, USA

Host Strains

For the transformation with linearized integrative plasmid DNA vectors containing putative new UPO gene sequences the *Pichia pastoris* platform strain BSYBG11 was used. Compared to the wild type strain BSYBG10, this strain has an AOX1 gene knock out leading to a slow growth phenotype, when methanol is used as carbon source.

TABLE 6

Information on the origin strains used to generate <i>Pichia pastoris</i> biocatalysts.		
Species/Host strain:		
	<i>Komagataella phaffii</i> BSYBG10 (Syn.: <i>Pichia pastoris</i>)	<i>Komagataella phaffii</i> BSYBG11 (Syn.: <i>Pichia pastoris</i>)
BT Culture collection number of IMBT at TU Graz	7287	7288

Media, Buffers and Solutions

Media used during the thesis are conventional media. If not mentioned else the amounts are given for 1 L of media and the media is autoclaved.

Plasmids

Plasmids were kindly provided by Bisy GmbH (Austria) and are listed in Table 7.

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TABLE 7

Plasmids.		
Plasmid	Promoter	Additional Information
pBSY3Z	PDC	Contains <i>P. pastoris</i> is CTA1 promoter
pBSY3S1Z	PDC	contains additional coding region for short variant of mating factor alpha signal
pBSY5Z	PDF	Contains <i>Hansenula polymorpha</i> FMD promoter variant
PBSY5S1Z	PDF	contains additional coding region for short variant of mating factor alpha signal

Results and Discussion

Evaluation of Synthetic Heme Thiolate Peroxygenase Genes

After vector digestion with LguI (SapI) the stuffer of the vectors pBSY3Z and pBSY3S1Z was cut out. For the vector pBSY3Z also a control digest was performed with EcoRI. Inserts coding for the peroxygenases were inserted into the vector backbones by recombination cloning and transformation of *E. coli* by electroporation. After plasmid isolation sequences were evaluated by Sanger sequencing. Table 8 shows the results of the sequence evaluation of cloned UPO and CPO genes synthesized by TWIST. In total 24 genes were sequenced, 19 of them proved to be correct. This corresponds to a validity of 79.17%.

The table 8 shows how many genes of each ordered construct were sequenced and how many of them were confirmed.

TABLE 8

Evaluation of synthetic genes.		
Ordered Gene	Correct genes	Genes sequenced
UPO 2	1	1
UPO 4	1	1
UPO 5	1	3
UPO 6	2	2
UPO 7	2	2
UPO 8	1	2
UPO 9	2	3
UPO 10	2	2
UPO 11	1	2
UPO 13	1	1
UPO 14	1	1
UPO 16	1	1
UPO 17	1	1
UPO 18	1	1
CPO 20	1	1
Total	19	24

The pairwise alignments show the identity of the selected new putative UPO candidates to previously known sequences. The alignment was made with Clustal Omega using the full available sequence length. The identity shows the percentage of amino acid sequence identity as given by the “percent identity matrix” created by clustal2.1. An overview of the identities can be found in Table 9 and Table 10.

Due to very low sequence identity the analysis performed with the CPO sequence (CPO19 & 20) is not included in this table.

TABLE 9

Identities of the new sequences with some previously described sequences in the NCBI patent sequence database "pat".								
<i>Agrocybe aegerita</i> GM831938	<i>Agrocybe aegerita</i> GM831940	<i>laccaria bicolor</i> GM831942	<i>Coprinopsis cinerea</i> okayama7#130_ GM831944	<i>Coprinopsis cinerea</i> okayama7#130 GM831946	<i>Coprinopsis cinerea</i> okayama7#130 GM831948	<i>Coprinopsis cinerea</i> okayama7#130 GM831950	<i>Coprinopsis cinerea</i> okayama7#130 GM831952	<i>Coprinellus radians</i> GM831952
UPO 2	66.58	69.73	69.09	63.27	60.40	60.22	59.64	64.32
UPO 3	59.79	61.46	62.63	58.11	55.91	56.12	57.14	62.11
UPO 4	63.41	62.6	67.74	58.33	56.2	56.18	62.35	63.44
UPO 5	63.51	63.69	94.41	59.68	56.69	57.95	60.12	67.84
UPO 6	59.57	61.35	71.73	59.68	57.68	57.68	60.24	63.44
UPO 7	60.27	60.16	69.33	57.8	58.6	54.72	57.1	63.88
UPO 8	57.84	59.62	68.00	54.84	56.81	55.80	57.40	62.11
UPO 9	26.02	27.31	28.78	27.68	27.52	27.04	25.19	27.67
UPO 10	28.79	29.12	31.42	27.59	28.79	30.27	31.01	32.20
UPO 11	33.33	32.05	28.96	29.89	29.25	30.74	31.52	30.88
UPO 12	30.95	29.88	30.20	29.88	32.74	31.10	30.31	28.78
UPO 13	27.99	26.67	28.21	25.38	29.96	27.90	29.24	29.47
UPO 14	28.38	28.96	28.05	28.64	29.63	26.85	26.39	29.59
UPO 15	57.72	57.18	59.25	56.91	58.02	56.37	59.39	64.76
UPO 16	58.74	58.47	57.84	56.13	54.12	59.29	62.39	64.32
UPO 17	28.63	25.68	26.46	26.06	24.89	25.68	27.86	24.64
UPO 18	57.18	59.35	59.19	57.14	58.43	55.68	58.79	63.00

TABLE 10

Identities of the new sequences with some previously described patent sequences								
<i>Coprinellus radians</i> GM831954	<i>Sclerotinia sclerotiorum</i>	<i>Aspergillus carbonarius</i>	<i>Humicola insolens</i>	<i>Myceliophthora himnulea</i>	<i>Pestalotiopsis virgatula</i>	<i>Chaetomium yirens</i>	<i>Daldinia caldariorum</i>	
UPO 2	45.19	28.21	30.35	29.57	30.56	30.80	22.88	25.67
UPO 3	41.35	28.15	30.59	29.41	30.35	28.93	25.38	27.80
UPO 4	44.66	27.10	28.02	31.13	29.96	29.25	23.33	29.02
UPO 5	45.19	29.92	30.59	32.12	30.35	28.23	26.22	26.25
UPO 6	43.81	26.15	27.85	30.71	29.07	29.88	26.48	25.10
UPO 7	36.63	29.01	31.37	32.12	29.30	30.28	27.56	29.57
UPO 8	37.86	28.24	30.47	31.18	29.48	27.71	25.91	27.41
UPO 9	26.67	79.93	42.08	42.19	40.31	42.86	28.74	39.84
UPO 10	21.57	41.57	60.87	53.12	53.97	56.47	29.96	50.98
UPO 11	23.08	41.83	60.47	53.39	54.76	57.65	31.47	52.55
UPO 12	19.59	39.62	50.98	50.39	51.18	54.80	31.62	63.39
UPO 13	27.94	42.58	49.80	48.87	61.30	52.78	31.06	48.85
UPO 14	23.53	37.33	46.95	50.91	72.00	52.27	28.44	44.14
UPO 15	44.12	23.48	26.56	28.35	28.29	27.69	23.40	24.31
UPO 16	44.00	25.58	30.59	30.20	27.52	27.67	27.07	25.97
UPO 17	21.69	28.14	35.42	34.25	30.65	30.04	71.97	32.03
UPO 18	45.10	27.69	27.52	30.80	28.40	26.59	24.07	24.61

ABTS Assays

Activity Landscapes in 100 mM Buffer

All activity landscapes formed by measured activities of individual transformants were directly done with samples (culture supernatants) from deep well plate cultivation. The slope of absorption was in all cases calculated with Microsoft Excel's "slope()" function and is equivalent to $1.4 \Delta \text{AABS} \cdot \text{min}^{-1}$. The measurement was done with the plate reader.

In FIG. 1 the landscapes of the PaDa1 mutant transformants (mutant of AaeUPO1) at pH 3.5, 4.5 and 5.5 were compared and showed a maximum of conversion at pH 4.5.

FIG. 2 shows the measurement results of AaeUPO 1 mutant PaDa1 (indicated as UPO1 mut) as well as UPO 11 at pH 4.5. Supernatants of UPO 1 clones converted ABTS faster than UPO 11, nonetheless also the new wt enzyme UPO 11 converted ABTS very well, indicating good expression of the recombinant protein.

UPO 12 behaved similar to UPO 11 in ABTS peroxidase assays as shown in FIG. 3. In primary screenings UPO 11 showed a maximum conversion at pH 4.5 when tested at pH values 3.5, 4.5 and 5.5, UPO 12 in comparison showed similar behavior at pH 4.5 and 5.5, indicating a higher robustness of activity at different pH than the AaeUPO1 variant.

FIG. 4 shows the comparison of UPO 1, UPO 17 and UPO 17 without additional mating factor alpha—but the given native signal sequence. This given signal sequence is from *Podospora anserina* and increased the conversion of ABTS around 2-fold compared to the construct containing the short *Saccharomyces cerevisiae*'s mating factor alpha signal.

To confirm that the AaeUPO 1 with short mating factor alpha signal is converting ABTS better than with the native signal peptide was tested with given settings and the

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described expression system. As shown in FIG. 5 there was no mentionable activity measured for UPO 1 with the native signal peptide.

Furthermore, in this specific experiment the constructs containing PaDa 1 with mating factor alpha signal were converting ABTS poorly. Nonetheless, at least the mating factor alpha signal containing constructs were active and behaved better than the ones with native signal peptides.

Constructs with Measurable Activity in the Rescreening

TABLE 11

Results of the rescreening at pH 4.5 using the ABTS based peroxidase assay, indicating functional expression.		
Construct	Clone	Δ ABS(405 nm)/min
UPO 17	17-A0	349.3
UPO 17	17-B0	291.7
UPO 17	17-C0	285.2
UPO 12	12-A0	68.8
UPO 12	12-B0	68.5
PaDa 1	1-A0	66.8
UPO 12	12-C0	66.5
UPO 11	11-A0	65.2
UPO 11	11-B0	64.1
UPO 11	11-C0	62.1
PaDa 1	1-B0	57.9
PaDa 1	1-C0	57.6
CPO 19*		1.2
CPO 19*		1.2
CPO 19*		1.0
UPO 3	3-A0	0.9
UPO 2	2-A0	0.7
UPO 15	15-A0	0.7
UPO 15	15-B0	0.6
UPO 5	5-A0	0.6
UPO 3	3-B0	0.6
UPO 15	15-C0	0.5
UPO 4	4-A0	0.4
UPO 2	2-B0	0.4
UPO 3	3-C0	0.3
UPO 4	4-B0	0.2
UPO 16	16-A0	0.2
Negative control/BSYBG 11 without integrated expression vector		0.1

*Data from the primary screening

The Table shows the transformants that were most active in the rescreening with their medium change in absorption per minute, measured over 13 minutes.

As can be seen in Table 11 the rescreening of the most promising clones with the ABTS assay was successful and indicated peroxidase activity and functional expression for all tested genes. UPO17 showed higher activity than the evolved AaeUPO1 variant. Surprisingly also CPO transformants showed activity, indicating functional expression of CfuCPO by *P. pastoris*.

Assays after Bioreactor Cultivation

ABTS Assay

Constructs under the PDC Promoter showed increased activity after the bioreactor cultivation compared to 96-deepwell plate cultivation, as shown in Table 12. Surprisingly the activity of UPO 17 was far lower than the activities of the UPOs 11 and 12, indicating possible enzyme instabilities caused by long term cultivation. Compared to the benchmark, AaeUPO 1 variant PaDa1, activities up to 355-fold higher were seen.

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TABLE 12

The table lists the constructs cultivated in the bioreactor and the units per milliliter unconcentrated supernatant, measured with the ABTS assay. Furthermore, a comparison of the new UPOs compared to the benchmark, AaeUPO1 variant PaDa1 is listed.		
Construct	U * ml ⁻¹	Peroxidase activity compared to PaDa1
PaDa 1	0.13	1
UPO 11	21.55	166-fold
UPO 12	46.12	355-fold
UPO 17	6.61	51-fold

Determined by the Bradford Assay the protein concentrations of the supernatant are as listed in Table 13. Protein concentrations in the supernatant from non-optimized bioreactor cultivations were equal or mostly higher than for the benchmark clone, which was also made with the new expression vectors based on the PDC promoter.

TABLE 13

Protein concentrations of the supernatant of the bioreactor cultivations as a result of Bradford Assays. The amount of enzyme found in the supernatant of the cultivation of the PaDa1 expression clone confirmed previous data of Molina et al (2015)		
Construct	mg * mL ⁻¹	
PaDa 1	0.1769	
UPO 11	0.2455	
UPO 12	0.2031	
UPO 17	0.1651	

Naphthalene—Fast-Blue Assays

The naphthalene assay is suitable to measure peroxylgenase activity. As shown in FIG. 6, there are, beneath UPO 1, six UPOs that clearly showed activity in this assay. Two of the new constructs, UPO 12 and UPO 11, converted the substrate nearly twice as fast as the known benchmark. UPO 1.

The rescreened clones that showed activity in the Naphthalene-Fast Blue-Assay are listed in Table 14, including Δ ABS/min values at 520 nm.

TABLE 14

The table lists the clones that were active at the rescreening at the naphthalene-fast blue Assay and their average change in absorption per minute at a measurement over 13 minutes.		
Construct	Clone	Δ ABS(520 nm)/min
UPO 12	12-B0	62.7
UPO 12	12-C0	62.0
UPO 12	12-A0	61.6
UPO 11	11-A0	59.3
UPO 11	11-C0	59.2
UPO 11	11-B0	55.5
PaDa 1	1-B0	31.6
PaDa 1	1-C0	28.2
PaDa 1	1-A0	24.2
UPO 5	5-A0	19.0
UPO 5	5-C0	18.2
UPO 5	5-B0	15.1
UPO 7	7-A0	11.2
UPO 2	2-A0	8.9
UPO 2	2-B0	8.5
UPO 7	7-B0	8.5
UPO 7	7-C0	8.0
UPO 2	2-C0	7.1
UPO 18	18-A0	6.4
UPO 18	18-B0	5.5

TABLE 14-continued

The table lists the clones that were active at the rescreening at the naphthalene-fast blue Assay and their average change in absorption per minute at a measurement over 13 minutes.		
Construct	Clone	ΔABS(520 nm)/min
UPO 18	18-C0	5.0
UPO 4	4-B0	2.8
UPO 4	4-C0	2.8
UPO 4	4-A0	2.5
UPO 9	9-1	1.5
UPO 16	16-A0	0.9
UPO 16	16-C0	0.8
UPO 8	8-1	0.7
UPO 9	9-2	0.7
UPO 9	9-3	0.7
UPO 10	10-1	0.7
UPO 14	14-1	0.7
UPO 15	15-A0	0.6
UPO 14	14-2	0.6
Negative control/BSYBG 11		0.5

Clear naphthalene oxidation activity was found for most tested expression clones but not all of them. For UPO14 this might be explained by the fact that the used database sequence was wrongly annotated and the used was not correct according to Kiehist et al. (2017). Surprisingly new recombinant UPOs with a higher peroxygenase/peroxidase activity ration were identified by these rescreening experiments using the ABTS and naphthalene assay, indicating the high potential of the new recombinant heme thiolate peroxygenases and the diversity of catalytic properties with diverse substrates and chemical reactions.

Filter Assay

In the filter assay with horse radish peroxidase the promising results were obtained. A greenish zone was visible around every active colony.

Peroxidase Plate Assay

After incubation the plates were visually inspected. The plate assay is working with HRP as positive control. All positive controls showed green zones of converted ABTS, while none of the negative controls shows any visible conversion.

For the tested UPOs the plate assays did not show changes after one day. Therefore, the plates were stored for more than a week in the fridge. Surprisingly the color of the plate with the pH 4.5 buffer turned green at those plates, while the plate with the pH 6.0 buffer showed no changes. Color changes were expected for pH 4.5, because most UPOs are active at this pH value.

UPO12 Variants with Increased Peroxygenase Activity

In this Example, a UPO12 mutant library was screened for superior variants of UPO12 using ABTS, naphthalene and 2,6-DMP as substrates. Surprisingly, the largest group of improved variants were found to have a mutation at the C-terminus of the POX12 (UPO12) protein sequence (see FIG. 12).

Multiple variants of UPO12 (SEQ ID NO:12) were identified, also referred to herein as POX12, that showed improved activity on one or more of the tested substrates (ABTS, 2,6-DMP, naphthalene) or altered substrate profiles compared to UPO12 wild type (i.e. variants 23E12 (SEQ ID NO:30), 11G3 (SEQ ID NO:31), 8G3 (SEQ ID NO:32), 11H12 (SEQ ID NO:33), 13A2 (SEQ ID NO:34), 18G3 (SEQ ID NO:35) and 20H11 (SEQ ID NO:36)) (see FIG. 9).

As for UPO12, the corresponding genes of the 11 variants were cloned into the pBSY5S1Z integrative expression vector (containing a FMD promoter fragment of *Hansenula*

polymorpha) via BioXP™ after codon optimization and replacement of their native secretion signals by the alpha factor secretion signal variant (MataD, a deletion variant of the *S. cerevisiae* mating factor alpha signal sequence). The expression vector was introduced in *P. pastoris* for secretion of the variants.

Best results were obtained for variants 8G3 and 11H12. Variant 8G3 (C256S) had an amino acid exchange from cysteine (C) to serine (S) at position 256 which is just 5 amino acids prior the end of the protein. This exchange resulted in a doubling of peroxidase activity, i.e., twice as high activity on ABTS. Variant 8G3 also showed a 1.4-fold improvement on 2,6-DMP and 1.2-fold improvement on naphthalene. Also clone 11H12 showed twice as high activity on 2,6-DMP and a 1.5-fold higher activity on ABTS compared to the UPO12 reference clone. In agreement with the results from clone 8G3 and very surprising, also clone 11H12, showed a mutation at the very same position (C256X); however, a stop codon instead of cysteine.

Similarly, variants 20H11 (E249X), 13A2 (D253N), 18G3 (D253I), showed a C-terminal modification associated with higher activity, and showed an increase of at least 1.4-fold on 2,6-DMP.

Interestingly, variant 23E12 (524F) also had an amino acid exchange from polar serine to large hydrophobic phenylalanine just two amino acids further resulting in 1.3-fold and 1.6-fold higher activity on ABTS and 2,6-DMP, respectively.

Activities of UPO12 variants and their corresponding amino acid mutations compared to wild type UPO12 are summarized in FIG. 9 and Table 15.

TABLE 15

UPO12 variants amino acid sequence mutations. The variants are listed in groups related to the position of the mutation (N-terminal, middle, C-terminal or signal sequence), some clones were identified as WT for others sequencing was unambiguous ("n.s.r." = no sequencing results). (*) This clone did not show activity on any of the three substrates (ABTS, 2,6-DMP, naphthalene) in shake flask.		
Mutation grouping		
N-terminal	Middle/C-terminal	C-terminal
23E12 (S24F)	11G3(D145YY)(*)	8G3 (C256S) 11H12 (C256X) 13A2 (D253N) 18G3 (D253I) 20H11 (E249X)

Identification of New Highly Active Peroxygenase Biocatalysts

In this Example, novel peroxygenases were identified by BLAST search using the UPO12 protein sequence (SEQ ID NO:12) as reference. Identified candidates were expressed in *P. pastoris* and screened for activity on ABTS, naphthalene and 2,6-DMP.

Using the UPO12 wild type amino acid sequence (SEQ ID NO:12) and the online BLAST tool from NCBI, 17 homologous enzymes containing the PCP-motif were identified. As for UPO12 the corresponding genes were cloned into the pBSY5S1Z integrative expression vector (containing an FMD promoter fragment of *Hansenula polymorpha*) via BioXP™ after codon optimization and replacement of their native secretion signals by the alpha factor secretion signal variant (MataD, a deletion variant of the *S. cerevisiae* mating factor alpha signal sequence).

After *P. pastoris* transformation, screening of transformants identified four new UPOs with high activity on ABTS

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(POX27 (SEQ ID NO:37), POX32 (SEQ ID NO:39), POX34 (SEQ ID NO:40), POX39 (SEQ ID NO:41), see FIG. 11), three of them were also found active on 2,6-DMP and naphthalene (POX27, POX32, POX39). These UPOs were also studied in a reaction with ABTS using 2 mM H₂O₂ (8-fold concentration).

Surprisingly, another highly active UPO showing significantly higher activity on ABTS than the reference was identified (POX30 (SEQ ID NO:38), see FIG. 11). Novel peroxygenases showing significant activity on one or more of the tested substrates are summarized in FIG. 10 and FIG. 11.

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SEQUENCE LISTING

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<210> SEQ ID NO 1

<211> LENGTH: 371

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: UPO1 mut

<400> SEQUENCE: 1

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20          25          30

Leu Asp Ala Ile Ile Pro Thr Leu Glu Ala Arg Glu Pro Gly Leu Pro
35          40          45

Pro Gly Pro Leu Glu Asn Ser Ser Ala Lys Leu Val Asn Asp Glu Ala
50          55          60

His Pro Trp Lys Pro Leu Arg Pro Gly Asp Ile Arg Gly Pro Cys Pro
65          70          75          80

Gly Leu Asn Thr Leu Ala Ser His Gly Tyr Leu Pro Arg Asn Gly Val
85          90          95

Ala Thr Pro Ala Gln Ile Ile Asn Ala Val Gln Glu Gly Phe Asn Phe
100         105         110

Asp Asn Gln Ala Ala Ile Phe Ala Thr Tyr Ala Ala His Leu Val Asp
115         120         125

Gly Asn Leu Ile Thr Asp Leu Leu Ser Ile Gly Arg Lys Thr Arg Leu
130         135         140

Thr Gly Pro Asp Pro Pro Pro Ala Ser Val Gly Gly Leu Asn Glu
145         150         155         160
```

-continued

His	Gly	Thr	Phe	Glu	Gly	Asp	Ala	Ser	Met	Thr	Arg	Gly	Asp	Ala	Phe
			165						170					175	
Phe	Gly	Asn	Asn	His	Asp	Phe	Asn	Glu	Thr	Leu	Phe	Glu	Gln	Leu	Val
		180					185						190		
Asp	Tyr	Ser	Asn	Arg	Phe	Gly	Gly	Gly	Lys	Tyr	Asn	Leu	Thr	Val	Ala
	195					200						205			
Gly	Glu	Leu	Arg	Phe	Lys	Arg	Ile	Gln	Asp	Ser	Ile	Ala	Thr	Asn	Pro
	210				215						220				
Asn	Phe	Ser	Phe	Val	Asp	Phe	Arg	Phe	Phe	Thr	Ala	Tyr	Gly	Glu	Thr
225				230						235					240
Thr	Phe	Pro	Ala	Asn	Leu	Phe	Val	Asp	Gly	Arg	Arg	Asp	Asp	Gly	Gln
			245					250						255	
Leu	Asp	Met	Asp	Ala	Ala	Arg	Ser	Phe	Phe	Gln	Phe	Ser	Arg	Met	Pro
		260						265					270		
Asp	Asp	Phe	Phe	Arg	Ala	Pro	Ser	Pro	Arg	Ser	Gly	Thr	Gly	Val	Glu
	275						280						285		
Val	Val	Val	Gln	Ala	His	Pro	Met	Gln	Pro	Gly	Arg	Asn	Val	Gly	Lys
	290				295						300				
Ile	Asn	Ser	Tyr	Thr	Val	Asp	Pro	Thr	Ser	Ser	Asp	Phe	Ser	Thr	Pro
305				310						315					320
Cys	Leu	Met	Tyr	Glu	Lys	Phe	Val	Asn	Ile	Thr	Val	Lys	Ser	Leu	Tyr
			325					330						335	
Pro	Asn	Pro	Thr	Val	Gln	Leu	Arg	Lys	Ala	Leu	Asn	Thr	Asn	Leu	Asp
			340					345					350		
Phe	Leu	Phe	Gln	Gly	Val	Ala	Ala	Gly	Cys	Thr	Gln	Val	Phe	Pro	Tyr
	355					360					365				
Gly	Arg	Asp													
	370														

<210> SEQ ID NO 2
 <211> LENGTH: 373
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 2

<400> SEQUENCE: 2

Met	Arg	Gly	Ala	Ala	Arg	Phe	Ala	Val	Leu	Ile	Ala	Leu	Phe	Thr	His
1				5					10					15	
Ala	Ala	Ile	Ala	Phe	Pro	Ala	Tyr	Gly	Ser	Leu	Ala	Gly	Leu	Thr	Arg
		20					25						30		
Glu	Gln	Leu	Asp	Glu	Ile	Leu	Pro	Thr	Leu	Glu	Ile	Arg	Glu	Pro	Gly
		35				40					45				
Lys	Pro	Pro	Gly	Pro	Leu	Lys	Asp	Thr	Ser	Ala	Lys	Leu	Val	Asn	Asp
	50				55						60				
Lys	Ala	His	Pro	Trp	Lys	Pro	Val	Ala	Pro	Ala	Asp	Ile	Arg	Gly	Pro
65			70					75						80	
Cys	Pro	Gly	Leu	Asn	Thr	Leu	Ala	Ser	His	Gly	Trp	Leu	Pro	Arg	Asn
			85					90						95	
Gly	Ile	Ala	Ser	Pro	Ser	Glu	Ile	Ile	Thr	Ala	Val	Gln	Glu	Gly	Phe
		100					105						110		
Asn	Met	Asp	Asn	Gly	Leu	Ala	Ile	Phe	Val	Thr	Tyr	Ala	Ala	His	Leu
	115					120						125			
Val	Asp	Gly	Asn	Ile	Leu	Thr	Asp	Lys	Leu	Ser	Ile	Gly	Gly	Lys	Thr
	130					135					140				

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Gly Leu Thr Gly Pro Asn Pro Pro Ala Pro Ala Ile Val Gly Gly Leu
 145 150 155 160
 Asn Thr His Ala Val Phe Glu Gly Asp Thr Ser Met Thr Arg Gly Asp
 165 170 175
 Phe Phe Phe Gly Asn Asn His Asp Phe Asn Glu Thr Leu Phe Asp Glu
 180 185 190
 Phe Val Asp Phe Ser Asn Arg Phe Gly Ala Gly Lys Tyr Asn Leu Thr
 195 200 205
 Val Ala Gly Glu Phe Arg Trp Gln Arg Ile Gln Asp Ser Ile Ala Thr
 210 215 220
 Asn Pro Glu Phe Ser Phe Val Ser Pro Arg Phe Phe Thr Ala Tyr Ala
 225 230 235 240
 Glu Ser Thr Phe Pro Ile Asn Phe Phe Ile Asp Gly Arg Gln Thr Asp
 245 250 255
 Gly Gln Leu Asp Leu Thr Val Ala Arg Gly Phe Phe Gln Asn Ser Arg
 260 265 270
 Met Pro Asp Asp Phe His Arg Ala Asn Gly Thr Arg Gly Thr Glu Gly
 275 280 285
 Ile Asp Leu Val Ala Glu Ala His Pro Ile Glu Pro Gly Ser Asn Val
 290 295 300
 Gly Gly Val Asn Asn Tyr Val Val Asp Pro Thr Ser Ala Asp Phe Ser
 305 310 315 320
 Thr Phe Cys Leu Leu Tyr Glu Asn Phe Val Asn Lys Thr Val Lys Gly
 325 330 335
 Leu Tyr Pro Asn Pro Thr Gly Ala Leu Arg Lys Ala Leu Asn Thr Asn
 340 345 350
 Leu Gly Phe Phe Phe Ser Gly Ile Ser Asp Ser Gly Cys Thr Gln Val
 355 360 365
 Phe Pro Tyr Gly Lys
 370

<210> SEQ ID NO 3
 <211> LENGTH: 377
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 3

<400> SEQUENCE: 3

Met Leu Lys Leu Phe Phe Val Gln Thr Ala Leu Leu Ala Leu Ser Gly
 1 5 10 15
 Thr Thr Phe Ala Tyr Pro Ser His Met Ser Leu Ala Gly Leu Thr Arg
 20 25 30
 Glu Gln Leu Asp Gln Ile Val Pro Thr Leu Thr Phe Thr Pro Pro Pro
 35 40 45
 Pro Pro Pro Ala Pro Leu Asn Asp Thr Ser Ala Lys Leu Val Asn Asp
 50 55 60
 Pro Ala His Pro Trp Gln Pro Leu Arg Ala Gly Asp Ile Arg Gly Val
 65 70 75 80
 Cys Pro Gly Leu Asn Thr Leu Ala Ser His Gly Tyr Leu Pro Arg Asn
 85 90 95
 Gly Ile Val Thr Pro Asn Gln Ile Ile Glu Ala Ala Gln Asp Gly Phe
 100 105 110
 Asn Met Asp Asn Thr Leu Ala Arg Phe Leu Ala Tyr Gly Thr Phe Leu
 115 120 125

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Gly Val Ala Thr Pro Ala Gln Ile Ile Asn Ala Val Gln Glu Gly Phe
 100 105 110
 Asn Met Gly Asn Asp Leu Ala Val Phe Val Thr Tyr Ala Ala His Leu
 115 120 125
 Val Asp Gly Asn Gln Val Thr Asp Leu Leu Ser Ile Gly Gly Lys Thr
 130 135 140
 Pro Gln Thr Gly Pro Asp Pro Pro Gln Pro Ala Ile Val Gly Gly Leu
 145 150 155 160
 Asp Thr His Ala Val Phe Glu Gly Asp Ala Ser Met Thr Arg Gly Asp
 165 170 175
 Ala Phe Phe Gly Asp Asn His Ser Phe Asn Glu Thr Gln Phe Asp Glu
 180 185 190
 Phe Ser Ala Phe Ser Asn Lys Phe Gly Gly Gly Tyr Tyr Asn Leu Ser
 195 200 205
 Val Ala Ala Glu Phe Arg Trp Gln Arg Ile Gln Glu Ser Ile Ala Thr
 210 215 220
 Asn Pro Asn Phe Ser Phe Ile Ser Pro Arg Tyr Phe Thr Ala Tyr Ala
 225 230 235 240
 Glu Ser Val Phe Pro Leu Val Phe Phe Val Asp Gly Arg Val Ser Asp
 245 250 255
 Gly Arg Leu Ser Leu Pro Asn Ala Arg Gly Phe Phe Gln Asn Ser Gln
 260 265 270
 Met Pro Thr Asp Phe Phe Arg Pro Asn Gln Ser Ile Gly Leu Asp Val
 275 280 285
 Ile Gly Asp Gly Ile Ser Ala Ile Ala Ser Ala His Pro Ile Ala Pro
 290 295 300
 Gly Lys Asn Glu Gly Val Gly Asn Tyr Val Leu Asp Pro Thr Ser Ala
 305 310 315 320
 Asp Phe Asp His Phe Cys Leu Leu Tyr Ile Asn Phe Val Asn Gln Thr
 325 330 335
 Val Lys Ser Leu Tyr Pro Asn Pro Thr Gly Val Leu Arg Asp Ala Leu
 340 345 350
 Lys Arg Asn Leu Asp Asn Phe Tyr Ser Pro Leu Asn Gly Ser Asp Cys
 355 360 365
 Val Gln Ile Phe Pro Tyr Gly Lys
 370 375

<210> SEQ ID NO 6
 <211> LENGTH: 376
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 6

<400> SEQUENCE: 6

Met Val Gln Phe Thr Val Ile Leu Ser Leu Leu Leu Ala Thr Gly Lys
 1 5 10 15
 Ala Leu Ala Phe Pro Gln Tyr Gly Ser Leu Ala Gly Leu Ser Glu Arg
 20 25 30
 Glu Leu Glu Asp Ile Leu Pro Arg Leu His Ala Val Lys Pro Pro Pro
 35 40 45
 Pro Pro Gly Pro Leu Asn Asp Thr Ser Thr Lys Leu Val Asn Asn Pro
 50 55 60
 Ala His Pro Phe Leu Pro Gln Arg Asn Gly Asp Met Arg Gly Pro Cys
 65 70 75 80

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Pro Gly Leu Asn Thr Leu Ala Ser His Gly Tyr Leu Pro Arg Asn Gly
      85                      90                      95

Ile Ala Thr Pro Ala Gln Ile Ile Asn Ala Val Gln Glu Gly Phe Asn
      100                    105                    110

Met Gly Asn Asp Leu Ala Val Phe Val Thr Tyr Ala Ala Phe Leu Val
      115                    120                    125

Asp Gly Asn Gln Val Thr Asn Leu Leu Ser Ile Gly Gly Lys Ser Ser
      130                    135                    140

Leu Thr Gly Pro Asp Pro Pro Lys Pro Ala Ile Val Gly Gly Leu Asp
      145                    150                    155                    160

Thr His Ala Val Phe Glu Gly Asp Ala Ser Met Thr Arg Gly Asp Ala
      165                    170                    175

Phe Phe Gly Asp Asn His Ser Phe Lys Glu Asn Gln Phe Asp Glu Phe
      180                    185                    190

Ile Ala Phe Ser Asn Lys Phe Gly Gly Gly Lys Tyr Asn Leu Thr Val
      195                    200                    205

Ala Ser Glu Phe Arg Trp Gln Arg Ile Gln Glu Ser Thr Ala Thr Asn
      210                    215                    220

Pro Asn Phe Ser Phe Ile Ser Pro Arg Tyr Phe Thr Ala Tyr Ala Glu
      225                    230                    235                    240

Ser Thr Phe Pro Ile Thr Phe Phe Val Asp Gly Arg Asn Glu Asp Gly
      245                    250                    255

Ala Leu Ser Leu Asp Val Ala Arg Gly Phe Phe Gln Asp Ser Arg Met
      260                    265                    270

Pro Asn Gly Phe Phe Arg Ala Asn Gln Ser Ile Gly Leu Asp Ile Ile
      275                    280                    285

Gly Ser Leu Ile Asp Phe Ile Phe Glu Pro His Pro Ile Gln Pro Gly
      290                    295                    300

Gly Asn Gln Gly Arg Val Asn Ser Tyr Thr Val Asp Pro Asn Ser Ala
      305                    310                    315                    320

Asn Phe Ser Gln Phe Cys Gln Leu Tyr Gln Asp Phe Val Asn Asn Thr
      325                    330                    335

Val Lys Gly Leu Tyr Pro Asn Pro Gln Gly Val Leu Arg Asp Asn Leu
      340                    345                    350

Asn Thr Asn Leu Gly Phe Phe Phe Ser Pro Leu Gln Gly Ser Gly Cys
      355                    360                    365

Pro Gln Val Phe Pro Phe Gly Gln
      370                    375

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<210> SEQ ID NO 7
<211> LENGTH: 376
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: UPO 7

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<400> SEQUENCE: 7

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Met Arg Phe Phe Ser His Leu Ser Ile Ile Pro Leu Leu Ser Leu His
1          5                      10                      15

Gly Val Leu Ala Phe Pro Ser Tyr Gly Thr Leu Ala Gly Leu Ser Arg
      20                    25                    30

Ser Glu Leu Glu Ala Ile Leu Pro Thr Leu Lys Pro Gly Val Pro Glu
      35                    40                    45

Ser Pro Pro Gly Pro Leu Asn Asp Thr Ser Ala Lys Leu Val Asn Asp
      50                    55                    60

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Lys 65	Lys	His	Pro	Trp	Lys 70	Pro	Ala	Gly	Lys 75	Lys	Asp	Ile	Arg	Gly	Pro 80
Cys	Pro	Gly	Leu	Asn 85	Thr	Leu	Ala	Ser	His 90	Gly	Trp	Leu	Pro	Arg	Asp 95
Gly	Val	Ala	Thr	Pro	Ala	Gln	Ile	Val	Asn 105	Ala	Val	Gln	Glu	Gly	Phe 110
Asn	Met	Gly	Asn	Asp	Leu	Ala	Val	Phe	Val	Thr	Tyr	Ala	Ala	His	Leu 115
Val	Asp	Gly	Asn	Leu	Ile	Thr	Asn	Leu	Leu	Ser	Ile	Gly	Gly	Lys	Thr 130
Asp	Arg	Thr	Gly	Pro	Asn	Pro	Pro	Pro	Pro	Ala	Ile	Val	Gly	Gly	Leu 145
Asn	Thr	His	Ala	Val	Phe	Glu	Gly	Asp	Ala	Ser	Thr	Thr	Arg	Ala	Asp 160
Phe	Phe	Phe	Gly	Asp	Asn	His	Ser	Phe	Asn 185	Glu	Thr	Leu	Phe	Asp	Glu 180
Leu	Thr	Ala	Phe	Ser	Asn	Lys	Phe	Gly	Gly	Gly	Phe	Tyr	Asn	Leu	Ser 195
Val	Ala	Ala	Glu	Phe	Arg	Phe	Gln	Arg	Ile	Gln	Asp	Ser	Ile	Ala	Thr 210
Asn	Pro	Gln	Phe	Asp	Phe	Ile	Ser	Pro	Arg	Tyr	Phe	Thr	Ala	Tyr	Ala 225
Glu	Ser	Ile	Phe	Pro	Leu	Thr	Phe	Phe	Ile	Asp	Gly	Arg	Asp	Lys	Ser 245
Leu	His	Leu	Asp	Met	Asn	Val	Ala	Arg	Gly	Phe	Phe	Gln	Asn	Ser	Arg 260
Phe	Pro	Asp	Gly	Phe	Phe	Arg	Ser	Asn	Thr	Ser	Ile	Thr	Leu	Asp	Val 275
Ile	Gly	Gly	Gly	Ile	Asp	Tyr	Ile	Phe	Ser	Lys	His	Pro	Val	Pro	Pro 290
Gly	Ser	Asn	Asn	Gly	Thr	Val	Asn	Ser	Tyr	Thr	Pro	Asn	Pro	Asn	Ser 305
Ala	Asp	Phe	Thr	Gln	Phe	Cys	Lys	Leu	Tyr	Thr	Asp	Phe	Val	Asn	Ile 325
Thr	Ile	Arg	Gly	Leu	Tyr	Pro	Asn	Ala	Lys	Gly	Ala	Leu	Leu	Thr	Ala 340
Leu	Asn	Lys	Asn	Leu	Glu	Tyr	Phe	Tyr	Ser	Pro	Leu	Val	Gly	Ser	Gly 355
Cys	Pro	Gln	Val	Pro	Pro	Phe	Val								

<210> SEQ ID NO 8
 <211> LENGTH: 376
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 8

<400> SEQUENCE: 8

Met	Ala	Arg	Val	Phe	Phe	Ala	Ile	Ala	Ala	Leu	Leu	Leu	Ala	Ala	Lys
1				5						10					15
Asp	Val	Val	Ser	Phe	Pro	Asn	Tyr	Ala	Ser	Leu	Ala	Gly	Leu	Ser	Glu
			20					25					30		
Arg	Glu	Leu	Asp	Glu	Ile	Ile	Pro	Gln	Leu	Thr	Val	Arg	Thr	Leu	Glu
	35						40					45			

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Lys	Pro	Pro	Gly	Gln	Met	Lys	Asn	Thr	Leu	Thr	Lys	Leu	Val	Asn	Asp
50						55					60				
Pro	Ala	His	Pro	Trp	Ile	Ala	Pro	Ala	Pro	Asp	Asp	Gln	Arg	Asp	Pro
65					70					75					80
Cys	Pro	Gly	Leu	Asn	Thr	Leu	Ala	Asn	His	Gly	Tyr	Leu	Pro	Arg	Asp
				85					90					95	
Gly	Ile	Ala	Thr	Pro	Ala	Gln	Ile	Val	Asn	Ala	Val	Gln	Glu	Gly	Phe
			100					105					110		
Asn	Met	Ala	Asn	Asp	Ile	Ala	Val	Phe	Val	Thr	Tyr	Ala	Ala	His	Leu
		115					120					125			
Val	Asp	Gly	Asn	Leu	Leu	Thr	Asp	Leu	Leu	Ser	Ile	Gly	Gly	Lys	Ser
	130					135					140				
Ala	Lys	Thr	Gly	Pro	Asn	Pro	Pro	Ser	Pro	Ala	Ile	Val	Gly	Gly	Leu
145					150					155					160
Asp	Thr	His	Ala	Val	Phe	Glu	Gly	Asp	Ala	Ser	Thr	Thr	Arg	Gly	Asp
				165					170					175	
Ala	Phe	Phe	Gly	Asp	Asn	His	Ser	Phe	Asn	Glu	Ser	Leu	Phe	Asp	Glu
			180					185					190		
Leu	Thr	Ala	Phe	Ser	Asn	Lys	Phe	Gly	Ala	Gly	Phe	Tyr	Asn	Leu	Ser
		195					200					205			
Val	Ala	Thr	Glu	Phe	Arg	Phe	Gln	Arg	Ile	Gln	Asp	Ser	Ile	Ala	Thr
	210					215					220				
Asn	Pro	Gln	Phe	Ser	Leu	Ile	Ser	Pro	Arg	Tyr	Tyr	Thr	Ala	Tyr	Ala
225					230					235					240
Glu	Ser	Val	Phe	Pro	Val	Ala	Phe	Phe	Val	Asp	Gly	Arg	Glu	Thr	Asn
			245						250					255	
Gly	Ser	Leu	Asn	Met	Thr	Val	Ala	Arg	Gly	Phe	Phe	Gln	Asp	Gly	Arg
			260					265					270		
Met	Pro	Asn	Asp	Phe	Phe	Arg	Ser	Asn	Ile	Ser	Trp	Gly	Leu	Asp	Leu
		275					280					285			
Ile	Gly	Glu	Gly	Ile	Gly	Phe	Ile	Phe	Thr	Pro	His	Pro	Ile	Glu	Pro
	290					295					300				
Gly	Thr	Asn	Asn	Gly	Thr	Leu	Asn	Ser	Tyr	Thr	Leu	Asp	Pro	Asn	Ser
305					310					315					320
Ala	Asp	Phe	Ser	Asp	Phe	Cys	Lys	Leu	Tyr	Thr	Asp	Phe	Val	Asn	Val
				325					330					335	
Thr	Val	Arg	Gly	Leu	Tyr	Pro	Asn	Ala	Thr	Gly	Pro	Leu	Leu	Asn	Ala
			340					345					350		
Leu	Asn	Gln	Asn	Leu	Asp	Phe	Phe	Phe	Gly	Pro	Leu	Gly	Asp	Gln	Gly
		355					360					365			
Cys	Thr	Gln	Val	Pro	Ala	Phe	Val								
	370					375									

<210> SEQ ID NO 9

<211> LENGTH: 275

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: UPO 9

<400> SEQUENCE: 9

Met	Lys	Leu	Asn	Ile	Phe	Ser	Thr	Thr	Leu	Ala	Ile	Gly	Leu	Val	Ser
1				5					10					15	

Ala	Gly	Ala	His	Tyr	His	Gln	Gln	Asp	Val	Val	Ala	Asn	Gly	Thr	Glu
			20					25					30		

Gly	Val	Trp	Ile	Ala	Pro	Thr	Asp	Thr	Asp	Tyr	Arg	Gly	Pro	Cys	Pro
354045															
Met	Met	Asn	Thr	Leu	Ala	Asn	His	Gly	Phe	Leu	Pro	Arg	Asp	Gly	Lys
505560															
Asn	Leu	Thr	Glu	Tyr	Asn	Val	Val	Lys	Gly	Leu	Asn	Asp	Gly	Leu	Asn
65707580															
Phe	Asn	Lys	Ser	Leu	Ala	Thr	Ile	Met	Phe	Gln	Gln	Ala	Ile	Pro	Ala
859095															
Ser	Pro	Ala	Tyr	Pro	Asn	Ala	Thr	Phe	Phe	Thr	Leu	Asn	Asp	Leu	Asn
100105110															
Arg	His	Asn	Val	Leu	Glu	His	Asp	Gly	Ser	Ile	Ser	Arg	Ser	Asp	Ala
115120125															
Tyr	Tyr	Gly	Asn	Asn	Tyr	Ile	Phe	Asn	Gln	Thr	Ile	Phe	Asp	Thr	Thr
130135140															
Lys	Ala	Tyr	Trp	Pro	Ser	Glu	Thr	Leu	Thr	Ala	Gln	His	Leu	Ile	Asp
145150155160															
Gly	Lys	Met	Phe	Arg	Gln	Ile	Val	Ser	Arg	Ser	Thr	Asn	Pro	Asn	Tyr
165170175															
Thr	Phe	Ser	Ala	Thr	Thr	Gln	Gln	Phe	Ser	Leu	Gly	Glu	Met	Ala	Ala
180185190															
Pro	Ile	Val	Ala	Phe	Gly	Asp	Lys	Tyr	Val	Val	Thr	Ala	Asn	Arg	Thr
195200205															
Leu	Val	Glu	Ser	Trp	Ile	Glu	Asn	Glu	Arg	Leu	Pro	Thr	Glu	Leu	Gly
210215220															
Trp	Arg	Lys	Pro	Val	Glu	Glu	Ile	Leu	Leu	Ser	Asp	Ile	Thr	Tyr	Val
225230235240															
Thr	Glu	Val	Leu	Gly	Asn	Leu	Thr	Ser	Leu	Tyr	Ser	Thr	Val	Ile	Ile
245250255															
Thr	Pro	Asn	Pro	Asp	Ser	Leu	Ala	Lys	Arg	Gln	Met	Gly	His	Trp	Gly
260265270															
Gln	Ser	Ile													
275															

<400> SEQUENCE: 10

Met	Lys	Thr	Thr	Thr	Leu	Leu	Cys	Leu	Ala	Ala	Ala	Leu	Thr	Gln	Thr
1				5					10					15	
Tyr	Ala	Phe	Pro	Gln	Gln	Gly	Ala	Pro	His	Pro	Leu	Pro	Trp	Ser	Pro
			20				25					30			
Pro	Gly	Pro	Asn	Asp	Val	Arg	Ala	Pro	Cys	Pro	Met	Leu	Asn	Thr	Leu
		35					40					45			
Ala	Asn	His	Gly	Tyr	Leu	Pro	His	Asn	Gly	Lys	Asp	Ile	Thr	Glu	Arg
	50					55					60				
His	Thr	Ile	Asn	Ala	Leu	Tyr	Asn	Ala	Leu	Gly	Ile	Glu	Glu	Glu	Leu
65					70					75					80
Ala	Ile	Tyr	Leu	His	Gln	Glu	Ala	Val	Thr	Thr	Asn	Pro	Ala	Pro	Asn
				85				90						95	
Ala	Thr	Thr	Phe	Ser	Leu	Asn	Asp	Leu	Ser	Arg	His	Asp	Ile	Leu	Glu
			100					105					110		

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His Asp Ala Ser Leu Ser Arg Gln Asp Ala Tyr Phe Gly Asp Asn His
 115                      120                      125

Asp Phe Asn Gln Thr Ile Phe Asp Glu Thr Arg Ser Tyr Trp Thr Ser
 130                      135                      140

Pro Ile Ile Asp Val Lys Gln Ala Ala Val Ser Arg Gln Ala Arg Val
 145                      150                      155                      160

Asn Thr Ser Met Ala Thr Asn Pro Asn Tyr Thr Met Ser Glu Leu Gly
                      165                      170                      175

Asp Ser Phe Ser Tyr Gly Glu Thr Ala Ala Tyr Ile Ile Val Leu Gly
                      180                      185                      190

Asp Lys Glu Lys Gly Leu Val Asn Arg Ser Arg Val Glu Tyr Leu Phe
 195                      200                      205

Glu Asn Glu Arg Leu Pro Leu Asp Leu Gly Trp Ser Arg Ala Lys Glu
 210                      215                      220

Asn Ile Thr Phe Asp Asp Leu Ser Thr Met Leu Gln Arg Ile Ile Asn
 225                      230                      235                      240

Ala Thr Gly Gly Glu Met Asp Phe Arg Ala Thr Ile Ala Leu Pro Arg
                      245                      250                      255

Leu Val Tyr Ile Tyr Tyr Glu Glu Ala
 260                      265

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<210> SEQ ID NO 11
<211> LENGTH: 263
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: UPO 11

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<400> SEQUENCE: 11

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Met Lys Thr Thr Thr Leu Leu Phe Leu Val Gly Ala Leu Thr Gln Thr
 1                      5                      10                      15

His Ala Phe Pro Gln Gln Gly Val Pro His Pro Leu Pro Trp Ser Pro
                      20                      25                      30

Pro Gly Pro Asn Asp Val Arg Ala Pro Cys Pro Met Leu Asn Thr Leu
                      35                      40                      45

Ala Asn His Gly Phe Leu Pro His Asn Gly Lys Asn Ile Thr Gln Gln
 50                      55                      60

His Thr Ile Asn Ala Leu Tyr Asn Ala Leu Gly Ile Asp Ala Glu Leu
 65                      70                      75                      80

Ala Thr Tyr Leu His Gln Glu Ala Val Thr Thr Asn Pro Val Pro Asn
                      85                      90                      95

Ala Thr Thr Phe Ser Leu Asn Asp Leu Ser Arg His Asp Ile Leu Glu
                      100                      105                      110

His Asp Ala Ser Leu Ser Arg Gln Asp Ala Phe Phe Gly Asp Asn His
 115                      120                      125

Asp Phe Asn Gln Thr Ile Phe Asn Gln Thr Arg Ser Tyr Trp Thr Ser
 130                      135                      140

Pro Ile Ile Asp Val Lys Gln Ala Ala Leu Ala Arg Gln Ala Arg Val
 145                      150                      155                      160

Asn Thr Ser Met Ala Thr Asn Pro Asn Tyr Thr Met Ser Glu Leu Gly
                      165                      170                      175

Asp Ala Phe Ser Tyr Gly Glu Thr Ala Ala Tyr Ile Ile Val Leu Gly
                      180                      185                      190

Asp Lys Glu Ala Gly Leu Val Asn Arg Ser Arg Val Glu Tyr Leu Phe
 195                      200                      205

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Glu Asn Glu Arg Leu Pro Val Glu Leu Gly Trp Ser Arg Ala Arg Glu
 210 215 220

Asn Ile Thr Phe Asp Asp Leu Ser Thr Met Leu Asn Lys Ile Ile Asn
 225 230 235 240

Ala Thr Gly Gly Glu Ser Glu Phe Glu Arg Glu Leu Ala Lys Arg Gly
 245 250 255

Gly Val His Val Gly Trp Arg
 260

<210> SEQ ID NO 12
 <211> LENGTH: 261
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 12

<400> SEQUENCE: 12

Met Lys Ser Leu Ser Phe Ser Leu Ala Leu Gly Phe Gly Ser Thr Leu
 1 5 10 15

Val Tyr Ser Ala Pro Ser Pro Ser Ser Gly Trp Gln Ala Pro Gly Pro
 20 25 30

Asn Asp Val Arg Ala Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His
 35 40 45

Gly Phe Leu Pro His Asp Gly Lys Gly Ile Thr Val Asn Lys Thr Ile
 50 55 60

Asp Ala Leu Gly Ser Ala Leu Asn Ile Asp Ala Asn Leu Ser Thr Leu
 65 70 75 80

Leu Phe Gly Phe Ala Ala Thr Thr Asn Pro Gln Pro Asn Ala Thr Phe
 85 90 95

Phe Asp Leu Asp His Leu Ser Arg His Asn Ile Leu Glu His Asp Ala
 100 105 110

Ser Leu Ser Arg Gln Asp Ser Tyr Phe Gly Pro Ala Asp Val Phe Asn
 115 120 125

Glu Ala Val Phe Asn Gln Thr Lys Ser Phe Trp Thr Gly Asp Ile Ile
 130 135 140

Asp Val Gln Met Ala Ala Asn Ala Arg Ile Val Arg Leu Leu Thr Ser
 145 150 155 160

Asn Leu Thr Asn Pro Glu Tyr Ser Leu Ser Asp Leu Gly Ser Ala Phe
 165 170 175

Ser Ile Gly Glu Ser Ala Ala Tyr Ile Gly Ile Leu Gly Asp Lys Lys
 180 185 190

Ser Ala Thr Val Pro Lys Ser Trp Val Glu Tyr Leu Phe Glu Asn Glu
 195 200 205

Arg Leu Pro Tyr Glu Leu Gly Phe Lys Arg Pro Asn Asp Pro Phe Thr
 210 215 220

Thr Asp Asp Leu Gly Asp Leu Ser Thr Gln Ile Ile Asn Ala Gln His
 225 230 235 240

Phe Pro Gln Ser Pro Gly Lys Val Glu Lys Arg Gly Asp Thr Arg Cys
 245 250 255

Pro Tyr Gly Tyr His
 260

<210> SEQ ID NO 13
 <211> LENGTH: 283
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:

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<223> OTHER INFORMATION: UPO 13

<400> SEQUENCE: 13

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Met Lys Thr Leu Phe Leu Leu Thr Leu Ala Ala Phe Thr Pro Val Phe
 1           5           10           15

Ala Gly Phe Asp Thr Trp Ala Pro Pro Gly Pro Tyr Asp Val Arg Ala
 20           25           30

Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His Gly Phe Leu Pro His
 35           40           45

Asp Gly His Glu Ile Thr Arg Glu Gln Thr Glu Asn Ala Leu Phe Asp
 50           55           60

Ala Leu His Ile Asp Lys Met Leu Gly Ser Ser Leu Phe Asp Phe Ala
 65           70           75           80

Met Thr Thr Asn Pro Val Ala Asn Ser Thr Thr Phe Ser Leu Asn Asp
 85           90           95

Leu Gly Asn His Asn Val Leu Glu His Asp Ala Ser Leu Ser Arg Ser
100           105           110

Asp Ala Tyr Phe Gly Asn Thr Leu Thr Phe Asn Gln Thr Val Phe Asp
115           120           125

Glu Thr Lys Ser Tyr Trp Thr Asp Glu Thr Val Thr Ile Glu Met Ala
130           135           140

Ser Asn Ala Arg Leu Ala Arg Ile Lys Thr Ser Asn Ala Thr Asn Pro
145           150           155           160

Thr Tyr Ser Met Ser Glu Leu Gly Asn Gly Phe Thr Lys Gly Glu Ser
165           170           175

Ala Ala Tyr Val Val Ile Phe Gly Asp Lys Ile Ser Gly Thr Val Pro
180           185           190

Arg Ala Trp Val Glu Trp Leu Phe Glu Ile Ala Leu Lys Thr Gln Pro
195           200           205

Ser Thr Pro Ser Ile Lys Pro Thr Gln Thr Pro Ser Ser Pro Thr Arg
210           215           220

Leu Leu Leu Lys Arg Leu Gly Arg Gln Leu Met Leu Ile Val Pro Arg
225           230           235           240

Pro Ile Arg Leu Arg Val Leu Arg Asn Thr Pro Pro Leu Arg Leu Ile
245           250           255

Thr Lys Asn Lys Pro Arg Glu Met Ala Pro Asn Leu Leu Ile Leu Ala
260           265           270

Val His Lys Arg Ala Thr Ser Met Gln Lys Arg
275           280

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<210> SEQ ID NO 14

<211> LENGTH: 225

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: UPO 14

<400> SEQUENCE: 14

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Met Arg Thr Ser Leu Leu Pro Ala Leu Ala Ala Val Ser Pro Val Leu
 1           5           10           15

Ala Gly Phe Asp Thr Trp Ala Pro Pro Gly Pro Tyr Asp Val Arg Gly
 20           25           30

Pro Cys Pro Met Leu Asn Thr Leu Thr Asn His Gly Phe Phe Pro His
 35           40           45

Asp Gly Gln Asp Ile Asp Arg Glu Thr Thr Glu Asn Ala Leu Phe Asp
 50           55           60

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Ala Leu His Val Asn Lys Thr Leu Ala Ser Phe Leu Arg Ala Asp Ala
65 70 75 80

Tyr His Gly Ser Val Leu Ala Phe Asn His Thr Ile Phe Glu Glu Thr
85 90 95

Lys Ser Tyr Trp Thr Asp Glu Thr Val Thr Leu Lys Met Ala Ala Asp
100 105 110

Ala Arg Tyr Tyr Arg Ile Lys Ser Ser Gln Ala Thr Asn Pro Thr Tyr
115 120 125

Gln Met Ser Glu Leu Gly Asp Ala Phe Thr Tyr Gly Glu Ser Ala Ala
130 135 140

Tyr Val Val Leu Phe Gly Asp Lys Glu Ser Gln Thr Val Pro Arg Ser
145 150 155 160

Trp Val Glu Trp Leu Phe Glu Lys Glu Gln Leu Pro Gln His Leu Gly
165 170 175

Trp Lys Arg Pro Ala Thr Ser Phe Glu Leu Asn Asp Leu Asp Lys Phe
180 185 190

Met Ala Leu Ile Gln Asn Tyr Thr Gln Glu Ile Glu Glu Pro Ser Cys
195 200 205

Glu Ser Arg Lys Gln Arg Arg Lys Pro Arg Gly Pro Ser His Phe Gly
210 215 220

Phe
225

<210> SEQ ID NO 15
 <211> LENGTH: 375
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 15

<400> SEQUENCE: 15

Met Ala Lys Phe Ser Thr Leu Phe Ala Phe Ser Ala Leu Ala Ile Gln
1 5 10 15

Ala Ile Ala Leu Pro Gln Tyr Arg Ser Leu Ala Gly Leu Ser Glu Arg
20 25 30

Glu Leu Glu Gly Ile Leu Pro Arg Leu Asn Val Val Thr Pro Pro Pro
35 40 45

Ser Pro Pro Gly Pro Pro Asn Asp Thr Ser Val Lys Leu Val Asn Asp
50 55 60

Ala Ala His Pro Phe Met Pro Leu Gln Asp Gly Asp Ile Arg Gly Pro
65 70 75 80

Cys Pro Gly Leu Asn Thr Leu Ala Ser His Gly Tyr Leu Pro Arg Asn
85 90 95

Gly Ile Ala Thr Pro Ala Gln Ile Ile Asn Ala Val Gln Asp Gly Phe
100 105 110

Ser Met Asp Asn Gly Leu Ala Thr Leu Leu Ala Tyr Ala Thr Met Leu
115 120 125

Val Asp Gly Asn Pro Leu Thr Asn Leu Met Ser Ile Gly Gly Lys Ser
130 135 140

Pro Leu Thr Gly Met Asp Pro Pro Gln Pro Ala Ile Val Gly Gly Leu
145 150 155 160

Asp Thr His Ala Val Phe Glu Gly Asp Ala Ser Met Thr Arg Ala Asp
165 170 175

Phe Phe Phe Gly Asp Asn His Ser Phe Asn Gln Thr Leu Phe Asn Gln
180 185 190

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Phe Ala Asn Phe Ser Asn Gln Phe Gly Asp Gly Asn Tyr Asn Leu Thr
 195 200 205
 Thr Ala Glu Glu Tyr Arg Phe Phe Arg Ile Gln Gln Ser Ile Ala Glu
 210 215 220
 Asn Pro Gln Phe Ser Phe Ile Ser Pro Arg Phe Phe Thr Ala Tyr Phe
 225 230 235 240
 Glu Ser Ala Phe Pro Leu Val Phe Phe Val Asp Gly Arg Gln Ala Asp
 245 250 255
 Gly Gln Leu Ser Val Glu Asn Ala Thr Ser Phe Phe Arg Asp Met Gln
 260 265 270
 Phe Pro Asp Asp Phe His Arg Ala Asp Gly Ser Gln Thr Ala Asp Leu
 275 280 285
 Val Asn Asn Ala Ala Thr Ala Ile Phe Ser Ala His Pro Met Gln Pro
 290 295 300
 Gly Gly Asn Asn Gly Thr Val Asn Ser Tyr Thr Phe Asp Pro Asn Ser
 305 310 315 320
 Ala Asn Phe Thr Glu Gly Cys Lys Leu Tyr Thr Asp Phe Val Asn Asn
 325 330 335
 Val Val Val Pro Leu Tyr Pro Thr Pro Gln Gly Ala Leu Lys Val Asn
 340 345 350
 Leu Asn Ala Asn Leu Gly Phe Leu Phe Ser Thr Phe Ser Asn Cys Thr
 355 360 365
 Gln Val Phe Pro Tyr Gly Gln
 370 375

<210> SEQ ID NO 16
 <211> LENGTH: 371
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 16

<400> SEQUENCE: 16

Met Ala Lys Phe Ser Thr Leu Leu Ala Leu Ser Val Leu Ala Ile Gln
 1 5 10 15
 Ala Val Ala Phe Pro Gln His Gln Pro Leu Ala Gly Leu Thr Glu Arg
 20 25 30
 Glu Leu Glu Asp Leu Leu Pro Arg Phe Lys Pro Val Val Pro Pro Pro
 35 40 45
 Pro Pro Gly Pro Pro Lys Asp Thr Ser Val Lys Leu Val Asn Asp Lys
 50 55 60
 Asp His Pro Tyr Glu Pro Leu Arg Lys Gly Asp Ile Arg Gly Pro Cys
 65 70 75 80
 Pro Gly Leu Asn Thr Leu Ala Ser His Gly Tyr Leu Pro Arg Asn Gly
 85 90 95
 Val Val Thr Pro Ala Gln Ile Ile Asn Ala Val Gln Asp Gly Phe Gly
 100 105 110
 Met Asp Asn Glu Leu Ala Ile Leu Leu Ala Tyr Ser Thr Met Leu Thr
 115 120 125
 Asp Gly Asn Val Val Thr Asn Leu Met Ser Ile Gly Gln Lys Thr Pro
 130 135 140
 Leu Thr Gly Pro Asp Pro Pro Ala Pro Ala Ile Val Gly Gly Leu Asn
 145 150 155 160
 Thr His Gly Thr Phe Glu Gly Asp Ala Gly Leu Thr Arg Ala Asp Phe
 165 170 175

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Phe Phe Gly Asp Asn His Ser Phe Asn Gln Thr Leu Phe Asn Glu Phe
 180 185 190
 Val Glu Phe Ser Asn Lys Phe Gly Gly Gly Val Tyr Asn Gln Thr Val
 195 200 205
 Ala Ala Glu Tyr Arg Phe Phe Arg Ile Gln Gln Ser Thr Ala Glu Asn
 210 215 220
 Pro Thr Phe Thr Phe Val Thr Pro Arg Phe Val Thr Ala Tyr Arg Glu
 225 230 235 240
 Ser Val Phe Pro Phe Ile Phe Phe Val Asp Gly Arg Lys Ala Asp Gly
 245 250 255
 Gln Leu Ser Met Lys Asp Ala Phe Gly Phe Phe Asn Glu Ser Arg Met
 260 265 270
 Pro Asp Gly Phe His Arg Ala Asp Gly Ser Lys Thr Ala Asp Leu Val
 275 280 285
 Gly Asn Ala Ser Asp Ala Ile Phe Ala Ala His Pro Val Gln Pro Gly
 290 295 300
 Ala Asn Ala Gly Lys Val Asn Thr Tyr Thr Pro Asp Pro Asn Ser Pro
 305 310 315 320
 Thr Asp Asp Cys Gly Leu Tyr Glu Thr Phe Val Asn Leu Met Val Lys
 325 330 335
 Gln Tyr Pro Asn Pro Gln Gly Val Leu Arg Thr Asn Leu Asn Leu Asn
 340 345 350
 Leu Gly Phe Phe Phe Gln Gly Phe Pro Gly Cys Thr Gln Leu Phe Pro
 355 360 365
 Phe Gly Gln
 370

<210> SEQ ID NO 17
 <211> LENGTH: 268
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 17

<400> SEQUENCE: 17

Met Leu Gly Ile Arg Leu Val Ser Leu Leu Ala Phe Thr Gly Ser Ala
 1 5 10 15
 Leu Ala Glu Leu Asp Phe Ser Lys Trp Lys Thr Arg Gln Pro Gly Glu
 20 25 30
 Leu Arg Ala Pro Cys Pro Ala Met Asn Ser Leu Ala Asn His Gly Phe
 35 40 45
 Ile Gln Arg Asp Gly Lys Asn Ile Thr Val Glu Gly Leu Thr Pro Val
 50 55 60
 Leu Lys Glu Val Phe His Leu Ser His Glu Leu Ala Phe Thr Val Ser
 65 70 75 80
 Gln Leu Gly Leu Phe Thr Ala Leu Asp Pro Ser Lys Gly Val Phe Thr
 85 90 95
 Leu Gln Asp Leu Thr Asp Arg His Asn Val Phe Glu His Asp Ala Ser
 100 105 110
 Leu Ser Arg Glu Asp Ala Lys Phe Gly Gly Asp Gln Ser Val Leu His
 115 120 125
 Lys Gly Gln Phe Gln Lys Phe Met Asp His Phe Lys Gly Glu Lys Tyr
 130 135 140
 Ile Ser Phe Glu Ala Ala Ala Lys Ala Arg Tyr Ala Met Val Gln Asp
 145 150 155 160

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Ser Arg Lys Arg Asn Pro Asp Phe Thr Tyr Asp Val Thr His Arg Ile
      165                      170                      175

Thr Ser Tyr Gly Glu Thr Ile Lys Tyr Leu Arg Thr Ile Val Glu Pro
      180                      185                      190

Ser Thr Gly Lys Cys Pro Val Asp Trp Ile Lys Ile Leu Phe Glu Gln
      195                      200                      205

Glu Arg Leu Pro Tyr Asn Glu Gly Trp Arg Pro Pro Thr Asn Glu Leu
      210                      215                      220

Ser Gly Phe Ser Leu Ala Ser Glu Val Leu Glu Leu Ala Leu Ile Thr
      225                      230                      235                      240

Pro Glu Lys Leu Pro Val Asp Glu Cys Leu Gly Lys Gly Lys Gly Lys
      245                      250                      255

Gly Asn Cys Lys Arg Arg Arg Ser Tyr Leu Gly Ile
      260                      265

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<210> SEQ ID NO 18
<211> LENGTH: 371
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: UPO 18

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<400> SEQUENCE: 18

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Met Asn Pro Phe Leu Lys Leu Ala Val Leu Ala Leu Val Thr Ala Pro
 1      5                      10                      15

Leu Ala Gly Ala Phe Pro Ser His Arg Ser Leu Gly Gly Leu Ser Ser
      20                      25                      30

Glu Gln Leu Asp Arg Ile Phe Pro Thr Leu Lys Val Ala Pro Pro Glu
      35                      40                      45

Gly Pro Pro Pro Pro Gln Asp Asp Thr Ser Thr Arg Leu Val Asp Asp
      50                      55                      60

Ala Asp His Pro Phe Met Pro Ala Gly Pro Asn Asp Met Arg Gly Pro
      65                      70                      75                      80

Cys Pro Gly Leu Asn Thr Leu Ala Ser His Gly Tyr Leu Pro Arg Asn
      85                      90                      95

Gly Ile Ala Thr Pro Ala Gln Val Ile Asn Ala Thr Met Gln Gly Phe
      100                     105                     110

Asn Met Glu Phe Ser Leu Ala Lys Phe Val Thr Tyr Ala Ala Phe Leu
      115                     120                     125

Val Asp Gly Asn Pro Ile Thr Asn Leu Met Ser Ile Gly Gly Lys Ser
      130                     135                     140

Asp Leu Thr Gly Glu Asp Pro Pro Asp Pro Ala Thr Val Gly Gly Leu
      145                     150                     155                     160

Asn Thr His Ala Val Phe Glu Gly Asp Ala Ser Met Thr Arg Ala Asp
      165                     170                     175

Ala Phe Phe Gly Asp Asn His Ser Phe Asn Gln Thr Leu Trp Asp Gly
      180                     185                     190

Phe Val Asp Phe Ser Asn Arg Phe Gly Ala Gly Lys Tyr Asn Leu Thr
      195                     200                     205

Val Ala Thr Glu Leu Arg Ile Gln Arg Ile Gln Asp Ser Ile Ala Thr
      210                     215                     220

Asn Pro Gln Phe Ser Phe Val Ser Pro Arg Phe Ile Thr Ala Tyr Ala
      225                     230                     235                     240

Glu Ser Thr Phe Pro Ile Asn Phe Phe Ile Asp Gly Arg Gln Gln Asp
      245                     250                     255

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Gly Gln Leu Asp Leu Asp Ala Ala Ile Ser Phe Phe Arg Asp Met Arg
 260 265 270

Tyr Pro Ser Gly Phe Phe Arg Ala Pro Lys Pro Met Gly Val Glu Gly
 275 280 285

Ile Glu Thr Ile Ile Ala Ala His Pro Ile Pro Ala Gly Ala Asn Asn
 290 295 300

Gly Ala Val Asn Thr Tyr Thr Pro Asp Pro His Ser Gly Asp Phe Asn
 305 310 315 320

Ser Phe Cys Thr Val Tyr Thr Asn Phe Val Asn Glu Thr Ile Arg Gly
 325 330 335

Leu Tyr Pro Ser Pro Thr Gly Ile Leu Lys Asp Ser Leu Asn Arg Asn
 340 345 350

Leu Asp Phe Leu His Asp Phe Val Ser Gly Cys Pro Gln Ile Phe Pro
 355 360 365

Trp Gly Arg
 370

<210> SEQ ID NO 19
 <211> LENGTH: 321
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: CPO 19

<400> SEQUENCE: 19

Met Phe Ser Lys Val Leu Pro Phe Val Gly Ala Val Ala Ala Leu Pro
 1 5 10 15

His Ser Val Arg Gln Glu Pro Gly Ser Gly Ile Gly Tyr Pro Tyr Asp
 20 25 30

Asn Asn Thr Leu Pro Tyr Val Ala Pro Gly Pro Thr Asp Ser Arg Ala
 35 40 45

Pro Cys Pro Ala Leu Asn Ala Leu Ala Asn His Gly Tyr Ile Pro His
 50 55 60

Asp Gly Arg Ala Ile Ser Arg Glu Thr Leu Gln Asn Ala Phe Leu Asn
 65 70 75 80

His Met Gly Ile Ala Asn Ser Val Ile Glu Leu Ala Leu Thr Asn Ala
 85 90 95

Phe Val Val Cys Glu Tyr Val Thr Gly Ser Asp Cys Gly Asp Ser Leu
 100 105 110

Val Asn Leu Thr Leu Leu Ala Glu Pro His Ala Phe Glu His Asp His
 115 120 125

Ser Phe Ser Arg Lys Asp Tyr Lys Gln Gly Val Ala Asn Ser Asn Asp
 130 135 140

Phe Ile Asp Asn Arg Asn Phe Asp Ala Glu Thr Phe Gln Thr Ser Leu
 145 150 155 160

Asp Val Val Ala Gly Lys Thr His Phe Asp Tyr Ala Asp Met Asn Glu
 165 170 175

Ile Arg Leu Gln Arg Glu Ser Leu Ser Asn Glu Leu Asp Phe Pro Gly
 180 185 190

Trp Phe Thr Glu Ser Lys Pro Ile Gln Asn Val Glu Ser Gly Phe Ile
 195 200 205

Phe Ala Leu Val Ser Asp Phe Asn Leu Pro Asp Asn Asp Glu Asn Pro
 210 215 220

Leu Val Arg Ile Asp Trp Trp Lys Tyr Trp Phe Thr Asn Glu Ser Phe
 225 230 235 240

Pro	Tyr	His	Leu	Gly	Trp	His	Pro	Pro	Ser	Pro	Ala	Arg	Glu	Ile	Glu
				245					250					255	
Phe	Val	Thr	Ser	Ala	Ser	Ser	Ala	Val	Leu	Ala	Ala	Ser	Val	Thr	Ser
			260					265					270		
Thr	Pro	Ser	Ser	Leu	Pro	Ser	Gly	Ala	Ile	Gly	Pro	Gly	Ala	Glu	Ala
		275					280					285			
Val	Pro	Leu	Ser	Phe	Ala	Ser	Thr	Met	Thr	Pro	Phe	Leu	Leu	Ala	Thr
	290					295					300				
Asn	Ala	Pro	Tyr	Tyr	Ala	Gln	Asp	Pro	Thr	Leu	Arg	Pro	Gln	Arg	Gln
305					310					315					320

Ala

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<210> SEQ ID NO 20
<211> LENGTH: 376
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CPO 20
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<400> SEQUENCE: 20

Met 1	Phe	Ser	Lys	Ile 5	Leu	Pro	Leu	Val	Gly 10	Val	Ala	Ala	Ala	Leu 15	Pro
His	Trp	Leu	Gln 20	Leu	Arg	Gln	Glu	Pro 25	Asn	Ser	Gly	Ile	Gly 30	Tyr	Pro
Tyr	Asp	Asn	His 35	Thr	Lys	Pro	Tyr 40	Val	His	Pro	Gly	Pro 45	His	Asp	Ser
Arg	Ala	Pro	Cys 50	Pro	Ala	Leu 55	Asn	Ala	Leu	Ala	Asn 60	His	Gly	Tyr	Ile
Pro 65	His	Asn	Gly	Arg 70	Ala	Ile	Thr	Lys	Glu	Asn 75	Leu	Gln	Asn	Ala	Phe 80
Leu	Glu	His	Met 85	Gly	Ile	Gly	Asn	Ser	Val 90	Ile	Ala	Leu	Ala 95	Leu	Thr
Asn	Ala	Phe	Val 100	Val	Cys	Glu	Tyr	Val 105	Thr	Gly	Gln	Asp 110	Cys	Gly	Asp
Thr	Leu	Val	Asn 115	Leu	Thr	Leu	Leu 120	Ser	Glu	Pro	His 125	Ala	Phe	Glu	His
Asp	His 130	Ser	Phe	Ser	Arg	Lys 135	Asp	Tyr	Lys	Gln	Gly 140	Val	Ser	Asn	Phe
Asn 145	Glu	Ile	Val	Asp 150	Asn	Arg	Asn	Phe	Asp	Leu 155	Ser	Thr	Phe	Glu	Thr 160
Ser	Leu	Asp	Val 165	Val	Ala	Gly	Gln	Thr	His 170	Phe	Gly	Tyr	Ala	Glu	Met 175
Asn	Gln	Ile	Arg 180	Leu	Gln	Arg	Glu	Ser 185	Leu	Ser	Asn	Glu 190	Ala	Asp	Phe
Pro	Gly 195	Trp	Phe	Thr	Glu	Ser	Lys 200	Pro	Ile	Gln	Glu	Val 205	Glu	Ala	Gly
Phe 210	Ile	Phe	Ala	Leu	Val	Ser 215	Asp	Phe	Asn	Leu	Pro 220	Asp	Asn	Asp	Glu
Asn 225	Pro	Leu	Val	Arg 230	Val	Asp	Trp	Trp	Lys 235	Tyr	Trp	Phe	Ile	Asn	Glu 240
Ser	Phe	Pro	Tyr 245	His	Leu	Gly	Trp	His 250	Pro	Pro	Thr	Pro	Ala	Arg	Glu
Ile	Glu	Phe	Val 260	Thr	Ser	Ala	Ser 265	Ser	Ala	Ile	Leu	Ala 270	Ala	Ala	Val

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Thr	Ser	Thr	Pro	Ser	Ser	Leu	Pro	Ser	Gly	Ala	Ile	Gly	Pro	Gly	Ala
		275					280					285			
Glu	Ala	Val	Pro	Leu	Ser	Phe	Ala	Ser	Thr	Met	Thr	Pro	Phe	Leu	Leu
	290					295					300				
Ala	Thr	Asp	Ile	Pro	Tyr	Phe	Ala	His	Pro	Thr	Leu	Gly	Pro	Asn	Asp
305					310					315					320
Lys	Arg	Glu	Ala	Ala	Pro	Ala	Pro	Ala	Ala	Thr	Thr	Ser	Thr	Ala	Thr
				325					330					335	
Phe	Lys	Asn	Pro	Tyr	Leu	Glu	Pro	Ile	Gly	Thr	Gln	Asp	Ile	Lys	Asn
			340					345					350		
Gln	Gln	Ala	Tyr	Val	Ser	Ser	Lys	Ala	Ala	Ala	Met	Ser	Ser	Ala	Met
		355					360					365			
Ala	Val	Asn	Lys	Ala	Arg	Ser	Leu								
	370					375									

<210> SEQ ID NO 21
 <211> LENGTH: 371
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 21

<400> SEQUENCE: 21

Met	Lys	Tyr	Phe	Pro	Leu	Phe	Pro	Thr	Leu	Val	Phe	Ala	Ala	Arg	Val
1				5					10					15	
Val	Ala	Phe	Pro	Ala	Tyr	Ala	Ser	Leu	Ala	Gly	Leu	Ser	Gln	Gln	Glu
			20					25					30		
Leu	Asp	Ala	Ile	Ile	Pro	Thr	Leu	Glu	Ala	Arg	Glu	Pro	Gly	Leu	Pro
		35					40					45			
Pro	Gly	Pro	Leu	Glu	Asn	Ser	Ser	Ala	Lys	Leu	Val	Asn	Asp	Glu	Ala
		50			55						60				
His	Pro	Trp	Lys	Pro	Leu	Arg	Pro	Gly	Asp	Ile	Arg	Gly	Pro	Cys	Pro
65					70					75					80
Gly	Leu	Asn	Thr	Leu	Ala	Ser	His	Gly	Tyr	Leu	Pro	Arg	Asn	Gly	Val
			85						90					95	
Ala	Thr	Pro	Val	Gln	Ile	Ile	Asn	Ala	Val	Gln	Glu	Gly	Leu	Asn	Phe
			100					105					110		
Asp	Asn	Gln	Ala	Ala	Val	Phe	Ala	Thr	Tyr	Ala	Ala	His	Leu	Val	Asp
		115					120					125			
Gly	Asn	Leu	Ile	Thr	Asp	Leu	Leu	Ser	Ile	Gly	Arg	Lys	Thr	Arg	Leu
	130					135					140				
Thr	Gly	Pro	Asp	Pro	Pro	Pro	Pro	Ala	Ser	Val	Gly	Gly	Leu	Asn	Glu
145					150					155				160	
His	Gly	Thr	Phe	Glu	Gly	Asp	Ala	Ser	Met	Thr	Arg	Gly	Asp	Ala	Phe
			165					170						175	
Phe	Gly	Asn	Asn	His	Asp	Phe	Asn	Glu	Thr	Leu	Phe	Glu	Gln	Leu	Val
			180					185					190		
Asp	Tyr	Ser	Asn	Arg	Phe	Gly	Gly	Gly	Lys	Tyr	Asn	Leu	Thr	Val	Ala
		195						200				205			
Gly	Glu	Leu	Arg	Phe	Lys	Arg	Ile	Gln	Asp	Ser	Ile	Ala	Thr	Asn	Pro
	210					215					220				
Asn	Phe	Ser	Phe	Val	Asp	Phe	Arg	Phe	Phe	Thr	Ala	Tyr	Gly	Glu	Thr
225					230					235					240
Thr	Phe	Pro	Ala	Asn	Leu	Phe	Val	Asp	Gly	Arg	Arg	Asp	Asp	Gly	Gln
				245					250					255	

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Leu Asp Met Asp Ala Ala Arg Ser Phe Phe Gln Phe Ser Arg Met Pro
 260 265 270
 Asp Asp Phe Phe Arg Ala Pro Ser Pro Arg Ser Gly Thr Gly Val Glu
 275 280 285
 Val Val Ile Gln Ala His Pro Met Gln Pro Gly Arg Asn Val Gly Lys
 290 295 300
 Ile Asn Ser Tyr Thr Val Asp Pro Thr Ser Ser Asp Phe Ser Thr Pro
 305 310 315 320
 Cys Leu Met Tyr Glu Lys Phe Val Asn Ile Thr Val Lys Ser Leu Tyr
 325 330 335
 Pro Asn Pro Thr Val His Val Arg Lys Ala Leu Asn Thr Asn Leu Asp
 340 345 350
 Phe Phe Phe Gln Gly Val Ala Ala Gly Cys Thr Gln Val Phe Pro Tyr
 355 360 365
 Gly Arg Asp
 370

<210> SEQ ID NO 22
 <211> LENGTH: 262
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO22

<400> SEQUENCE: 22

Met Lys Leu Val Tyr Leu Ser Ser Ala Val Ala Phe Gly Ser Ala Ile
 1 5 10 15
 Ala Asp Thr Ala Pro Trp Glu Gly Pro Gly Pro Asn Asp Val Arg Gly
 20 25 30
 Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His Gly Phe Leu Pro His
 35 40 45
 Asp Gly Lys Asn Ile His Val Asn Lys Thr Val Asp Ala Leu Ser Ser
 50 55 60
 Ala Leu Asn Ile Asp Pro Glu Leu Gly Ser Phe Leu His Ser Phe Ala
 65 70 75 80
 Val Thr Ala Asn Pro Gln Pro Asn Ala Thr Trp Trp Asn Leu Asp His
 85 90 95
 Leu Ser Arg His Asn Ile Leu Glu His Asp Ala Ser Leu Ser Arg Gln
 100 105 110
 Asp Ala Tyr Phe Gly Ala Pro Asp Val Phe Asn Glu Ala Val Phe Asn
 115 120 125
 Gln Thr Lys Ser Tyr Trp Thr Gly Asp Val Ile Thr Leu Gln Met Ala
 130 135 140
 Ala Asn Ala Arg Leu Ala Arg Leu Met Thr Ser Asn Leu Thr Asn Pro
 145 150 155 160
 Glu Tyr Ser Met Ser Asp Leu Gly Ser Ser Phe Ser Ile Gly Glu Ser
 165 170 175
 Val Ala Tyr Val Ala Ile Leu Gly Ser Lys Glu Thr Arg Thr Val Pro
 180 185 190
 Lys Ala Tyr Val Glu Tyr Leu Phe Glu Lys Glu Arg Leu Pro Tyr Glu
 195 200 205
 Leu Gly Phe Lys Lys Ala Glu Thr Pro Met Thr Glu Thr Asp Leu Gly
 210 215 220
 Asn Leu Met Asp Glu Leu Ile Ser Leu Gln His Phe Pro Gln Ser Pro
 225 230 235 240

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Tyr Ala Phe Pro Gln Gln Gly Ala Pro His Pro Leu Pro Trp Ser Pro
 20 25 30
 Pro Gly Pro Asn Asp Val Arg Ala Pro Cys Pro Met Leu Asn Thr Leu
 35 40 45
 Ala Asn His Gly Tyr Leu Pro His Asn Gly Lys Asp Ile Thr Glu Arg
 50 55 60
 His Thr Ile Asn Ala Leu Tyr Asn Ala Leu Gly Ile Glu Glu Glu Leu
 65 70 75 80
 Ala Ile Tyr Leu His Gln Glu Ala Val Thr Thr Asn Pro Ala Pro Asn
 85 90 95
 Ala Thr Thr Phe Ser Leu Asn Asp Leu Ser Arg His Asp Ile Leu Glu
 100 105 110
 His Asp Ala Ser Leu Ser Arg Gln Asp Ala Tyr Phe Gly Asp Asn His
 115 120 125
 Asp Phe Asn Gln Thr Ile Phe Asp Glu Thr Arg Ser Tyr Trp Thr Ser
 130 135 140
 Pro Ile Ile Asp Val Lys Gln Ala Ala Val Ser Arg Gln Ala Arg Val
 145 150 155 160
 Asn Thr Ser Met Ala Thr Asn Pro Asn Tyr Thr Met Ser Glu Leu Gly
 165 170 175
 Asp Ser Phe Ser Tyr Gly Glu Thr Ala Ala Tyr Ile Ile Val Leu Gly
 180 185 190
 Asp Lys Glu Lys Gly Leu Val Asn Arg Ser Arg Val Glu Tyr Leu Phe
 195 200 205
 Glu Asn Glu Arg Leu Pro Leu Asp Leu Gly Trp Ser Arg Ala Lys Glu
 210 215 220
 Asn Ile Thr Phe Asp Asp Leu Ser Thr Met Leu Gln Arg Ile Ile Asn
 225 230 235 240
 Ala Thr Gly Gly Glu Ser Glu Phe Asp Arg Glu Leu Ala Lys Arg Gly
 245 250 255
 Gly Val His Val Gly Ser Trp Arg Gly
 260 265

<210> SEQ ID NO 25
 <211> LENGTH: 267
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO25

<400> SEQUENCE: 25

Met Lys Thr Thr Pro Leu Leu Phe Phe Ala Ala Gly Leu Ala Gln Thr
 1 5 10 15
 His Ala Phe Pro Ser Gln Gly Gly Ala Pro His Pro Leu Pro Trp Ser
 20 25 30
 Pro Pro Gly Pro Asn Asp Val Arg Ala Pro Cys Pro Met Leu Asn Thr
 35 40 45
 Leu Ala Asn His Gly Tyr Leu Pro His Asn Gly Lys Asp Ile Thr Glu
 50 55 60
 Gln His Thr Ile Asn Ala Leu Tyr Asn Ala Leu Gly Ile Asp Ala Glu
 65 70 75 80
 Leu Ala Thr Tyr Leu His Gln Glu Ala Val Thr Thr Asn Pro Ala Pro
 85 90 95
 Asn Ala Thr Thr Phe Ser Leu Asn Asp Leu Ser Arg His Asp Ile Leu
 100 105 110

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Glu His Asp Ala Ser Leu Ser Arg Gln Asp Ala Phe Phe Gly Asp Asn
 115 120 125
 His Asp Phe Asn Gln Thr Ile Phe Asp Glu Thr Arg Ser Tyr Trp Thr
 130 135 140
 Ser Pro Ile Ile Asp Val Met Gln Ala Ala Leu Ser Arg Gln Ala Arg
 145 150 155 160
 Val Asp Thr Ser Met Ala Thr Asn Pro Asn Tyr Thr Met Ser Glu Leu
 165 170 175
 Gly Ala Ser Phe Ser Tyr Gly Glu Thr Ala Ala Tyr Ile Ile Val Leu
 180 185 190
 Gly Asp Lys Glu Asn Gly Leu Val Asn Arg Ser Arg Val Glu Tyr Leu
 195 200 205
 Phe Glu Asn Glu Arg Leu Pro Leu Asp Leu Gly Trp Thr Arg Ala Lys
 210 215 220
 Glu Asn Ile Thr Phe Asp Asp Leu Ser Thr Met Leu Asn Arg Ile Val
 225 230 235 240
 Asn Ala Thr Gly Gly Glu Ser Glu Phe Asp Arg Glu Leu Ala Lys Arg
 245 250 255
 Gly Gly Val His Val Gly Lys Trp Arg Gly Tyr
 260 265

<210> SEQ ID NO 26
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Conserved Motif

<400> SEQUENCE: 26

Tyr Asp Asn Asn Thr Leu Pro Tyr Val Ala Pro Gly Pro Thr Asp Ser
 1 5 10 15
 Arg Ala Pro Cys Pro Ala Leu Asn Ala Leu Ala Asn His Gly Tyr Ile
 20 25 30
 Pro His Asp Gly Arg Ala Ile Ser Arg Glu Thr Leu Gln Asn Ala Phe
 35 40 45
 Leu Asn His Met Gly Ile Ala Asn Ser Val Ile Glu
 50 55 60

<210> SEQ ID NO 27
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Conserved Motif

<400> SEQUENCE: 27

Val Asn Asp Lys Asp His Pro Trp Lys Pro Leu Arg Pro Gly Asp Ile
 1 5 10 15
 Arg Gly Pro Cys Pro Gly Leu Asn Thr Leu Ala Ser His Gly Tyr Leu
 20 25 30
 Pro Arg Asn Gly Val Ala Thr Pro Ala Gln Ile Ile Asn Ala Val Gln
 35 40 45
 Glu Gly Phe Asn Met Asp Asn Ser Val Ala Leu
 50 55

<210> SEQ ID NO 28
 <211> LENGTH: 48
 <212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Conserved Motif

<400> SEQUENCE: 28

Leu Ala Leu Thr Asn Ala Phe Val Val Cys Glu Tyr Val Thr Gly Ser
1          5          10         15

Asp Cys Gly Asp Ser Leu Val Asn Leu Thr Leu Leu Ala Glu Pro His
20         25         30

Ala Phe Glu His Asp His Ser Phe Ser Arg Lys Asp Tyr Lys Gln Gly
35         40         45

<210> SEQ ID NO 29
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Conserved Motif

<400> SEQUENCE: 29

Phe Ala Thr Tyr Glu Ala His Leu Met Val Gly Asn Leu Leu Thr Asp
1          5          10         15

Leu Leu Ser Ile Gly Arg Lys Thr Pro Leu Thr Gly Pro Asp Leu Pro
20         25         30

Pro Pro Ala Asn Ile Gly Gly Leu Ser Glu His Gly Leu Phe Glu Gly
35         40         45

Asp Ala Ser Met Thr Arg Gly Asp Ala Phe Phe Gly
50         55         60

<210> SEQ ID NO 30
<211> LENGTH: 261
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: UPO 12 variant

<400> SEQUENCE: 30

Met Lys Ser Leu Ser Phe Ser Leu Ala Leu Gly Phe Gly Ser Thr Leu
1          5          10         15

Val Tyr Ser Ala Pro Ser Pro Phe Ser Gly Trp Gln Ala Pro Gly Pro
20         25         30

Asn Asp Val Arg Ala Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His
35         40         45

Gly Phe Leu Pro His Asp Gly Lys Gly Ile Thr Val Asn Lys Thr Ile
50         55         60

Asp Ala Leu Gly Ser Ala Leu Asn Ile Asp Ala Asn Leu Ser Thr Leu
65         70         75         80

Leu Phe Gly Phe Ala Ala Thr Thr Asn Pro Gln Pro Asn Ala Thr Phe
85         90         95

Phe Asp Leu Asp His Leu Ser Arg His Asn Ile Leu Glu His Asp Ala
100        105        110

Ser Leu Ser Arg Gln Asp Ser Tyr Phe Gly Pro Ala Asp Val Phe Asn
115        120        125

Glu Ala Val Phe Asn Gln Thr Lys Ser Phe Trp Thr Gly Asp Ile Ile
130        135        140

Asp Val Gln Met Ala Ala Asn Ala Arg Ile Val Arg Leu Leu Thr Ser
145        150        155        160

Asn Leu Thr Asn Pro Glu Tyr Ser Leu Ser Asp Leu Gly Ser Ala Phe
165        170        175

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Ser Ile Gly Glu Ser Ala Ala Tyr Ile Gly Ile Leu Gly Asp Lys Lys
 180 185 190

Ser Ala Thr Val Pro Lys Ser Trp Val Glu Tyr Leu Phe Glu Asn Glu
 195 200 205

Arg Leu Pro Tyr Glu Leu Gly Phe Lys Arg Pro Asn Asp Pro Phe Thr
 210 215 220

Thr Asp Asp Leu Gly Asp Leu Ser Thr Gln Ile Ile Asn Ala Gln His
 225 230 235 240

Phe Pro Gln Ser Pro Gly Lys Val Glu Lys Arg Gly Asp Thr Arg Cys
 245 250 255

Pro Tyr Gly Tyr His
 260

<210> SEQ ID NO 31
 <211> LENGTH: 261
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 12 variant

<400> SEQUENCE: 31

Met Lys Ser Leu Ser Phe Ser Leu Ala Leu Gly Phe Gly Ser Thr Leu
 1 5 10 15

Val Tyr Ser Ala Pro Ser Pro Ser Ser Gly Trp Gln Ala Pro Gly Pro
 20 25 30

Asn Asp Val Arg Ala Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His
 35 40 45

Gly Phe Leu Pro His Asp Gly Lys Gly Ile Thr Val Asn Lys Thr Ile
 50 55 60

Asp Ala Leu Gly Ser Ala Leu Asn Ile Asp Ala Asn Leu Ser Thr Leu
 65 70 75 80

Leu Phe Gly Phe Ala Ala Thr Thr Asn Pro Gln Pro Asn Ala Thr Phe
 85 90 95

Phe Asp Leu Asp His Leu Ser Arg His Asn Ile Leu Glu His Asp Ala
 100 105 110

Ser Leu Ser Arg Gln Asp Ser Tyr Phe Gly Pro Ala Asp Val Phe Asn
 115 120 125

Glu Ala Val Phe Asn Gln Thr Lys Ser Phe Trp Thr Gly Asp Ile Ile
 130 135 140

Tyr Val Gln Met Ala Ala Asn Ala Arg Ile Val Arg Leu Leu Thr Ser
 145 150 155 160

Asn Leu Thr Asn Pro Glu Tyr Ser Leu Ser Asp Leu Gly Ser Ala Phe
 165 170 175

Ser Ile Gly Glu Ser Ala Ala Tyr Ile Gly Ile Leu Gly Asp Lys Lys
 180 185 190

Ser Ala Thr Val Pro Lys Ser Trp Val Glu Tyr Leu Phe Glu Asn Glu
 195 200 205

Arg Leu Pro Tyr Glu Leu Gly Phe Lys Arg Pro Asn Asp Pro Phe Thr
 210 215 220

Thr Asp Asp Leu Gly Asp Leu Ser Thr Gln Ile Ile Asn Ala Gln His
 225 230 235 240

Phe Pro Gln Ser Pro Gly Lys Val Glu Lys Arg Gly Asp Thr Arg Cys
 245 250 255

Pro Tyr Gly Tyr His
 260

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<210> SEQ ID NO 32
<211> LENGTH: 261
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: UPO 12 variant

<400> SEQUENCE: 32
Met Lys Ser Leu Ser Phe Ser Leu Ala Leu Gly Phe Gly Ser Thr Leu
1          5          10          15
Val Tyr Ser Ala Pro Ser Pro Ser Ser Gly Trp Gln Ala Pro Gly Pro
20          25          30
Asn Asp Val Arg Ala Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His
35          40          45
Gly Phe Leu Pro His Asp Gly Lys Gly Ile Thr Val Asn Lys Thr Ile
50          55          60
Asp Ala Leu Gly Ser Ala Leu Asn Ile Asp Ala Asn Leu Ser Thr Leu
65          70          75          80
Leu Phe Gly Phe Ala Ala Thr Thr Asn Pro Gln Pro Asn Ala Thr Phe
85          90          95
Phe Asp Leu Asp His Leu Ser Arg His Asn Ile Leu Glu His Asp Ala
100         105         110
Ser Leu Ser Arg Gln Asp Ser Tyr Phe Gly Pro Ala Asp Val Phe Asn
115         120         125
Glu Ala Val Phe Asn Gln Thr Lys Ser Phe Trp Thr Gly Asp Ile Ile
130         135         140
Asp Val Gln Met Ala Ala Asn Ala Arg Ile Val Arg Leu Leu Thr Ser
145         150         155         160
Asn Leu Thr Asn Pro Glu Tyr Ser Leu Ser Asp Leu Gly Ser Ala Phe
165         170         175
Ser Ile Gly Glu Ser Ala Ala Tyr Ile Gly Ile Leu Gly Asp Lys Lys
180         185         190
Ser Ala Thr Val Pro Lys Ser Trp Val Glu Tyr Leu Phe Glu Asn Glu
195         200         205
Arg Leu Pro Tyr Glu Leu Gly Phe Lys Arg Pro Asn Asp Pro Phe Thr
210         215         220
Thr Asp Asp Leu Gly Asp Leu Ser Thr Gln Ile Ile Asn Ala Gln His
225         230         235         240
Phe Pro Gln Ser Pro Gly Lys Val Glu Lys Arg Gly Asp Thr Arg Ser
245         250         255
Pro Tyr Gly Tyr His
260

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<210> SEQ ID NO 33
<211> LENGTH: 255
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: UPO 12 variant

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<400> SEQUENCE: 33
Met Lys Ser Leu Ser Phe Ser Leu Ala Leu Gly Phe Gly Ser Thr Leu
1          5          10          15
Val Tyr Ser Ala Pro Ser Pro Ser Ser Gly Trp Gln Ala Pro Gly Pro
20          25          30
Asn Asp Val Arg Ala Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His

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35					40					45					
Gly	Phe	Leu	Pro	His	Asp	Gly	Lys	Gly	Ile	Thr	Val	Asn	Lys	Thr	Ile
50						55					60				
Asp	Ala	Leu	Gly	Ser	Ala	Leu	Asn	Ile	Asp	Ala	Asn	Leu	Ser	Thr	Leu
65					70					75					80
Leu	Phe	Gly	Phe	Ala	Ala	Thr	Thr	Asn	Pro	Gln	Pro	Asn	Ala	Thr	Phe
				85					90					95	
Phe	Asp	Leu	Asp	His	Leu	Ser	Arg	His	Asn	Ile	Leu	Glu	His	Asp	Ala
			100					105					110		
Ser	Leu	Ser	Arg	Gln	Asp	Ser	Tyr	Phe	Gly	Pro	Ala	Asp	Val	Phe	Asn
		115					120					125			
Glu	Ala	Val	Phe	Asn	Gln	Thr	Lys	Ser	Phe	Trp	Thr	Gly	Asp	Ile	Ile
	130					135					140				
Asp	Val	Gln	Met	Ala	Ala	Asn	Ala	Arg	Ile	Val	Arg	Leu	Leu	Thr	Ser
145					150					155					160
Asn	Leu	Thr	Asn	Pro	Glu	Tyr	Ser	Leu	Ser	Asp	Leu	Gly	Ser	Ala	Phe
				165					170					175	
Ser	Ile	Gly	Glu	Ser	Ala	Ala	Tyr	Ile	Gly	Ile	Leu	Gly	Asp	Lys	Lys
		180						185					190		
Ser	Ala	Thr	Val	Pro	Lys	Ser	Trp	Val	Glu	Tyr	Leu	Phe	Glu	Asn	Glu
		195					200					205			
Arg	Leu	Pro	Tyr	Glu	Leu	Gly	Phe	Lys	Arg	Pro	Asn	Asp	Pro	Phe	Thr
	210					215					220				
Thr	Asp	Asp	Leu	Gly	Asp	Leu	Ser	Thr	Gln	Ile	Ile	Asn	Ala	Gln	His
225					230					235					240
Phe	Pro	Gln	Ser	Pro	Gly	Lys	Val	Glu	Lys	Arg	Gly	Asp	Thr	Arg	
				245					250					255	

<210> SEQ ID NO 34

<211> LENGTH: 261

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: UPO 12 variant

<400> SEQUENCE: 34

Met	Lys	Ser	Leu	Ser	Phe	Ser	Leu	Ala	Leu	Gly	Phe	Gly	Ser	Thr	Leu
1				5						10				15	
Val	Tyr	Ser	Ala	Pro	Ser	Pro	Ser	Ser	Gly	Trp	Gln	Ala	Pro	Gly	Pro
			20						25				30		
Asn	Asp	Val	Arg	Ala	Pro	Cys	Pro	Met	Leu	Asn	Thr	Leu	Ala	Asn	His
		35					40					45			
Gly	Phe	Leu	Pro	His	Asp	Gly	Lys	Gly	Ile	Thr	Val	Asn	Lys	Thr	Ile
	50					55					60				
Asp	Ala	Leu	Gly	Ser	Ala	Leu	Asn	Ile	Asp	Ala	Asn	Leu	Ser	Thr	Leu
65					70					75					80
Leu	Phe	Gly	Phe	Ala	Ala	Thr	Thr	Asn	Pro	Gln	Pro	Asn	Ala	Thr	Phe
				85					90					95	
Phe	Asp	Leu	Asp	His	Leu	Ser	Arg	His	Asn	Ile	Leu	Glu	His	Asp	Ala
			100					105					110		
Ser	Leu	Ser	Arg	Gln	Asp	Ser	Tyr	Phe	Gly	Pro	Ala	Asp	Val	Phe	Asn
		115					120					125			
Glu	Ala	Val	Phe	Asn	Gln	Thr	Lys	Ser	Phe	Trp	Thr	Gly	Asp	Ile	Ile
	130					135					140				
Asp	Val	Gln	Met	Ala	Ala	Asn	Ala	Arg	Ile	Val	Arg	Leu	Leu	Thr	Ser

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145	150	155	160
Asn Leu Thr Asn Pro Glu Tyr Ser Leu Ser Asp Leu Gly Ser Ala Phe	165	170	175
Ser Ile Gly Glu Ser Ala Ala Tyr Ile Gly Ile Leu Gly Asp Lys Lys	180	185	190
Ser Ala Thr Val Pro Lys Ser Trp Val Glu Tyr Leu Phe Glu Asn Glu	195	200	205
Arg Leu Pro Tyr Glu Leu Gly Phe Lys Arg Pro Asn Asp Pro Phe Thr	210	215	220
Thr Asp Asp Leu Gly Asp Leu Ser Thr Gln Ile Ile Asn Ala Gln His	225	230	235
Phe Pro Gln Ser Pro Gly Lys Val Glu Lys Arg Gly Asn Thr Arg Cys	245	250	255
Pro Tyr Gly Tyr His	260		

<210> SEQ ID NO 35
 <211> LENGTH: 261
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: UPO 12 variant

<400> SEQUENCE: 35

Met Lys Ser Leu Ser Phe Ser Leu Ala Leu Gly Phe Gly Ser Thr Leu	1	5	10	15
Val Tyr Ser Ala Pro Ser Pro Ser Ser Gly Trp Gln Ala Pro Gly Pro	20	25	30	
Asn Asp Val Arg Ala Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His	35	40	45	
Gly Phe Leu Pro His Asp Gly Lys Gly Ile Thr Val Asn Lys Thr Ile	50	55	60	
Asp Ala Leu Gly Ser Ala Leu Asn Ile Asp Ala Asn Leu Ser Thr Leu	65	70	75	80
Leu Phe Gly Phe Ala Ala Thr Thr Asn Pro Gln Pro Asn Ala Thr Phe	85	90	95	
Phe Asp Leu Asp His Leu Ser Arg His Asn Ile Leu Glu His Asp Ala	100	105	110	
Ser Leu Ser Arg Gln Asp Ser Tyr Phe Gly Pro Ala Asp Val Phe Asn	115	120	125	
Glu Ala Val Phe Asn Gln Thr Lys Ser Phe Trp Thr Gly Asp Ile Ile	130	135	140	
Asp Val Gln Met Ala Ala Asn Ala Arg Ile Val Arg Leu Leu Thr Ser	145	150	155	160
Asn Leu Thr Asn Pro Glu Tyr Ser Leu Ser Asp Leu Gly Ser Ala Phe	165	170	175	
Ser Ile Gly Glu Ser Ala Ala Tyr Ile Gly Ile Leu Gly Asp Lys Lys	180	185	190	
Ser Ala Thr Val Pro Lys Ser Trp Val Glu Tyr Leu Phe Glu Asn Glu	195	200	205	
Arg Leu Pro Tyr Glu Leu Gly Phe Lys Arg Pro Asn Asp Pro Phe Thr	210	215	220	
Thr Asp Asp Leu Gly Asp Leu Ser Thr Gln Ile Ile Asn Ala Gln His	225	230	235	240
Phe Pro Gln Ser Pro Gly Lys Val Glu Lys Arg Gly Ile Thr Arg Cys				

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245	250	255
Pro Tyr Gly Tyr His		
260		
<210> SEQ ID NO 36 <211> LENGTH: 248 <212> TYPE: PRT <213> ORGANISM: Artificial Sequence <220> FEATURE: <223> OTHER INFORMATION: UPO 12 variant <400> SEQUENCE: 36		
Met Lys Ser Leu Ser Phe Ser Leu Ala Leu Gly Phe Gly Ser Thr Leu		
1 5 10 15		
Val Tyr Ser Ala Pro Ser Pro Ser Ser Gly Trp Gln Ala Pro Gly Pro		
20 25 30		
Asn Asp Val Arg Ala Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His		
35 40 45		
Gly Phe Leu Pro His Asp Gly Lys Gly Ile Thr Val Asn Lys Thr Ile		
50 55 60		
Asp Ala Leu Gly Ser Ala Leu Asn Ile Asp Ala Asn Leu Ser Thr Leu		
65 70 75 80		
Leu Phe Gly Phe Ala Ala Thr Thr Asn Pro Gln Pro Asn Ala Thr Phe		
85 90 95		
Phe Asp Leu Asp His Leu Ser Arg His Asn Ile Leu Glu His Asp Ala		
100 105 110		
Ser Leu Ser Arg Gln Asp Ser Tyr Phe Gly Pro Ala Asp Val Phe Asn		
115 120 125		
Glu Ala Val Phe Asn Gln Thr Lys Ser Phe Trp Thr Gly Asp Ile Ile		
130 135 140		
Asp Val Gln Met Ala Ala Asn Ala Arg Ile Val Arg Leu Leu Thr Ser		
145 150 155 160		
Asn Leu Thr Asn Pro Glu Tyr Ser Leu Ser Asp Leu Gly Ser Ala Phe		
165 170 175		
Ser Ile Gly Glu Ser Ala Ala Tyr Ile Gly Ile Leu Gly Asp Lys Lys		
180 185 190		
Ser Ala Thr Val Pro Lys Ser Trp Val Glu Tyr Leu Phe Glu Asn Glu		
195 200 205		
Arg Leu Pro Tyr Glu Leu Gly Phe Lys Arg Pro Asn Asp Pro Phe Thr		
210 215 220		
Thr Asp Asp Leu Gly Asp Leu Ser Thr Gln Ile Ile Asn Ala Gln His		
225 230 235 240		
Phe Pro Gln Ser Pro Gly Lys Val		
245		

<210> SEQ ID NO 37
 <211> LENGTH: 266
 <212> TYPE: PRT
 <213> ORGANISM: Hypoxylon

<400> SEQUENCE: 37

Met Lys Ser Val Gln Leu Ser Ala Leu Ile Ala Phe Gly Ala Lys Ala		
1 5 10 15		
Val Tyr Ser Phe Pro Ser Ala Asn Ala Pro Trp Ser Gly Pro Gly Thr		
20 25 30		
Asp Asp Val Arg Gly Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His		
35 40 45		

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Gly Phe Leu Pro His Ser Gly Lys Gly Ile Thr Val Asn Lys Thr Ile
 50 55 60
 Asp Ala Leu Asn Ala Gly Leu Asn Met Glu Ala Asp Leu Ala Ala Leu
 65 70 75 80
 Leu Phe Asp Phe Ala Val Thr Thr Asn Pro Thr Pro Asn Ala Ser Tyr
 85 90 95
 Phe Asp Leu Asp His Leu Thr Arg His Asn Ile Leu Glu His Asp Ala
 100 105 110
 Ser Ile Ser Arg Gln Asp Ser Tyr Phe Gly Arg Ala Asp Ile Leu Asn
 115 120 125
 Glu Ala Val Phe Asn Gln Thr Lys Ser Tyr Trp Thr Gly Glu Leu Val
 130 135 140
 Asp Ile Gln Met Ala Ala Asn Ala Arg Val Ala Arg Leu Met Thr Ser
 145 150 155 160
 Asn Leu Thr Asn Pro Glu Tyr Ser Leu Ser Asp Thr Gly Ser Val Phe
 165 170 175
 Ser Ile Gly Glu Ser Ala Ala Tyr Val Gly Ile Leu Gly Asp Lys Val
 180 185 190
 Ser Gly Thr Val Pro Lys Thr Trp Leu Ile Tyr Leu Phe Glu Gln Glu
 195 200 205
 Arg Leu Pro Tyr Glu Leu Gly Phe Lys Arg Pro Val Asp Pro Phe Thr
 210 215 220
 Glu Asp Asp Leu Phe Asn Met Ser Glu Ala Ile Arg Asp Ala Gln His
 225 230 235 240
 Phe Pro Gln Asp Ile Gly Lys Val Thr Lys Arg Gly Asn Lys Ala Arg
 245 250 255
 Cys Pro His Gly Tyr Cys Ile Glu Val Leu
 260 265

<210> SEQ ID NO 38

<211> LENGTH: 250

<212> TYPE: PRT

<213> ORGANISM: Rosellinia necatrix

<400> SEQUENCE: 38

Met Lys Leu Thr Thr Leu Leu Phe Pro Ala Val Val Leu Gly Ala Ala
 1 5 10 15
 Cys Pro Tyr Gly Thr Phe Lys Pro Glu Glu Pro Thr Asp Thr Arg Gly
 20 25 30
 Val Cys Pro Met Leu Asn Ala Leu Ala Asn His Gly Phe Leu Pro Arg
 35 40 45
 Asp Gly Arg Asn Ile Asn Glu Asn Gln Thr Val Thr Ala Leu Asn Asn
 50 55 60
 Ala Leu Asn Leu Thr Pro Asp Phe Gly Arg Phe Leu Phe Thr Ala Gly
 65 70 75 80
 Arg Leu Ser Asn Pro Lys Pro Asn Ser Thr Thr Phe Asp Leu Asn His
 85 90 95
 Leu Asp Arg His Asn Leu Phe Glu His Asp Gly Ser Leu Ser Arg Gln
 100 105 110
 Asp Ala His Phe Gly Gln Trp Ser Arg Phe Asn Gln Thr Val Trp Asn
 115 120 125
 Trp Thr Met Gln Tyr Trp Thr Gly Asp Ile Leu Asp Val Gln Met Val
 130 135 140
 Ala Asn Gly Arg Ala Gln Arg His Thr Arg Ser Asn Leu Thr Asn Pro

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145	150	155	160
Asp Tyr Ala Leu Ser Val Val Gly Tyr Asp Phe Ser Val Ala Glu Asn			
	165	170	175
Ala Ala Leu Leu Ser Ile Ile Gly Asp Lys Val Thr Gln Thr Cys Pro			
	180	185	190
Lys Lys Phe Val Asp Tyr Leu Phe Val Asn Glu Glu Leu Pro Tyr Ser			
	195	200	205
Val Gly Trp Lys Lys Ser Glu Leu Pro Ile Ala Leu Glu Asp Leu Ile			
	210	215	220
Arg Thr Phe Arg Asp Ile Glu Leu Ala Thr Ala Phe Pro Ala Pro Pro			
225	230	235	240
Pro Pro Asp Asn Ser Gly Glu Ile Phe Ala			
	245	250	

<210> SEQ ID NO 39

<211> LENGTH: 263

<212> TYPE: PRT

<213> ORGANISM: Daldinia sp. EC12

<400> SEQUENCE: 39

Met Lys Leu Thr Phe Met Ser Ser Val Val Thr Leu Gly Ser Ala Val			
1	5	10	15
Ala Ala Tyr Pro Thr Ser Trp Glu Ala Pro Gly Pro Asn Asp Val Arg			
	20	25	30
Gly Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His Gly Phe Leu Pro			
	35	40	45
His Asp Gly Lys Asn Ile Asn Val Asn Asn Thr Ala Glu Ala Leu Ser			
	50	55	60
Lys Gly Leu Asn Leu Ala Trp Glu Leu Gly Val Asp Leu His Asp Phe			
65	70	75	80
Ala Val Met Ala Asn Pro Gln Pro Asn Ala Thr Thr Phe Asp Leu Asp			
	85	90	95
His Leu Ser Arg His Asn Val Leu Glu His Asp Gly Ser Leu Ser Arg			
	100	105	110
Gln Asp Ala His Phe Gly Pro Pro Asp Val Phe Asn Glu Ala Val Phe			
	115	120	125
Asn Gln Thr Val Ser Tyr Trp Thr Gly Asp Val Val Thr Met Gln Met			
	130	135	140
Ala Ala Asn Ala Arg Leu Ala Arg Leu Met Thr Ser Asn Leu Thr Asn			
145	150	155	160
Pro Glu Tyr Ser Leu Ser Asp Leu Gly Ser Gly Phe Ser Ile Gly Glu			
	165	170	175
Ser Val Val Tyr Leu Leu Val Leu Gly Asn Lys Asp Thr Ala Glu Ala			
	180	185	190
Pro Lys Asn Tyr Leu Glu Tyr Trp Phe Arg Asn Glu Arg Leu Pro Tyr			
	195	200	205
Glu Leu Gly Trp Glu Arg Pro Asn Val Ile Met Thr Gly Asp Asp Leu			
	210	215	220
Gly Asn Ala Met Asp Lys Leu Val Thr Leu Gln His Phe Pro Gln Ser			
225	230	235	240
Pro Gly Lys Ile Thr Ser Asp Pro Glu Lys Ala Ser Ala Lys Leu Ala			
	245	250	255
Gly Lys Arg His Leu Phe His			
	260		

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<210> SEQ ID NO 40
 <211> LENGTH: 265
 <212> TYPE: PRT
 <213> ORGANISM: Monosporascus

<400> SEQUENCE: 40

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Met Lys Phe Glu Leu Ala Ala Thr Ile Leu Ala Ala Gly Thr Ala Ser
1           5           10           15
Ala Phe Arg Leu Lys Ala Arg Asp Thr Tyr Asp Trp His Pro Pro Ala
20          25          30
Tyr Gly Asp Val Arg Gly Pro Cys Pro Met Leu Asn Thr Leu Ala Asn
35          40          45
His Gly Tyr Leu Pro Arg Asn Gly Lys Asp Ile Thr Glu Asn Arg Thr
50          55          60
Ile Glu Ala Leu Gly Thr Ala Leu Ser Ile Asp Ser Glu Leu Ala Gln
65          70          75          80
Leu Leu Phe Glu Gln Ala Ile Thr Thr Asn Pro Ala Pro Asn Ala Thr
85          90          95
Thr Phe Ser Leu Asn Asp Leu Val Arg His Asn Ile Leu Glu His Asp
100         105         110
Ala Ser Leu Ser Arg Val Asp Phe Tyr Phe Gly Asn Pro Gln Pro Phe
115         120         125
Asn Gln Thr Val Phe Ala Gln Thr Arg Ser Tyr Trp Thr Thr Pro Ile
130         135         140
Ile Asp Val Gln Gln Ala Ala Asn Ala Arg Trp Ala Arg Val Glu Thr
145         150         155         160
Ser Asn Ala Thr Asn Pro Asn Phe Thr Leu Ser Thr Leu Gly Glu Arg
165         170         175
Phe Ser Tyr Gly Glu Ser Ala Ala Tyr Ile Val Ile Leu Gly Asn Lys
180         185         190
Ile Thr Gly Thr Val Pro Arg Asp Trp Val Glu Tyr Leu Phe Glu Asn
195         200         205
Glu Arg Leu Pro Leu Glu Ile Gly Trp Thr Arg Arg Thr Gly Ser Ile
210         215         220
Thr Arg Asn Asp Leu Glu Asp Val Met Gln Gln Ile Tyr Ala Ala Thr
225         230         235         240
Pro Asn Asn Asn Ala Thr Thr Asn Ser Trp Arg Gly Asn Pro Arg Ala
245         250         255
Leu His Met Ala Val Arg Ala Ser Ala
260         265

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<210> SEQ ID NO 41
 <211> LENGTH: 273
 <212> TYPE: PRT
 <213> ORGANISM: Aspergillus aculeatus

<400> SEQUENCE: 41

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Met Arg Tyr Phe Val Leu Ala Cys Ala Pro Leu Leu Tyr Ala Val Thr
1           5           10           15
Leu Ala Phe Pro Arg Ala Asp Tyr Val Ser Glu Gly Lys Leu Pro Ala
20          25          30
Gly His Pro Pro Leu Asp Trp Lys Pro Ala Gly Leu Gly Asp Ala Arg
35          40          45
Ala Pro Cys Pro Met Leu Asn Thr Leu Ala Asn His Gly Tyr Leu Pro
50          55          60

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His	Asp	Gly	Lys	Asp	Ile	Thr	Lys	Ala	His	Thr	Ile	Ala	Ala	Leu	His	65	70	75	80
Ser	Ala	Leu	Asn	Ile	Asp	Arg	Glu	Leu	Ala	Gln	Tyr	Leu	Phe	Gln	Glu	85	90	95	
Ala	Leu	Thr	Thr	Asn	Pro	Ala	Ala	Asn	Ala	Thr	Thr	Phe	Ser	Leu	Asn	100	105	110	
Asp	Leu	Ser	Arg	His	Asn	Ile	Leu	Glu	His	Asp	Ala	Ser	Leu	Ser	Arg	115	120	125	
Leu	Asp	Tyr	Tyr	Phe	Gly	Asp	Asn	His	Asp	Phe	Asn	Gln	Ala	Ile	Phe	130	135	140	
Asp	Gln	Thr	Arg	Gln	His	Trp	Pro	Asp	Pro	Ile	Ile	Thr	Val	Gln	Ala	145	150	155	160
Ala	Ala	Asn	Ala	Arg	Glu	Ala	Arg	Val	Arg	Thr	Ser	Asn	Ala	Thr	Asn	165	170	175	
Pro	Thr	Phe	Thr	Leu	Ser	Glu	Leu	Gly	Thr	Ala	Phe	Gly	Tyr	Gly	Glu	180	185	190	
Thr	Ala	Ala	Tyr	Ile	Ile	Ile	Leu	Gly	Asn	Lys	Thr	Ser	Gly	Leu	Val	195	200	205	
Asp	Arg	Ser	Trp	Val	Glu	Tyr	Leu	Phe	Glu	Asn	Glu	Arg	Leu	Pro	Val	210	215	220	
Glu	Leu	Gly	Trp	Thr	Arg	His	Glu	Glu	Ala	Val	Ser	Met	Asp	Asp	Leu	225	230	235	240
Glu	Gly	Met	Met	Gln	Glu	Val	Ile	Asn	Ala	Thr	Gly	His	Ala	Glu	Glu	245	250	255	
Val	Lys	Arg	Glu	Leu	Val	Arg	Arg	Gly	Asp	Leu	His	Val	Gly	Arg	Arg	260	265	270	

Ala

The invention claimed is:

1. A modified unspecific peroxygenase (UPO) comprising an amino acid sequence having at least 75% sequence identity to the polypeptide of SEQ ID NO: 12 and having increased peroxygenase activity as compared to unmodified wild-type UPO12, which is the polypeptide of SEQ ID NO:12, wherein the modification is a modification of at least one amino acid selected from D145, E249, D253, and/or C256 of the polypeptide of SEQ ID NO:12, and wherein the peroxygenase activity is increased by 1.3-fold or more when measured in an ABTS assay or a 2,6-DMP assay.

2. The modified UPO of claim 1, comprising at least a mutation corresponding to D145Y, D253N, D253I, C256S, and/or introduction of a stop codon at a position corresponding to C256 or E249 of SEQ ID NO:12.

3. The modified UPO of claim 1, wherein the peroxygenase activity is 1.5-fold, 2.0-fold, or more increased when measured in an ABTS assay or a 2,6-DMP assay.

4. The modified UPO of claim 1, comprising SEQ ID NO: 31, SEQ ID NO:32, SEQ ID NO:33, SEQ ID NO:34, SEQ

ID NO:35, or SEQ ID NO: 36, or an amino acid sequence having at least 80%, 85%, 90%, 95%, 97%, 98%, or 99% identity to SEQ ID NO:31, SEQ ID NO:32, SEQ ID NO:33, SEQ ID NO:34, SEQ ID NO: 35, or SEQ ID NO:36.

5. A modified unspecific peroxygenase (UPO) having increased peroxygenase activity as compared to the unmodified wild-type UPO12 of SEQ ID NO: 12, wherein the modified UPO comprises SEQ ID NO:30, or an amino acid sequence having at least 90% sequence identity to SEQ ID NO:30 and comprising an amino acid modification of the amino acid corresponding to S24 of SEQ ID NO:12.

6. A method of performing a chemical reaction on a substrate, comprising reacting the substrate with the modified UPO of claim 1.

7. The method of claim 6, wherein the chemical reaction is an organic synthesis process, a polymerization process, a pharmaceutical production process, an environmental application, an application in consumer products, or a surface modification.

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